Part V

Credit

Credit risk is priced twice over: once as the chance a borrower fails, and again as the cost of facing a counterparty who might. This part runs from default modeling through the CDS market and the securitization machinery that failed in 2007–2009, and ends with the counterparty and capital regime built in response. The crisis chapter is history with a purpose—most of what Chapters 8, 9, and 27 require exists because of something in it.

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hazard rates, structural models, and the credit triangle

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Chapter 24
Credit Risk and Default Modeling

Chapter 9 computed a credit valuation adjustment and took two inputs as given: a default rate read off a CDS spread through the credit triangle, assumed flat, and a recovery rate assumed to be 40%. It pointed here for the machinery behind both. This chapter supplies it, and then asks what the inputs actually mean.

Credit risk is among the oldest financial risks and the largest one most banks carry. Bank loan books, the corporate and sovereign bond markets, the credit default swaps of Chapter 25, the securitizations of Chapter 26, and the counterparty exposures of Chapter 27 all price off the two quantities this chapter is about, and anyone who lends, holds a bond, or faces a counterparty is exposed to them whether or not they model them. Section 24.16 says who those holders are and how the risk has moved between them since the reforms.

The subject divides cleanly. Credit risk is two questions (whether a borrower defaults, and how much is recovered if it does), and almost every difficulty in this part of the book comes from market prices identifying only their product. Sections 24.1 to 24.4 establish that, with the credit triangle and the hazard-curve bootstrap on the way. Sections 24.5 to 24.8 are the two families of model that try to say more: structural models, which explain default as an option finishing out of the money and fail at short maturities, and reduced-form models, which assume it and fit anything. Sections 24.9 to 24.12 are the gap between what prices imply and what happens: risk-neutral against historical default rates, ratings and migration, and the sovereign case in which default is a choice. Section 24.13 introduces the correlation machinery, Section 24.14 turns it into the credit VaR against which a loan book is capitalized, Section 24.15 reads a hazard rate from a bond price and sets it against the CDS, and Section 24.17 records where the models came from.

24.1 THE TWO QUESTIONS

A defaultable bond promises \$100 and delivers either \$100 or something less. Valuing it requires a probability and a payoff.

Default probability. The default probability is not a single number but a curve: the chance of default by each future date. Write τ for the random time of default and $Q(t) = \mathbb{Q}(\tau > t)$ for the **survival probability**, the chance that the borrower is still performing at t. The convention is to work not with Q itself but with the **hazard rate** $\lambda(t)$, the probability of defaulting in the next instant given survival to t, per unit of time:

$$\lambda(t) = \lim_{\Delta t \to 0} \frac{\mathbb{Q}(t < \tau \le t + \Delta t \mid \tau > t)}{\Delta t} = -\frac{Q'(t)}{Q(t)} = -\frac{d}{dt} \ln Q(t) \tag{24.1}$$

The second equality is the definition of a conditional probability: the probability of defaulting in $(t, t + \Delta t]$ is $Q(t) - Q(t + \Delta t)$, and dividing by the probability $Q(t)$ of being there to default gives the conditional rate. Integrating the last form gives the survival probability to time T,

$$Q(T) = \exp\left(- \int_0^T \lambda(u)\, du \right) \tag{24.2}$$

For a constant hazard this is $e^{-\lambda T}$, and the density of default at t is $\lambda e^{-\lambda t}$.

The hazard rate is not the default probability. A 3% hazard rate is an intensity of 3% per year: *given survival to the start of a year*, the chance of defaulting during it is $1 - e^{-0.03} = 2.96\%$, and over ten years the cumulative default probability is $1 - e^{-0.3} = 25.9\%$, not 30%. The distinction matters as soon as the horizon is long or the borrower is weak.

Under which probability? Two versions of every quantity above will appear in this chapter, and they are not close. The one that prices a contract is the risk-neutral probability of Chapters 13 and 28 (written $\mathbb{Q}$ there and here), and it is the one the market reveals: a CDS spread read through the next two sections gives a risk-neutral λ, because the spread is a price. The one that forecasts is the **real-world** probability ($\mathbb{P}$), estimated from default records. Everything up to Section 24.8 is risk-neutral, and should be read as pricing rather than prediction; Section 24.9 sets the two side by side and finds them apart by a factor of several.

Recovery. Recovery is what the claim is worth after default, conventionally quoted as a fraction R of face value, with $1 - R$ the **loss given default**. Forty percent is the market's standard assumption for senior unsecured corporate debt, and Section 24.4 explains why that number is used far beyond the evidence supporting it. Everything else in this chapter is a way of estimating one of these two, or an admission that only their product is observable.

24.2 THE CREDIT TRIANGLE

A credit default swap pays $1 - R$ per unit of notional if the **reference entity** (the borrower the contract is written on) defaults before maturity T, in exchange for a spread s per year paid by the protection buyer until default or maturity. Its par spread is the s at which the two legs have the same value today, and both legs are integrals over the risk-neutral survival curve of Section 24.1. Write $P(0,t)$ for the risk-free discount factor and take the premium to be paid continuously for the moment. The premium leg collects $s\,dt$ over each instant the borrower is still alive, so its value is

$$\text{premium leg} = s \int_0^T P(0,t)\, Q(t)\, dt.$$

The protection leg pays $1 - R$ at the moment of default, and by (24.1) the probability of that moment falling in $(t, t + dt]$ is $\lambda(t)Q(t)\,dt$, the hazard rate times the probability of having survived to t, so

$$\text{protection leg} = (1 - R) \int_0^T P(0,t)\, \lambda(t)\, Q(t)\, dt.$$

If the hazard rate and the interest rate are both flat, $Q(t) = e^{-\lambda t}$ and $P(0,t) = e^{-rt}$, the integrand decays at $r + \lambda$, and

$$\text{protection leg} = (1 - R)\,\lambda \int_0^T e^{-(r+\lambda)t}\, dt = (1 - R)\,\lambda\, \frac{1 - e^{-(r+\lambda)T}}{r + \lambda}.$$

For a flat hazard rate on any discount curve, λ comes outside the integral, the same integral $\int_0^T P(0,t)Q(t)\,dt$ appears in both legs, and setting them equal cancels it:

$$\boxed{s = \lambda\,(1 - R)} \tag{24.3}$$

Neither the discount curve nor the level of the survival probability remains in the answer; the par spread is the default rate times the loss when default happens, which is what the price of insurance ought to be. The result is usually called the **credit triangle**, because any two of s, λ and $1 - R$ give the third. Reading a spread as a hazard rate therefore requires only a recovery assumption, not a discount curve, which is why traders can do the conversion mentally; and a change in the discount curve alone barely moves the par spread, which is why CDS

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spreads did not move mechanically when the discounting convention changed in the reforms of Chapter 4. The triangle is usually presented as an approximation valid for small λ, and the standard statement puts the error in the wrong place.

Example 24.1 The credit triangle is exact for a continuous premium, and the error people quote is the accrued premium

Price a five-year CDS with a flat hazard rate $\lambda = 1\%$, discounting at $r = 3\%$, recovery $R = 40\%$, three ways: with the premium paid continuously; quarterly, with the premium accrued since the last payment date paid on default, as real contracts do; and quarterly with no accrual. The protection leg is the same in all three; from the flat-rate formula above, with $r + \lambda = 0.04$, it is $0.6 \times 0.01 \times (1 - e^{-0.2})/0.04 = 0.027190$ per unit of notional. The premium legs are the spread s times an annuity, and the annuities differ: 4.5317, 4.5148 and 4.5091 ([TN25] works them out one quarter at a time). The par spread sets each premium leg equal to the protection leg:

Continuous premium	$4.5317\, s = 0.027190$	$\Rightarrow$	$s = 60.00$ bp
Quarterly, with accrual	$4.5148\, s = 0.027190$	$\Rightarrow$	$s = 60.23$ bp
Quarterly, no accrual	$4.5091\, s = 0.027190$	$\Rightarrow$	$s = 60.30$ bp

The first is $\lambda(1 - R) = 0.01 \times 0.6$ exactly, as the derivation of (24.3) says it must be. The second is 0.376% higher, because a premium paid at the end of a quarter is discounted for half a quarter longer, on average, than the same money paid continuously through it; to first order the excess is $r\Delta/2$ for a payment interval $\Delta = 0.25$, and Chapter 25 carries the same effect as about a basis point on a 250 bp contract. The third drops the accrued premium, and that is the "approximation error" of the textbook statement: a tenth of a percent here. It is the value of a contract feature, not an error in the triangle, and unlike the 0.376% it grows with the hazard rate: at $\lambda = 20\%$ the three spreads are 1200.00, 1204.47 and 1235.17 bp, and the missing accrual is worth 2.5%.

24.3 BOOTSTRAPPING A HAZARD CURVE

Spreads are quoted at several maturities and they are not flat. Extracting the implied hazard curve is the credit analog of Chapter 4's zero-curve bootstrap: work outward, taking each quote as determining the hazard rate over the interval that quote is the first to cover.

Example 24.2 From a spread curve to a hazard curve

An issuer's CDS curve is quoted at the maturities below; recovery is assumed at 40% and rates at 3%:

Maturity	Spread (bp)	Bootstrapped λ	Credit-triangle guess
1y	80	1.333% (0–1y)	1.333%
3y	130	2.617% (1–3y)	2.167%
5y	170	3.998% (3–5y)	2.833%
7y	190	4.222% (5–7y)	3.167%
10y	200	3.885% (7–10y)	3.333%

The second row shows the step. The one-year quote fixes $\lambda_1 = 0.0080/0.6 = 1.333\%$ exactly by (24.3), because a single bucket is the constant-hazard case. The three-year quote must then be reproduced by a contract whose first year runs at λ_1 and whose next two years run at an unknown λ_2. With a continuously paid premium both legs are integrals of $e^{-rt}Q(t)$, and the integral splits at the knot: write $A_1 = \int_0^1 e^{-(r+\lambda_1)t}\,dt = 0.9786$ for the first year and $A_2 = \int_1^3 e^{-rt}Q(t)\,dt = Q(1)\,e^{-r}\int_0^2 e^{-(r+\lambda_2)u}\,du$ for the next two (on $[1,3]$, $Q(t) = Q(1)\,e^{-\lambda_2(t-1)}$; substitute $u = t - 1$), where $Q(1) = e^{-\lambda_1} = 0.9868$ is the probability of reaching the second bucket alive. The par

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condition is

$$s_3\,(A_1 + A_2) = (1 - R)\,(\lambda_1 A_1 + \lambda_2 A_2),$$

one equation in λ_2, since A_2 depends on it. At $\lambda_2 = 2.617\%$, $A_2 = 1.8115$; the premium leg is $0.0130 \times 2.7902 = 0.036272$ and the protection leg is $0.6 \times (0.013049 + 0.047405) = 0.036272$, so the equation balances. Each later bucket repeats the step with one more known piece and one unknown. The credit triangle applied to the three-year spread gives $0.0130/0.6 = 2.167\%$ instead, close to the time-weighted average of the two bucket rates, $(1.333\% + 2 \times 2.617\%)/3 = 2.19\%$, but not the rate for years one to three.

The last column applies (24.3) directly to each quoted spread. It is the same as the bootstrapped rate for the first bucket and different after that, most of all in the 3–5 year bucket, and the reason is the reason a swap rate is not a forward rate: a spread is an average over the whole life of the contract, and the hazard rate for a later bucket must exceed that average when the curve is upward sloping.

At the 3–5 year bucket the difference is large: 3.998% against 2.833%, a factor of 1.41. Anyone using the five-year spread to price something that pays off only in years three to five will understate the hazard rate for those years by almost 30%.

The implied cumulative default probabilities are

	1y	3y	5y	7y	10y
Cumulative default probability	1.32%	6.36%	13.55%	20.55%	29.29%

Note the shape of the hazard curve: rising through seven years and then falling. An upward-sloping spread curve is the norm for **investment-grade** issuers, those rated BBB (Baa) or higher; an issuer rated below that line is **speculative grade**. The upward slope reflects the same mean reversion that Section 19.6 found in volatility: a currently healthy firm is more likely to deteriorate than to improve. The fall after seven years is arithmetic: the ten-year spread is only 10 bp above the seven-year one, so the last bucket cannot carry a hazard rate as high as the one before it. Distressed issuers show the same effect in extreme form, with an inverted spread curve, for the same reason inverted volatility curves appear in stress: if the firm survives the next year, the danger has largely passed and the hazard falls.

Figure 24.1 draws the table's last two columns.

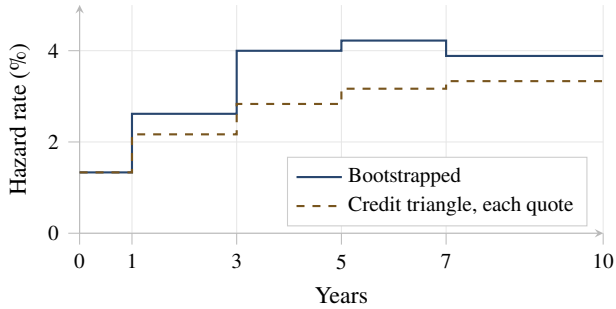


Figure 24.1: Hazard rates bootstrapped from the spread curve of Example 24.2, against the credit triangle applied to each quoted spread directly.

24.4 RECOVERY: THE OTHER UNKNOWN

The bootstrap above assumed $R = 40\%$. Where does that come from, and what happens if it is wrong? The answer to the first question is convention, supported by an average that conceals almost everything of interest.

Recovery varies by seniority. Table 24.1 gives Moody's long-run averages measured two ways: the market's bid for the defaulted debt shortly after default, close to what a CDS auction

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Table 24.1: Average recovery rates by priority of claim (%), issuer-weighted for trading prices. Moody's, March 2023 study.

	Trading price, 1983–2022	Ultimate, 1987–2022
First-lien bank loan	65.7	71.7
Second-lien bank loan	33.3	–
Senior unsecured bank loan	46.3	–
First-lien bond	55.0	61.2
Second-lien bond	44.2	–
Senior unsecured bond	37.7	47.1
Senior subordinated bond	31.0	–
Subordinated bond	32.0	27.8
Junior subordinated bond	22.3	–

sets, and the ultimate recovery when the default is resolved, usually a year or two later (North American nonfinancial issuers; the figures for all term loans and all senior secured bonds stand in the first-lien rows). The ranking broadly follows priority, from about two-thirds on first-lien loans to about a fifth on junior subordinated bonds, though not strictly. Senior unsecured bonds, the class the 40% convention describes, recovered 37.7% at the trading price and 47.1% at resolution; the two come from different samples, so the gap is not a measured increase.

Recovery is also correlated with the default rate. Recoveries are lowest in the years when defaults are highest (Altman, Brady, Resti, and Sironi 2005), because a firm defaulting in a recession is selling its assets into a market full of other distressed sellers.

Example 24.3 Hazard and recovery are not separately identified

A five-year CDS trades at 170 bp. Which (λ, R) pair produced it?

Recovery	Implied λ	5-year default probability
20%	2.125%	10.08%
40%	2.833%	13.21%
60%	4.250%	19.14%
80%	8.500%	34.62%

All four price the CDS identically. The market observes only the product $\lambda(1 - R)$, so the recovery assumption is not extracted from the spread—it is imposed on it.

Usually this does not matter, because the thing being priced also depends only on the product. When it does not, the split matters a great deal, and three cases show how much.

A digital default swap pays a fixed amount on default regardless of recovery, so its value depends on λ alone. The table's four rows price it roughly in proportion to the last column, a range of more than three to one; between the 40% and 60% rows alone a five-year digital is worth 45% more. The contract has only ever traded thinly, mostly between dealers separating a view on default from a view on recovery; its use here is as the sharpest test of the split.

A first-to-default basket pays $1 - R$ on the first default among a group of borrowers. To first order its value is the sum of the borrowers' $\lambda_i(1 - R)$, the product again, so most of the ambiguity cancels; what survives comes from the probability of at least one default being concave in the λ_i. On five independent borrowers each quoted at 170 bp, the basket prices about 14% lower at 60% recovery than at 40%, where the digital prices 45% higher.

And any calculation of expected loss for capital or provisioning needs the split, because regulation asks for probability of default and loss given default separately.

The practical convention is to fix R by seniority and sector and let λ absorb everything, which is defensible for pricing and indefensible for risk measurement, for the reasons the case study below sets out.

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Case Study 24.1 Why the 40% convention survives evidence against it

The recovery assumption of Section 24.4 is used with a confidence the data do not support, and the gap follows a pattern that recurs throughout this part of the book.

Realized recoveries are enormously dispersed. The 40% figure is a long-run average across senior unsecured corporate bonds. Year by year the average moves a good deal: senior unsecured bonds recovered 48.7% in 2021 and 31.1% in 2022, when defaults by Russian issuers (recovering 15.5%) and Chinese property developers (25.3%) pulled it down.

The dispersion is systematic, not noise. Recovery depends on seniority, on whether the debt is secured and by what, on the industry's asset tangibility, on the jurisdiction's insolvency regime, and on where in the credit cycle the default occurs. An asset-heavy firm in a creditor-friendly jurisdiction defaulting in a boom is a different proposition from an asset-light firm in a debtor-friendly one defaulting in a recession, and the two have almost nothing in common but the letter on the rating.

Recovery moves with the thing it is multiplied by. A portfolio model that fixes recovery at 40% and lets only the default rate vary understates tail losses twice over: once because more borrowers default, and once because each one recovers less. In a bad year in which twice the average number of borrowers default and recovery falls from 40% to 25%, the loss is $2 \times 0.75/0.60 = 2.5$ times the average year's, where the fixed-recovery model says 2.

So why does the convention hold? Because for pricing a CDS it does not matter. Example 24.3 showed that the spread identifies only $\lambda(1 - R)$, so a desk that fixes R and lets λ absorb the residual prices the contract correctly whatever R truly is. The convention is not a belief about recovery; it is a coordinate choice, and within its domain it is harmless.

The damage is done when the coordinate is exported. Multiplying the extracted λ back by the same 60% loss given default is harmless: it returns the spread, whatever the true recovery. The harm comes from pairing that λ with a recovery from somewhere else (a risk system that takes the λ extracted at 40% and applies its own loss-given-default estimate of 75% has used two recoveries for one borrower), or from feeding λ into a calculation that is nonlinear in the default probability, such as the worst-case default rate of Section 24.14, where the product no longer carries through. A capital calculation that requires probability of default and loss given default separately is asking for a decomposition the market never provided.

The general lesson is one this book has met before. Section 21.10 observed that a calibration encodes an assumption about what the model is for; Section 22.6 observed that VIX is well defined and is not the price of anything. A number can be exactly right for the purpose that produced it and badly wrong one step away, and credit is the area where that distance is shortest and least visible.

24.5 STRUCTURAL MODELS: MERTON

The two families of model differ in whether default is explained or assumed. Merton's insight is that a firm's equity is a call option on its assets. If the firm's assets are worth V_T at the date its debt of face value D matures, shareholders receive $\max(V_T - D, 0)$: they keep the surplus and walk away from the shortfall. The strike is the face value of the debt.

Figure 24.2 shows the consequence for the other side. Debt holders receive $\min(V_T, D)$, which is D less a put struck at D: lending is selling a put on the borrower's assets, and the credit spread is the premium on that put.

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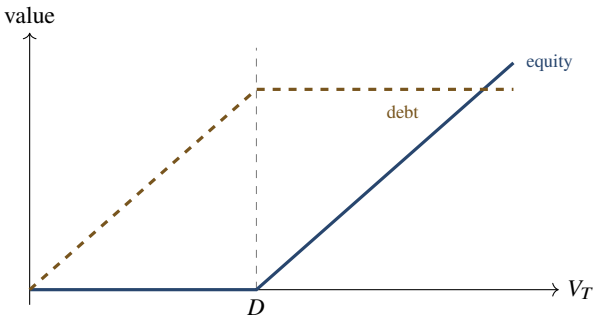


Figure 24.2: Equity and debt payoffs at the debt's maturity.

This is the sharpest available statement of the conflict Section 17.11 flagged. Equity is a call, so equity holders gain from volatility; debt is short a put, so debt holders lose from it. An executive paid in options is paid in a claim whose value rises with firm risk, and every increase in that risk, at a given firm value, is a transfer from creditors. Nothing in the payoff structure aligns the two, and no amount of governance language changes the derivative.

24.6 CALIBRATING MERTON FROM WHAT IS OBSERVABLE

Firm asset value and asset volatility are not observable. Equity value and equity volatility are, and two equations link them.

Write V for the firm's asset value today and σ_V for its volatility, with D and T the face value and maturity of the debt as in Section 24.5. The first equation is Chapter 15's formula applied to the call that equity is:

$$E = V\,N(d_1) - De^{-rT}N(d_2), \qquad d_1 = \frac{\ln(V/D) + \left(r + \tfrac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}}, \qquad d_2 = d_1 - \sigma_V\sqrt{T} \tag{24.4}$$

The second follows from applying Itô's lemma to E as a function of V, which is Chapter 14's calculation: equity volatility is asset volatility times the call's elasticity to the firm, $N(d_1)V/E$, the percentage change in equity value for a one percent change in asset value:

$$\sigma_E E = N(d_1)\,\sigma_V V \tag{24.5}$$

Two equations, two unknowns, and everything on the left-hand side is quoted. There is no closed form for V and σ_V; the pair is solved numerically, which is all Example 24.4 does.

The output the model is actually used for is the argument of the second normal:

$$\mathrm{DD} = d_2 = \frac{\ln(V/D) + \left(r - \tfrac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}} \tag{24.6}$$

This is the **distance to default** (how many standard deviations of asset value separate the firm from its debt), and the risk-neutral default probability is

$$\mathbb{Q}(V_T < D) = 1 - N(d_2) = N(-\mathrm{DD}).$$

Example 24.4 Distance to default from a share price

A firm has equity worth 40 with 55% volatility, and debt of face value 80 maturing in five years. Rates are 5%.

Solving the two equations simultaneously gives:

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Implied asset value	95.87
Implied asset volatility	27.2%
Distance to default	0.41 standard deviations
Risk-neutral 5-year default probability	34.26%
Implied credit spread	218 bp

The last row follows from the first. Debt is the firm less its equity, $95.87 - 40 = 55.87$, against $80e^{-0.25} = 62.30$ if it were riskless, and the continuously compounded spread that discounts one to the other is

$$s = -\frac{1}{T}\ln\frac{55.87}{62.30} = 2.18\%.$$

The first two rows can be checked by substitution. At the solution, before rounding, $d_1 = 1.0131$ and $d_2 = 0.4054$, so (24.4) gives $95.87 \times 0.8445 - 62.30 \times 0.6574 = 80.96 - 40.96 = 40.00$, the observed equity value, and (24.5) gives $\sigma_E = 0.8445 \times 0.272 \times 95.87/40 \approx 55\%$, the observed volatility. The same $d_2 = 0.4054$ is the third row, and $N(-0.4054) = 34.26\%$ is the fourth.

The distance to default of (24.6) is the model's most durable output. What Merton's model establishes is a monotonic relationship between distance to default and default probability: a firm further from its **default point** (the asset value at which it defaults, the face value D in Merton's model), measured in standard deviations of its own asset value, defaults less often. That ordering is more robust than the default probability that follows from it, because it does not require the normality assumption to be right, only the monotonicity, which is the license the commercial mapping below takes. Two observations about the numbers follow.

Equity is twice as volatile as the firm. The ratio of 55% to 27.2% is the equity's elasticity to the firm, $N(d_1)V/E = 0.8445 \times 95.87/40 = 2.02$: the leverage $V/E = 2.40$ scaled down by the call's delta. It rises as the firm approaches default, which is why distressed equities are so volatile and why Section 19.9's leverage explanation of the skew is this model in miniature.

And the implied recovery is high. The expected loss is 10.3% of the riskless value ($1 - 55.87/62.30$) against a 34.26% default probability, an implied recovery of $1 - 10.3/34.26$, near 70%. Merton recovers too much because, on default, the bondholders take over the whole of the firm's assets at no cost. With 27% asset volatility those assets are still worth on average about 70% of the debt's face when the firm defaults (55.88 of 80, a risk-neutral expectation, which is exactly the implied recovery), and the model has no bankruptcy costs or fire-sale discounts to reduce that.

What the commercial versions change. The framework became a product before it became a consensus, and the standard implementation is the one built by **KMV** (the firm Kealhofer, McQuown and Vasicek founded, named from their initials), now part of Moody's Analytics. It departs from Merton in three ways, each replacing one of the model's weakest assumptions with data rather than with more theory.

The default point is not the face value of debt. Firms do not fail the moment assets touch total liabilities, because long-dated debt need not be repaid yet. The default point is set at short-term debt plus half of long-term debt. That rule is not derived from the model; KMV chose it because, in its database of defaulted firms, asset value at the time of default was typically around that level.

The default probability does not come from the normal distribution. Instead of reading $N(-\text{DD})$, distance to default is mapped to an **expected default frequency** (EDF) through an empirical table built from a large default history, and the distance itself is computed with the firm's expected asset return in place of r, because the target is a real-world frequency rather than a price. This is the change that matters, because Section 24.7 is about to show that the normal tail is far too thin at short horizons. It is also why distance to default outlives the model that produced it: the mapping uses only the ordering, so (24.6) can be right about which firms are riskier while $N(-\text{DD})$ is wrong about how risky they are.

And the calibration is iterative. Rather than solving (24.4) and (24.5) once, the implied

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asset series is used to re-estimate σ_V and the system is re-solved to convergence, which is more stable when the equity volatility going in is itself an estimate.

The pattern recurs. A commercial version of an academic model is usually that model with its weakest assumption replaced by a fitted table, and the table holds both the value and the secrecy. Chapter 26 shows the same substitution made for default correlation, and made badly.

24.7 WHAT MERTON GETS WRONG

The model's characteristic failure is at the short end, and it can be measured precisely.

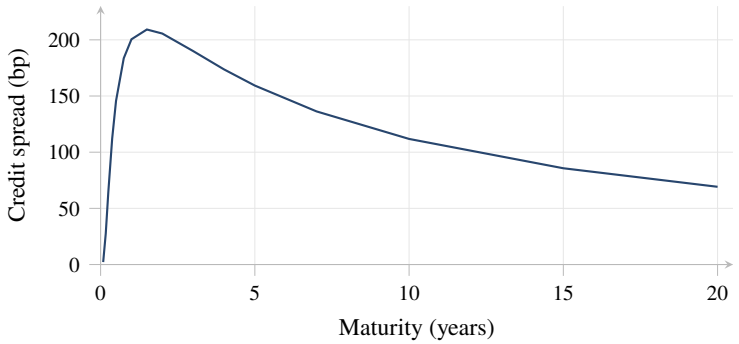


Figure 24.3: Credit spread by maturity in Merton's model. Assets 100, debt 80, asset volatility 25%, $r = 5\%$.

Figure 24.3 has the right shape in the middle and the wrong shape at the left edge.

Maturity	Spread	Default probability
1 week	0.0000 bp	6×10^{-9}%
1 month	2.2 bp	0.09%
3 months	64.6 bp	3.42%
1 year	200.5 bp	16.66%
2 years	205.6 bp	23.05%

The spread goes to zero as maturity shortens. A firm whose assets exceed its debt cannot default tomorrow if asset value follows a continuous path, because it would have to pass through every intermediate level first and there is no time for it to travel.

Real short-dated credit does not behave this way. One-year CDS on investment-grade issuers trades in the tens of basis points and does not collapse toward zero as maturity shortens; short-dated spreads on stressed issuers can be enormous.

This is the same defect, in the same place, for the same reason as Section 21.7's. There, a diffusive volatility model could not produce the observed short-dated skew because continuous paths cannot generate short-horizon tails. Here, a diffusive asset process cannot produce short-dated credit spreads. Two literatures discovered the same limitation of continuous paths independently, and the repairs are related: jumps in both fields and, in credit, default made to arrive as a surprise rather than as the outcome of a diffusion. That second repair is the reduced-form approach.

24.8 REDUCED-FORM MODELS

Rather than explain default, model it directly. Let the default time τ be the first arrival of a Poisson process whose intensity λ_t is itself a stochastic process: a **Cox process**, or doubly stochastic Poisson process. Conditional on the whole path of the intensity, survival to T has the

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probability (24.2) already gave; averaging over intensity paths,

$$Q(T) = \mathbb{E}\left[\exp\left(-\int_0^T \lambda_u \, du\right)\right] \tag{24.7}$$

which is (24.2) with the expectation restored. Default is a surprise: however healthy the firm looks, the probability that it defaults in the next instant is $\lambda_t \, dt$, not zero. That is what the diffusion of Section 24.7 could not deliver, and it is why this construction repairs the short end: with default arriving as a jump, a one-month contract carries a spread of $\lambda(1 - R)$ from its first day.

Now price a bond with the same expectation as (24.7), because that step is where the framework shows its value. A zero-coupon bond paying 1 at T and nothing if the issuer has defaulted is worth

$$\bar{P}(0,T) = \mathbb{E}\left[e^{-\int_0^T r_u \, du} \mathbf{1}_{\{\tau > T\}}\right] = \mathbb{E}\left[\exp\left(-\int_0^T (r_u + \lambda_u) \, du\right)\right] \tag{24.8}$$

By (24.8) the hazard rate enters exactly where the short rate does, and that is the whole reason the approach took over. A defaultable bond is a riskless bond discounted at $r + \lambda$, so everything built for short-rate models in Chapter 31 (the trees, the closed forms, the calibration machinery) transfers to credit unchanged, with the intensity playing the part of a second rate factor. Nothing else in credit modeling is this convenient, and convenience is why the market prices with it.

Recovery slots into the same place. Suppose that on default the claim is worth a fraction R of what it was worth just before, so the loss is $1 - R$ of its *market value*, not of its face. Follow the bond over a short interval dt, writing $\bar{P}_t$ for its value at t. With probability $1 - \lambda_t \, dt$ the borrower survives and the bond is worth $\bar{P}_{t+dt}$; with probability $\lambda_t \, dt$ it defaults and the bond is worth $R \, \bar{P}_{t+dt}$. Under the risk-neutral measure the bond must earn the riskless rate in expectation, so

$$\begin{aligned} \bar{P}_t &= e^{-r_t \, dt}\left[(1 - \lambda_t \, dt)\, \bar{P}_{t+dt} + \lambda_t \, dt \, R \, \bar{P}_{t+dt}\right] \\ &= e^{-r_t \, dt}\big(1 - \lambda_t (1 - R) \, dt\big)\bar{P}_{t+dt} \approx e^{-(r_t + \lambda_t (1-R)) \, dt} \, \bar{P}_{t+dt}. \end{aligned}$$

Figure 24.4 draws the step. Default has added $\lambda_t(1 - R)$ to the one-period discount rate: the intensity of the event times the fraction it destroys. Chaining these steps from 0 to T and taking the expectation over the paths of r and λ gives

$$\bar{P}(0,T) = \mathbb{E}\left[\exp\left(-\int_0^T \big(r_u + \lambda_u (1 - R)\big) \, du\right)\right] \tag{24.9}$$

so the credit spread is $\lambda(1 - R)$. That is (24.3), Section 24.2's credit triangle, arriving as the constant-intensity case of a general result rather than as a rule of thumb. It also makes Example 24.3's identification problem structural rather than accidental: λ and R enter the price only through their product, so under this convention no bond price and no CDS spread can separate them.

One convention changed along the way. Section 24.2 took recovery as a fraction of *face*, which is how a CDS settles and how the market quotes R; the derivation above takes it as a fraction of the bond's *market value* an instant before default, which is what makes the algebra close. For a bond trading near par the two conventions agree, because face and market value then coincide. They differ for a bond far from par (a bond at 50 that recovers 40 has recovered 40% of its face but 80% of its market value), and for a CDS on such a bond the face convention is the one that matters, since the contract pays $1 - R$ of face whatever the bond was worth. Chapter 25 keeps the face convention throughout for that reason; the market-value convention

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is used here because it is the one under which the hazard rate enters a bond price exactly as a spread.

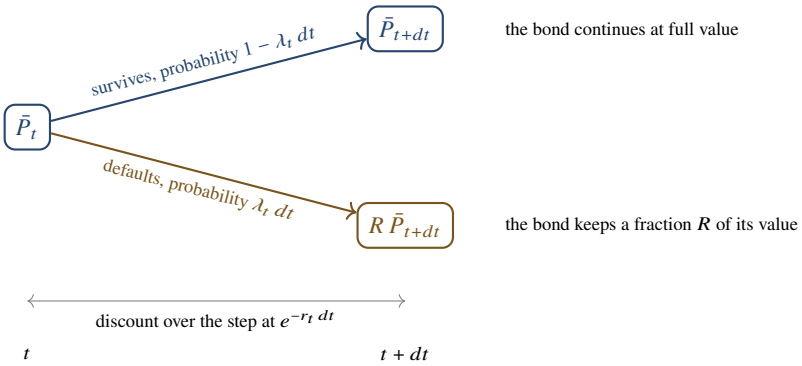


Figure 24.4: One step of the argument behind (24.9). The bond's value today is the discounted risk-neutral expectation of the two outcomes, which shortens to a discount rate of $r_t + \lambda_t(1 - R)$.

Each family has its strengths and its weaknesses.

Reduced-form models fit. Because λ is a free function, any observed spread curve can be reproduced exactly, which is Section 24.3's bootstrap, and it is the same property that makes local volatility the baseline in Chapter 21 for payoffs today's surface alone determines.

Structural models explain. They say why a spread is what it is, connect it to the equity market, and produce a hedge: if equity is a call on the firm, credit can be hedged with equity, and capital-structure arbitrage is that trade.

The two families also differ on correlation, where it matters most. In a structural model two firms default together because their asset values are correlated, a mechanism with an economic story attached. In a reduced-form model correlation must be imposed on the hazard rates, and the imposition is arbitrary. Chapter 26 is largely about what happened when the market chose a convenient imposition and treated it as a fact.

The market uses reduced-form for pricing and structural for thinking. A CDS desk bootstraps hazard rates; a credit analyst asks how far the firm is from default. The two are not in conflict, because they answer different questions.

24.9 REAL-WORLD AGAINST RISK-NEUTRAL

Section 9.2 computed CVA from the counterparty's CDS spread rather than from its historical default rate, and said that the default rate a spread implies has run several times the historical one. Here is how much.

Example 24.5 Most of a credit spread is not expected loss

The table sets five-year average cumulative default rates by rating for global corporate issuers, 1981–2024, from S&P Global Ratings' 2024 annual default study, against the risk-neutral probabilities implied by representative mid-cycle five-year spreads at 40% recovery. The spreads are illustrative; the historical rates are S&P's. AAA sits above AA only because few issuers were ever rated AAA, a small-sample artifact.

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Rating	Spread	Historical	Risk-neutral	Ratio
AAA	40 bp	0.34%	3.28%	9.6×
AA	55 bp	0.28%	4.48%	16.0×
A	80 bp	0.39%	6.45%	16.5×
BBB	130 bp	1.36%	10.27%	7.5×
BB	300 bp	5.75%	22.12%	3.8×
B	550 bp	15.60%	36.77%	2.4×

Decomposing the 130 bp BBB spread: expected loss at the historical default rate is the annual hazard $-\ln(1-0.0136)/5 = 0.274\%$ times $1-R$, or 16.4 bp, leaving 113.6 bp, so 87% of the spread is not expected loss.

The ratio of the two is largest for high-grade issuers. An AA or A issuer's risk-neutral default probability is about sixteen times its historical rate and a single-B issuer's under two and a half times; AAA is lower only because its small-sample historical rate exceeds AA's. Figure 24.5 draws the two columns on a log scale, where the vertical gap is the ratio, and the gap narrows steadily from A down to B. Whatever the excess compensates for, it does not scale with default risk.

Three things are in the residual, and separating them is a live question rather than a settled one. *A risk premium*, because defaults cluster in bad states and bearing them is unattractive for the reason Section 19.10 gave for volatility. *Illiquidity*, because corporate bonds trade rarely and the spread includes the cost of that. And *a premium for parameter uncertainty*, about the inputs themselves: the recovery of Example 24.3 is assumed, not known, and a seller of protection is being compensated for that too.

The practical consequence follows. Use risk-neutral probabilities to price and hedge, historical probabilities to forecast losses, and never mix them. An institution that provisions using risk-neutral probabilities will over-reserve by a factor of seven or more on investment-grade portfolios; one that prices using historical probabilities will systematically sell protection too cheaply and discover the difference when it tries to hedge.

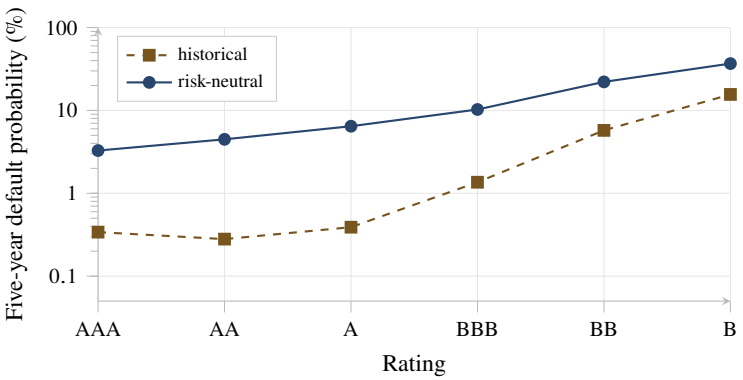


Figure 24.5: Five-year default probability by rating, historical and implied by spreads at 40% recovery, from Example 24.5. Log scale, so the vertical gap is the ratio.

24.10 RATINGS AND THEIR LIMITS

Ratings are an ordinal ranking of credit quality produced by agencies paid, for corporate and structured issues, by the issuer. Three properties determine how they should be used.

They rank well and calibrate poorly. The ordering has held up: over long samples, higher-rated issuers default less. But a corporate rating is defined as an ordinal opinion of

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relative credit risk, and no agency attaches a fixed probability to the letter. The mapping users work with is assembled after the fact from the agencies' own default studies (the source of Example 24.5), and it is much less stable than the ordering, drifting with the credit cycle and differing across asset classes. The one place an explicit letter-to-probability table did exist is structured finance. A **collateralized debt obligation** (CDO) is a pool of bonds or loans whose losses are divided into **tranches**, layers that absorb those losses in turn, and Moody's idealized default rates and the default-rate assumptions in S&P's CDO model were used to size the tranches. Chapter 26 is about what happened to that table: the same letter applied to a corporate bond and to a tranche described very different risks, because the tranche's rating depended on a correlation assumption that the letter did not disclose.

They are through-the-cycle by design. Agencies deliberately avoid re-rating on every market move, which makes ratings stable and makes them lag. An issuer whose spread has tripled may hold its rating for months. This is a feature for a bond mandate and a defect for a risk system, and confusing the two produces a risk report that says nothing has changed while the market says otherwise.

And they are used in regulation and contracts, which creates mechanical effects that have nothing to do with information. A downgrade below investment grade forces sales by mandate-constrained holders, triggers collateral calls under CSAs of the kind Chapter 8 described, and can raise funding costs discontinuously. A rating is therefore not merely an opinion about default; it is an event with contractual consequences, and the consequences can cause the deterioration the rating anticipated.

24.11 MIGRATION: HOW DEFAULT ACTUALLY HAPPENS

Firms rarely jump from health to default. They deteriorate, and the deterioration is what a rating records. Modeling that path requires a **transition matrix**: the probability of moving from each rating to each other rating over a year. Default is absorbing: an issuer that reaches it stays there.

Table 24.2: Illustrative one-year rating transition probabilities (%); rows are the starting rating, D is default.

	AAA	AA	A	BBB	BB	B	CCC	D
AAA	90.09	9.01	0.60	0.10	0.10	0.10	0.00	0.00
AA	0.60	90.59	7.91	0.60	0.10	0.20	0.00	0.00
A	0.05	2.10	91.40	5.50	0.60	0.20	0.10	0.05
BBB	0.02	0.20	4.20	89.90	4.30	0.90	0.20	0.28
BB	0.02	0.06	0.30	5.60	83.00	8.50	1.00	1.52
B	0.00	0.05	0.20	0.40	5.80	82.00	5.10	6.45
CCC	0.00	0.00	0.30	0.60	1.50	12.00	55.00	30.60

The matrix is illustrative, and from A down it is harsher than the long-run averages of Example 24.5: compounded to five years it gives default probabilities of 2.89% for BBB, 12.0% for BB and 31.9% for B, against S&P's 1.36%, 5.75% and 15.60%, so its pattern, not its levels, is what to carry over, and the two should not be mixed. Raising the matrix to the nth power gives n-year transitions, and the default column gives cumulative default probabilities. Horizons that are not whole years need one more step, which practitioners take and this chapter only names: find the **generator matrix** Λ with $P = e^{\Lambda}$, whose off-diagonal entries are the intensities of moving from one rating to another, extending the hazard rate of Section 24.1 from one exit to seven, and then $P(t) = e^{t\Lambda}$ gives the transition matrix for any horizon t. Not every annual matrix has a valid generator; Lando's book in the Further Reading treats the fitting.

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Example 24.6 Downgrade is the route to default

Table 24.2 places a BBB issuer after five years as follows:

	AAA	AA	A	BBB	BB	B	CCC	D
Probability (%)	0.10	1.37	14.64	62.17	12.90	5.06	0.87	2.89

So 78.3% is still investment grade, 18.8% has fallen to speculative grade (junk) without defaulting, and 2.89% has defaulted.

Now compare that 2.89% with the naive calculation. The one-year BBB default rate is 0.28%; compounding it as though each year were independent gives

$$1 - (1 - 0.0028)^5 = 1.39\%$$

against the matrix's 2.89%, a factor of 2.08.

The matrix gives more because default is reached through downgrade. A BBB issuer that is still alive in year four is disproportionately likely to be a BB or B issuer by then, facing a much higher hazard than the 0.28% it started with.

The route shows in the year-by-year numbers. Dividing each year's increase in the cumulative default probability by the probability of having survived to its start gives the **forward default probability** for that year, the discrete counterpart of the hazard rate of Section 24.1:

Year	1	2	3	4	5
Cumulative default probability (%)	0.28	0.72	1.31	2.03	2.89
Forward default probability (%)	0.28	0.44	0.59	0.74	0.87

The forward rate triples over five years although the starting rating has not changed, because the issuers still alive in year five are a different population from the ones that started: the ones that stayed BBB, plus the ones that fell to BB and B and have not yet defaulted. This is the same structure as the hazard curve of Example 24.2, arrived at from a different direction: the forward hazard rate of an investment-grade issuer rises with horizon, and here one can see the mechanism rather than merely the fitted curve.

Migration matters in two further ways. It is a risk even without default, because a downgrade widens spreads and triggers the contractual consequences of Section 24.10, so a portfolio marked to market bears it whether or not anything defaults. And the matrix is the natural input to a credit portfolio model; Section 24.14 builds one from it and measures that migration risk.

One caution follows: the matrix is not stable. Transition probabilities are strongly cyclical (downgrades cluster in recessions), so a matrix estimated over a long sample understates the risk of a bad year and overstates it in a good one, in the way Section 24.13's correlation is designed to capture.

24.12 SOVEREIGN CREDIT IS DIFFERENT IN KIND

A **sovereign** is a national government borrowing in its own name, and its debt comes in two kinds that the market names differently and this book has so far kept apart. **Local-currency debt** is issued in the government's own currency (the Treasuries of Chapters 4 to 6, and gilts; euro-area bonds such as Bunds are the borderline case, since the issuing government does not control the euro), and Part I treated it as the risk-free benchmark; it appeared later as the collateral everyone else posts. **External debt** (foreign-currency debt) is debt in a currency the government does not issue: dollars or euros borrowed by an emerging-market government from bondholders, banks, or official lenders. **Sovereign credit** is the risk that a government does not pay, and it lives mainly in the second kind, because a government can always print the currency it controls and cannot print anyone else's. CDS on emerging-market sovereigns reference that external debt; CDS on euro-area sovereigns, which do not control their currency, price the risk of redenomination as much as outright default. Everything above assumes a borrower that can

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be put through a bankruptcy process. A sovereign cannot, and the differences run deeper than the mechanics.

Ability and willingness are separate. A company defaults when it runs out of money. A country defaults when repaying costs more, politically, than not repaying—and it may have the money. Default is therefore a choice in a way that corporate default is not, which makes structural models of the kind in Section 24.5 inapplicable: there is no asset value crossing a debt threshold, because the relevant asset is the government's capacity to tax and its willingness to use it.

There is no court and no seniority. A defaulting sovereign cannot be liquidated, and the ranking of its obligations is whatever it chooses to respect. Restructurings are negotiated, protracted, and settled by a mixture of law and pressure. **Collective action clauses** (standard in New York-law sovereign bonds since Mexico's 2003 issue, and strengthened in 2014, after the Argentine holdout litigation, to a single vote aggregated across all of an issuer's series) bind dissenting holders to a majority agreement, and their presence or absence materially changes what a bond is worth in default.

The two kinds of debt are rated and priced apart. Default on local-currency debt is a policy choice with an inflationary alternative, so the market prices outright default and inflation together, as two routes to the same loss: a government that will not default outright may still repay in a currency worth less. Foreign-currency debt has no such escape, which is why the same issuer carries a local-currency and a foreign-currency rating, and why the second is usually the lower.

And the definition of default becomes contestable. A payment missed because the sovereign refuses is clearly a default; a payment missed because a third party blocked the transfer is less clear.

The pricing consequence is a wide and persistent basis between sovereign CDS and sovereign bond spreads, the difference Section 24.15 defines. The two should be close under the arbitrage that section describes, and they often are not, because the CDS references a definition of default while the bond references a cash flow, and because the natural sellers of sovereign protection are often institutions domiciled in the sovereign, which is Section 9.9's wrong-way risk in its purest form.

The point generalizes beyond sovereigns. Wherever default is a legal event rather than an economic one, the contract's definition matters as much as the borrower's finances, and Chapter 25 is largely about what happens when those two come apart.

24.13 DEFAULT CORRELATION

A portfolio of loans or bonds is not the sum of its parts, and the difference is entirely about correlation. State the problem first, because the device that solves it looks arbitrary until the problem is clear. Sections 24.2 to 24.4 deliver, for each borrower separately, the probability that it defaults by T; a portfolio's loss distribution needs the probability of every *combination* of defaults, and with n borrowers there are 2^n of them. Nothing in the data comes close to pinning that down: joint defaults are rare events, and for any particular group of borrowers a century of history holds few or none. So the joint distribution has to be built, and it has to be built under three constraints. It must reproduce each borrower's own default probability exactly, because the market quoted that and a dealer who disagrees with it has a position, not a model. It must carry dependence with a small number of parameters, because there is no data for more. And it must be computable for a portfolio of a hundred or more.

Correlating the default events themselves does not work. A default indicator is a zero-or-one variable, and the correlation of two such variables with different default probabilities is capped well below one. For probabilities $p_1 < p_2$ the ceiling is $\sqrt{p_1(1-p_2)/\big(p_2(1-p_1)\big)}$, which is

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0.43 at 2% and 10%: the rarer event sets it, because the two can default together no more often than the rarer one defaults at all. Nor can such correlations be chosen freely for more than two borrowers at once, since many matrices of pairwise indicator correlations correspond to no joint distribution at all. Merton's model of Section 24.5 suggests the way around: default happens when a continuous quantity, the firm's asset value, crosses a threshold, and continuous quantities are easy to correlate. So give each borrower a continuous latent variable standing in for its asset value, correlate the latent variables through the common state of the economy, and set each threshold so that the probability of crossing it equals that borrower's own market-implied default probability. The result is a Merton model calibrated to CDS curves instead of balance sheets, and it is what the credit market calls a **copula**: a function that supplies the dependence between variables whose marginal distributions are taken as given. Sklar showed in 1959 that any joint distribution can be split that way into its marginals and a copula, so the separation is always available; the choice made here is to take the copula from a multivariate normal.

The standard device is therefore the **one-factor Gaussian copula**. Give each borrower i a latent variable

$$X_i = \sqrt{\rho}\, M + \sqrt{1-\rho}\, Z_i, \qquad M, Z_1, \ldots, Z_n \text{ independent } N(0,1) \tag{24.10}$$

so that every X_i is standard normal and any two are correlated at ρ. Write $\pi_i(T) = 1 - Q_i(T)$ for borrower i's own probability of *defaulting* by T: the complement of the survival probability (24.2), and the marginal that Section 24.3's bootstrap produced from that borrower's CDS curve. Borrower i defaults when its latent variable falls below the threshold that marginal implies:

$$\text{default by } T \iff X_i < N^{-1}\big(\pi_i(T)\big) \tag{24.11}$$

The threshold is what makes this a copula. Each $\pi_i(T)$ is whatever the market said it was; the construction does not touch it. All (24.10) supplies is the *dependence* between the borrowers. Marginals from the market, dependence from an assumption: that separation is the entire appeal of a copula, and it is also what let a single number be asked to carry the joint behavior of the 125 issuers in a standard CDS index of Chapter 25.

The consequence is the one the section needs: conditional on the common factor, the borrowers are independent. In a homogeneous portfolio, where every borrower has the same default probability π by T, each defaults given M with probability

$$p(M) = N\left(\frac{N^{-1}(\pi) - \sqrt{\rho}\, M}{\sqrt{1-\rho}}\right) \tag{24.12}$$

so the number of defaults among n borrowers is a binomial averaged over the factor, weighted by the standard normal density ϕ of M:

$$\mathbb{P}(k \text{ defaults}) = \int_{-\infty}^{\infty} \binom{n}{k} p(M)^k \big(1 - p(M)\big)^{n-k}\, \phi(M)\, dM \tag{24.13}$$

One integral, and the whole portfolio is priced; Example 24.8 carries one case through it. That tractability, and not any evidence about how firms actually fail, is why this model became the market standard.

Two features of (24.12) recur in everything that follows. A bad common factor raises everyone's default probability at once: $p(M)$ rises as M falls, which is the clustering the model is built to produce. And the idiosyncratic channel diversifies away while the common one does not, so in a large portfolio the loss distribution is driven almost entirely by M.

The limit can be written down. Conditional on M the borrowers are independent with common default probability $p(M)$, so by the law of large numbers the fraction of a very large

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homogeneous portfolio that defaults is $p(M)$ itself:

$$\mathrm{DR}(M) = p(M), \tag{24.14}$$

a deterministic function of one standard normal draw, whose quantile Section 24.14 takes. Figure 24.6 shows what correlation does to the distribution of defaults, and Example 24.7 measures it.

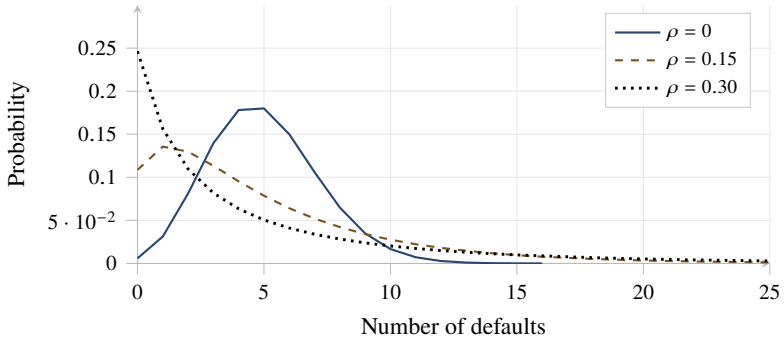


Figure 24.6: Distribution of defaults in a portfolio of 100 borrowers, each with 5% default probability. Expected defaults are 5 in every case.

Example 24.7 Correlation moves probability to both tails

Take one hundred borrowers, each with a 5% default probability over the horizon.

ρ	Expected	Std. dev.	P(no defaults)	P(20 or more)
0.00	5.00	2.18	0.59%	0.00%
0.05	5.00	3.22	3.13%	0.11%
0.15	5.00	4.89	10.86%	1.85%
0.30	5.00	7.12	24.62%	4.91%
0.60	5.00	11.56	53.02%	7.87%

The expected loss does not move at all. What correlation does is take probability out of the middle and put it at *both* ends, which is why the same expected loss can describe a safe portfolio or a dangerous one: at $\rho = 0.6$ there is a 53% chance that nothing defaults and an 8% chance that at least a fifth of the portfolio does.

That double-tailed behavior is what tranche pricing turns on. A senior tranche, one that suffers only if losses exceed some large threshold, is a bet on correlation and almost nothing else, because its expected loss is negligible at low correlation and material at high correlation while the portfolio's expected loss is unchanged. An equity tranche (the first-loss layer, which absorbs the portfolio's losses before any other tranche does) is the opposite: it benefits from correlation, because correlation raises the chance that nothing happens at all.

So a tranche is a correlation instrument that looks like a credit instrument, and pricing one requires a number, ρ, that no observable market quoted before tranches themselves began to trade. The circularity in that sentence is Chapter 26's subject.

Example 24.8 Applying the copula, one value of the common factor at a time

Here is how the $\rho = 0.15$ row of Example 24.7 is produced. With $\pi = 0.05$ the threshold in (24.11) is $N^{-1}(\pi) = -1.6449$, and conditional on the common factor M each of the 100 borrowers defaults

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independently with probability

$$p(M) = N\left(\frac{-1.6449 - \sqrt{0.15}\,M}{\sqrt{0.85}}\right),$$

which is (24.12). Given M, the number of defaults K is binomial with parameters 100 and $p(M)$, so every quantity of interest (the expected count $100\,p(M)$, the probability of no defaults $(1-p(M))^{100}$, the probability of twenty or more) is an elementary binomial calculation. The table evaluates them at six values of M, with the standard normal density $\phi(M)$ that weights each state.

M	$\phi(M)$	$p(M)$	$\mathbb{E}[K \mid M]$	$\mathbb{P}(K = 0 \mid M)$	$\mathbb{P}(K \geq 20 \mid M)$
−3	0.0044	30.0%	30.0	0.00%	99.1%
−2	0.0540	17.3%	17.3	0.00%	27.1%
−1	0.2420	8.6%	8.6	0.01%	0.03%
0	0.3989	3.7%	3.7	2.26%	0.00%
1	0.2420	1.4%	1.4	25.03%	0.00%
2	0.0540	0.4%	0.4	64.72%	0.00%

The unconditional numbers are each column integrated against $\phi(M)$, which is (24.13) applied to the event in question: expected defaults 5.000, no defaults 10.86%, twenty or more 1.85%. These are the $\rho = 0.15$ row. Two things are visible in the table that the row alone hides. The expected count comes out at five because $p(M)$ is raised above 5% in the bad states and lowered below it in the good ones by amounts that average out exactly. And the two tails live in different states: three-quarters of the probability of twenty or more defaults comes from $M \leq -2$, states with about 2% of the probability, while two-thirds of the probability of none at all comes from $M \geq 1$. Raising ρ steepens $p(M)$ in M, feeding both tails at once from the middle, which is the redistribution Figure 24.6 draws.

24.14 CREDIT VAR

Chapter 38 asks how much a trading book could lose over a few days. A loan book needs the same question asked differently, for three reasons. Its positions cannot be sold in a few days, so the horizon is the time it takes for losses to be realized and capital to be rebuilt, conventionally one year. Its loss distribution is nothing like a trading book's roughly symmetric profit and loss: in most years a few borrowers default and the loss is close to its average, and in rare years many default together, so the distribution is bounded on one side and long on the other. And its expected loss is not a risk in the sense that matters for capital. It is a cost, charged in the loan's spread and set aside in provisions: the expected credit loss of IFRS 9, and the current expected credit loss (CECL) standard in the United States. What capital has to absorb is the loss in excess of that cost.

Credit VaR is therefore defined as a high quantile of the one-year portfolio loss minus the expected loss, a difference called the **unexpected loss**. The confidence level is far higher than Chapter 38's: 99.9% for regulatory capital, and typically 99.95% or more for a bank's own **economic capital**, a level chosen so that the bank's probability of being wiped out in a year matches the default rate of the rating it wants to hold.

The large-portfolio limit of Section 24.13 delivers this quantity directly. In a large homogeneous portfolio whose borrowers share a one-year default probability π, the fraction that defaults within the year is $\mathrm{DR}(M) = p(M)$ of (24.14), a decreasing function of the common factor, so its quantile at confidence q is $p(M)$ evaluated at $M = -N^{-1}(q)$. That quantile is the **worst-case default rate** (WCDR), the default rate exceeded in only a fraction $1-q$ of years:

$$\mathrm{WCDR}(q) = N\left(\frac{N^{-1}(\pi) + \sqrt{\rho}\,N^{-1}(q)}{\sqrt{1-\rho}}\right) \tag{24.15}$$

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With exposure at default (EAD) and recovery R, the credit VaR is

$$\text{Credit VaR}(q) = \text{EAD}\,(1 - R)\,\big[\,\text{WCDR}(q) - \pi\,\big] \tag{24.16}$$

The feature that matters most is visible at $\rho = 0$, where (24.15) returns π at every confidence level. A large portfolio of independent loans has no credit VaR at all, because the law of large numbers pins its default rate to the mean. Every dollar of a granular portfolio's unexpected loss comes from correlation, which Example 24.9 puts in numbers.

Example 24.9 All of the credit VaR is correlation

A bank holds a \$100 million book of corporate loans, numerous and small enough to be granular, each with a one-year default probability of 1% and a loss given default of 40%, the Basel foundation value for a senior unsecured loan to a nonfinancial corporate. The expected loss is $100 \times 0.40 \times 0.01 = \$0.40$ million whatever the correlation. The worst-case default rate (24.15) and the credit VaR (24.16) are not:

ρ	WCDR, 99%	WCDR, 99.9%	Credit VaR, 99%	Credit VaR, 99.9%
0.00	1.00%	1.00%	\$0.00m	\$0.00m
0.05	3.19%	4.67%	\$0.88m	\$1.47m
0.10	4.68%	7.75%	\$1.47m	\$2.70m
0.20	7.53%	14.55%	\$2.61m	\$5.42m
0.30	10.43%	22.44%	\$3.77m	\$8.58m

Take the $\rho = 0.20$ row, close to the correlation the Basel formula assigns to a borrower with this default probability (Section 39.10). In the worst year in a thousand, 14.55% of the book defaults, against 1% on average, and the loss is $100 \times 0.40 \times 0.1455 = \$5.82$ million. Subtracting the \$0.40 million already priced leaves a credit VaR of \$5.42 million, 13.6 times the expected loss.

Two consequences follow. Correlation sets the ratio of credit VaR to expected loss, so two books with identical expected loss can need capital that differs by a factor of almost six (\$1.47 million at $\rho = 0.05$ against \$8.58 million at 0.30). And the confidence level matters as much as the model: moving from 99% to 99.9% raises the credit VaR to between 1.67 and 2.3 times its 99% value, more at higher correlation, because the tail is long and correlation is what lengthens it.

Two assumptions sit inside (24.15), and each understates the risk of a real book. The portfolio is taken to be infinitely granular. A book with a few large exposures keeps idiosyncratic risk that does not diversify, and its credit VaR is higher than the formula says, by an amount that grows with the concentration. And the dependence is the Gaussian copula, with the absence of tail dependence that Chapter 26 shows to matter and Section 38.12 treats formally; a copula with fatter joint tails puts more defaults in the bad years and raises the quantile further.

Migration, not only default. The measure above counts a loss only when a borrower defaults, which is right for a loan held at amortized cost. A portfolio marked to market loses money whenever a borrower is downgraded and its spread widens, which is the migration risk of Section 24.11. **CreditMetrics**, published by J.P. Morgan in 1997, measures it. It revalues each position at the horizon in every rating state, discounting the remaining cash flows on that rating's spread curve, and weights the values by a row of the transition matrix. To correlate migrations across borrowers it reuses the latent variable of (24.10), cutting that variable into bands rather than at a single default threshold: a BBB borrower defaults if X_i falls below $N^{-1}(0.28\%)$, ends the year at CCC or worse if it falls below $N^{-1}(0.48\%)$, and so on up the row. The same ρ then correlates downgrades as well as defaults, and a portfolio's distribution is found by simulating M and the Z_i.

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Example 24.10 Downgrade, not default, sets the 99% VaR of a BBB bond

A five-year BBB bond pays an annual coupon of 5% on \$100 face. A year from now it will have paid one coupon and have four years left, and its value then depends on its rating. The risk-free rate is 3% for every maturity, compounded annually, each rating trades at the spread in the table, the spreads of Example 24.5 extended to CCC, and the bond recovers \$40 if it defaults during the year, without the coupon. The probabilities are the BBB row of Table 24.2.

Rating in one year	Probability	Spread	Value
AAA	0.02%	40 bp	110.89
AA	0.20%	55 bp	110.32
A	4.20%	80 bp	109.38
BBB	89.90%	130 bp	107.52
BB	4.30%	300 bp	101.53
B	0.90%	550 bp	93.54
CCC	0.20%	1,200 bp	76.45
Default	0.28%	—	40.00

Each value is the coupon just received plus the remaining flows discounted at 3% plus the spread; at BBB it is $5 + 5/1.043 + 5/1.043^2 + 5/1.043^3 + 105/1.043^4 = 107.52$. The probability-weighted mean is 106.97.

Reading up from the bottom of the table, the bond ends the year in default with probability 0.28%, at CCC or worse with 0.48%, and at B or worse with 1.38%. The first percentile therefore falls in the B state, and the 99% credit VaR is $106.973 - 93.535 = 13.44$ per \$100 of face. The 0.1% quantile falls in default, and the 99.9% credit VaR is $106.973 - 40.00 = 66.97$.

Now drop migration and keep only default, so the bond is worth 107.52 unless it defaults. Its mean is 107.33, and since default has a probability below 1% the first percentile is 107.52 itself, above the mean: the default-only model reports no 99% credit VaR at all, where the migration model reports 13.44. At 99.9% the two agree to within half a dollar, because both quantiles sit in default.

So the moderate quantile of a high-grade bond is a question about downgrade and the extreme quantile is a question about default. That is why the Basel formula of Section 39.10 carries an adjustment for maturity, which charges a long loan for the downgrade risk it carries before it can default. It is also why the trading book's default charge of Section 39.3 leaves migration to the spread sensitivities, which already measure what a downgrade costs.

Who uses it. A bank's credit portfolio management function runs a model of this kind across the whole loan book to set concentration limits and to allocate economic capital loan by loan, and the allocation enters pricing: under **risk-adjusted return on capital** (RAROC), a loan must earn its hurdle return on the capital it consumes, so a loan to a borrower highly correlated with the rest of the book carries a wider spread than one that diversifies it. The allocation is a contribution to the portfolio quantile, the same decomposition Section 38.15 applies to market risk. Where the model finds a concentration, the bank buys CDS protection on it (Chapter 25) or transfers a tranche of the book in a synthetic securitization (Chapter 26), and Section 27.11 shows how much regulatory capital each of those relieves. Rating agencies used the same machinery on the pools behind structured credit. And supervisors turned (24.15) into the formula that sets most banks' credit capital, with a prescribed correlation and a maturity adjustment. They could do so because, as Gordy (2003) showed, in a one-factor model of a granular portfolio the portfolio's quantile is the sum of each loan's own worst-case loss, so (24.15) can be applied one loan at a time with each loan's own π. Section 39.10 takes the formula up beside the output floor that now limits it.

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24.15 THE CDS–BOND BASIS

Two instruments reference the same issuer: a bond, whose spread over the risk-free curve compensates for default risk, and a CDS, whose spread does the same. They should agree, and often they do not.

The bond's spread needs defining first, because there are several. The one that compares with a CDS is the **Z-spread**: the constant spread that, added to every rate on the risk-free curve, discounts the bond's cash flows to its market price. The curve is the SOFR swap curve, which is what a CDS dealer funds and discounts on; a spread over Treasuries mixes in the swap spread of Chapter 7. A close cousin is the par **asset-swap spread**, the spread over SOFR received when the bond's fixed coupons are swapped to floating, which is how a basis trade is packaged in practice and is close to the Z-spread for a bond near par. From the Z-spread, a hazard rate follows the same two ways it did for a CDS: roughly through the credit triangle, and exactly by pricing the bond.

Example 24.11 A hazard rate from a bond price

A five-year bond pays an annual coupon of 5% and trades at 98.55. The SOFR curve is flat at 3%, continuously compounded. Discounted on that curve alone the cash flows would be worth

$$5\,(e^{-0.03} + e^{-0.06} + e^{-0.09} + e^{-0.12}) + 105\,e^{-0.15} = 108.94,$$

and the bond's Z-spread, the z that makes the same sum at $3\% + z$ equal 98.55, is 220 bp.

The credit triangle reads the spread as $\lambda(1 - R)$, so at 40% recovery $\lambda = 0.022/0.6 = 3.67\%$.

Pricing the bond with a constant hazard rate, recovery of 40% of face paid at default, and coupons only while the issuer survives:

$$98.55 = \sum_{t=1}^{5} 5\,e^{-(0.03+\lambda)t} + 100\,e^{-(0.03+\lambda)5} + 40\,\lambda \int_0^5 e^{-(0.03+\lambda)t}\,dt$$

gives $\lambda = 3.63\%$, close to the triangle, as it should be for a bond near par. The gap of $108.94 - 98.55 = 10.39$ between the riskless and the actual price is the present value of the expected loss from default, and the equation above is that statement written out.

The same issuer's five-year CDS trades at 170 bp, which implies $\lambda = 2.83\%$. The bond and the CDS disagree about the issuer's hazard rate by about 0.8 percentage points, and Example 24.12 is about why that disagreement can persist.

The arbitrage argument appears complete. Buy the bond, buy protection on the issuer, and the combination has no credit risk: if the issuer defaults, the protection makes up the loss. So the bond's spread should equal the CDS spread, and the difference, the **basis**, should be zero.

$$\text{basis} = s_{\text{CDS}} - s_{\text{bond}} \tag{24.17}$$

Example 24.12 shows why a nonzero basis can persist.

Example 24.12 Why a negative basis is not an arbitrage

The bond of Example 24.11 trades at a Z-spread of 220 bp; the five-year CDS trades at 170 bp. The basis (24.17) is −50 bp. The trade suggests itself: buy the bond, buy protection, and collect $220 - 170 = 50$ bp on a package with no credit exposure.

Except that the bond must be financed. Buying it requires cash, raised in repo against the bond itself. A corporate bond does not repo at the risk-free rate: with a haircut and a funding spread the all-in cost might be 60 bp over the curve the 220 bp was measured against, and $50 - 60 = -10$ bp. The trade loses money. And it consumes balance sheet, which under the frameworks of Chapter 27 carries a capital charge of its own.

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So the negative basis is not an arbitrage but a measure of what balance sheet costs, and it persists while funding costs more than the basis pays. Several other factors move the basis, each in a known direction.

The cheapest-to-deliver option pushes the basis up. A CDS settles at the recovery price an auction sets for the cheapest qualifying obligations, so the protection buyer is in effect delivering the cheapest bond. That right raises the CDS spread by its value; it is the structure of the Treasury futures delivery option of Chapter 6.

Counterparty risk pushes it down. Protection is worth less than face if the seller might fail, and least when seller and reference entity are likely to fail together, which is Section 9.9's wrong-way risk.

And the contracts define default differently. A CDS pays on credit events that may include restructuring, a broader trigger than the missed payment that defaults a bond, so protection is worth more and the basis rises (Section 25.10).

At the trough of the 2008 crisis the basis on investment-grade issuers reached about −240 bp on average, and about −560 bp on speculative-grade issuers (Bai and Collin-Dufresne 2019), an enormous riskless profit on the arbitrage logic. It was not taken, because those who would have taken it could not fund the bond at any price, the limit of arbitrage of Section 5.15.

24.16 WHO BEARS CREDIT RISK

Banks originate it and, since the capital reforms of Sections 27.11 and 26.13, increasingly distribute it; what they retain is capitalized under the internal-ratings formula of Section 39.10, the regulatory form of credit VaR, and the charge often decides whether a loan is made. **Bond investors** hold it deliberately, paid by the spread of Example 24.5, mostly not for expected loss. **Insurers and pension funds** are the natural long-horizon holders, because they can collect the spread's illiquidity premium without being forced to sell. **Protection sellers in the CDS market** take it synthetically, through Chapter 25's contract.

Private credit funds have taken a growing share of what banks once held. The shift has two sides: the risk sits largely with locked-up long-horizon capital, an improvement (though retail semi-liquid vehicles rationed redemptions in 2026), but it is valued infrequently and by the holder, so its price is far harder to observe. Chapter 26 records what happened the last time credit risk migrated to holders whose marks no one could check.

24.17 WHERE THIS CAME FROM

Merton (1974) began the subject: equity is a call on the firm's assets, and debt a short put. Black and Cox (1976) added the covenants that let default happen before maturity, and Leland (1994) made the capital structure itself endogenous. The structural approach is illuminating but inaccurate: it predicts short-maturity spreads near zero. The reduced-form answer modeled the arrival of default instead: Jarrow and Turnbull (1995) and Duffie and Singleton (1999) founded it, Lando (1998) supplied the Cox-process formulation, and Hull and White (2000) gave Chapter 25's CDS valuation.

Altman (1968) had earlier combined accounting ratios in a discriminant score to predict bankruptcy. On the portfolio side, KMV commercialized distance to default in the 1990s; J.P. Morgan published CreditMetrics in 1997; Li (2000) put a Gaussian copula on default times; and Vasicek's large-portfolio limit became Basel II's risk-weight formula.

SUMMARY

Credit risk is two questions, default probability and recovery, and market prices identify only their product. The hazard rate is the instantaneous conditional default rate, so survival is $\exp\left(-\int_0^T \lambda(u)\,du\right)$ and default probabilities compound rather than add.

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The credit triangle $s = \lambda(1 - R)$ is exact for a continuously paid premium, because survival and discounting cancel between the legs. Hazard and recovery are not separately identified: the same CDS spread is reproduced by a low recovery with a low hazard or a high recovery with a high hazard, which is harmless for anything that depends only on the product and decisive for a digital default swap or a regulatory calculation that asks for the two separately. Recovery itself ranks with priority of claim and is lowest when defaults are highest, so a fixed recovery convention understates tail losses twice.

In Merton's model equity is a call on the firm's assets struck at the face value of debt, so lending is selling a put on those assets, and equity holders gain from the volatility that debt holders lose from. The model's spread goes to zero at short maturity, for the reason Section 21.7 found diffusive volatility models unable to make a short-dated skew. Reduced-form models repair it by making default a surprise with a free hazard function, which fits any curve and explains nothing.

Risk-neutral default probabilities exceed historical ones, and most of an investment-grade spread is not expected loss: price with risk-neutral probabilities, forecast losses with historical ones, and never mix them. Ratings rank well, calibrate poorly, and lag by design. Default is reached through downgrade, so a rating transition matrix says more than compounding the one-year rate, and for a high-grade bond the moderate quantile is a question about downgrade and the extreme one a question about default. A sovereign's default is partly a choice, so structural models do not apply and the contract's definition of default matters as much as the borrower's finances.

Correlation leaves a portfolio's expected loss unchanged and moves probability to both tails, which is why a senior tranche is a bet on correlation and why Chapter 26 is about a number no market quoted. The same mechanism sets credit VaR, the loan book's version of Chapter 38's question: a one-year horizon, a confidence level of 99.9% or more, and only the loss beyond the expected loss, which is priced and provisioned rather than held as capital. With independent defaults a large portfolio's worst-case default rate equals the average, so all of a granular book's credit VaR comes from correlation. The CDS–bond basis persists because closing it takes funded balance sheet, and it widens when that is scarce.

FURTHER READING

Merton (1974) is short and worth reading in the original: Merton states his assumptions plainly, and the model's limitations follow directly from them. Crosbie and Bohn (2003) document the KMV implementation of Section 24.6, including the default point and the mapping to expected default frequencies. Duffie and Singleton's *Credit Risk* (2003) is the standard treatment of reduced-form models and takes up the identification problem of Example 24.3; Lando's *Credit Risk Modeling* (2004) is the other standard, and the one to consult for rating-transition models and the generator matrix that Section 24.11 names. Vasicek's (2002) short note is the source of the large-portfolio limit (24.14). Gupton, Finger and Bhatia's *CreditMetrics—Technical Document* (1997) is the original statement of the migration approach of Section 24.14, and Gordy (2003) is the paper that shows why the large-portfolio quantile can be computed one loan at a time, the property the Basel formula rests on. On recovery, the empirical literature is larger than the 40% convention suggests, and less reassuring; the positive correlation between default rates and loss given default is the finding that recurs across samples. S&P Global Ratings' annual default study is the source of the historical figures in Example 24.5 and Moody's of Table 24.1; both are worth consulting directly, for the data and for the methodological notes that explain why the numbers move between vintages. On the decomposition of credit spreads, the "credit spread puzzle" literature is large and unresolved. Li (2000) is the paper that put the copula of Section 24.13 into the market, and is worth reading beside Chapter 26's account of

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what followed.

PRACTICE QUESTIONS

24.1 A borrower has a constant hazard rate of 4%. Compute the probability of default within five years and within ten, and explain why a 3% hazard rate does not mean a 30% probability over ten years.

24.2 A CDS trades at 240 bp with assumed recovery of 40%. Estimate the hazard rate, and explain why a change in the discount curve alone barely moves a par CDS spread.

24.3 Explain why the credit triangle is exact rather than approximate, and what the usually quoted error actually measures.

24.4 Using Example 24.2, explain why applying the credit triangle to a five-year spread understates the hazard rate over years three to five.

24.5 A five-year CDS trades at 170 bp. State the implied hazard rate at 40% recovery and at 60%, and name one instrument the two assumptions would price differently.

24.6 Explain why recoveries are lowest in the years when defaults are highest, and what this does to portfolio losses.

24.7 Explain why equity is a call option on the firm and debt is short a put, what this implies about the shareholder–creditor conflict, and what the ratio of equity to asset volatility measures as the firm approaches default.

24.8 Explain why Merton's credit spread goes to zero as maturity shortens, and name the chapter in which the same defect appeared in a different market.

24.9 Historical five-year default rates for BBB issuers are about 1.36% and the spread is about 130 bp. Compute the expected loss, state what fraction of the spread it accounts for, and explain why the ratio of risk-neutral to historical default probability is much larger for investment-grade than for single-B issuers.

24.10 A portfolio of 100 borrowers has 5% default probability per borrower. Explain why raising correlation leaves expected defaults unchanged but raises both the probability of zero defaults and the probability of twenty or more, and why a senior tranche is a bet on correlation while an equity tranche benefits from it.

24.11 Using Table 24.2, state the probability that a BBB issuer is still investment grade after five years and the probability it has fallen to speculative grade without defaulting, then explain why compounding a one-year default rate understates the five-year probability and give the factor from Example 24.6.

24.12 State two reasons a structural model cannot be applied to a sovereign borrower, and explain why the natural sellers of sovereign protection create wrong-way risk.

24.13 Explain why fixing recovery at 40% is harmless for pricing a CDS and damaging for a capital calculation.

24.14 A large, granular loan book has a one-year default probability of 2%, a loss given default of 45% and a copula correlation of 0.15. Compute the worst-case default rate at 99.9% and the credit VaR per $100 million, and explain why the credit VaR would be zero if the loans defaulted independently.

24.15 Using Example 24.10, explain why the 99% credit VaR of a BBB bond is set by downgrade and its 99.9% credit VaR by default, and state what a model that ignores migration reports at each level.

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24.16 Explain why credit VaR subtracts the expected loss while Chapter 38's VaR is measured from zero, and why its horizon is a year rather than a few days.

24.17 A five-year bond with a 5% annual coupon trades at 97.00 against the flat 3% SOFR curve of Example 24.11. Find its Z-spread and the hazard rate it implies at 40% recovery by the credit triangle, and state the CDS–bond basis if the issuer's five-year CDS trades at 170 bp. Explain why a basis trader would not simply buy the bond and buy protection.

FURTHER QUESTIONS

24.18 Example 24.1 shows the credit triangle is exact for a continuously paid premium and 0.376% off for a quarterly contract with accrual. Derive the cancellation, identify precisely which contract features are required for exactness, and determine how the relation changes for a contract that pays the premium annually in arrears with no accrual on default.

24.19 Example 24.3 shows that a CDS spread does not identify λ and R separately. Construct a pair of instruments whose prices together *would* identify them, explain why such instruments are illiquid or absent in practice, and discuss what this implies for the regulatory requirement to report probability of default and loss given default separately.

24.20 Section 24.7 argues that Merton's short-maturity failure is the same defect Section 21.7 found in diffusive volatility models. Make the parallel precise: state the shared mathematical property, show why it forces both spreads and skews to vanish at short horizons, and evaluate whether the repairs adopted in the two literatures are the same repair.

24.21 Example 24.5 finds that 87% of a BBB spread is not expected loss and that the ratio falls steeply from the high-grade ratings to single-B. Propose a decomposition of the residual into risk premium, illiquidity, and parameter uncertainty; state what data would identify each component; and explain why the rating-dependence of the ratio is evidence about which component dominates.

24.22 Section 24.5 states that lending is selling a put on the borrower's assets. Use this to derive the effect on debt value of an increase in asset volatility, a dividend payment to shareholders, and a debt-financed acquisition. Then explain which covenants in a real loan agreement correspond to which of these, and why covenants are the creditor's substitute for a hedge they cannot execute.

24.23 Example 24.7 shows correlation moving probability to both tails. Derive the limiting loss distribution as $\rho \to 1$, explain what it says about the meaning of a AAA rating on a senior tranche, and use this to state precisely what a rating agency was asserting when it rated such a tranche.

24.24 Section 24.16 argues that the migration of credit from banks to private funds genuinely has two sides. Set out the argument that it has made the system safer and the argument that it has made risk unobservable, identify what evidence would distinguish them, and explain why the absence of a market price is not the same as the absence of a loss.

24.25 Section 24.14's worst-case default rate assigns each loan a capital charge that depends only on that loan. Show that in the one-factor large-portfolio model the portfolio's loss quantile is the sum of the loans' own worst-case losses, explain why a regulator writing a capital rule needs this property, and identify two features of a real book that break it (a concentration and a second systematic factor), stating the direction in which each moves the true quantile.

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Chapter 25
Credit Derivatives and the CDS Market

Chapter 24 treated the credit default swap as a source of information: a spread from which a hazard rate could be extracted. This chapter treats it as a contract, and the difference matters more here than anywhere else in the book.

A CDS does not pay on default. It pays on a **credit event**, which is a list of things written into a legal document. Most of the time the two coincide and the distinction is pedantic. When they come apart, someone loses a great deal of money for reasons that have nothing to do with whether the borrower could pay—in March 2023 roughly CHF 16 billion of Credit Suisse Additional Tier 1 (AT1) instruments were written down to zero while shareholders kept about CHF 3 billion, and no credit event occurred at all. Sections 25.10 to 25.12 are about the occasions on which the two have come apart.

The chapter runs from the contract outward. Sections 25.1 to 25.5 are the single-name contract itself: its legs, its credit events, the 2009 standardization that made the risky annuity the whole of valuation, how a position is marked, and how a defaulted contract is settled. Sections 25.6 to 25.8 build portfolios out of it (index CDS and options on it, index tranches, and basket default swaps) and show that the same 125 issuers can be five different instruments depending on one correlation parameter, which is the bridge to Chapter 26. Section 25.9 is the total return swap, the other contract that separates economics from ownership. Sections 25.10 to 25.12 are the cases in which the definition and the economics came apart, and Section 25.13 turns them into a taxonomy of what a CDS actually insures. Sections 25.14 to 25.17 are the neighboring instruments: convertible bonds, credit-linked notes, hybrid structures, and sovereign CDS paid in a foreign currency. Sections 25.18 and 25.19 close with what the post-crisis reforms did to the market and who trades what is left of it.

Section 9.9 pointed here to show that its wrong-way risk example (sovereign protection bought from a bank in that country) was not hypothetical, and Section 25.17 takes it up. Section 10.10 promised convertible bonds here, and Section 25.14 supplies them, because a convertible is the other major instrument in which credit and an option are fused.

25.1 THE CONTRACT

A credit default swap has two legs on a notional N referencing an entity and a class of its obligations.

The premium leg. The buyer of protection pays the premium quarterly until maturity or a credit event, whichever comes first. The market calls the fixed rate at which it is paid the **coupon**, a word borrowed from bonds for a contract that is not one, and this chapter follows the market. The ISDA definitions do not: there the rate is the **Fixed Rate** and the protection buyer is the Fixed Rate Payer, so a reader who opens the legal document will find no coupon in it, and a confirmation that says Fixed Rate 100 basis points and a trader who says 100 coupon are describing the same term. On a credit event the buyer also pays the accrual from the last

payment date, a feature Example 24.1 valued at about a tenth of a percent of the spread for an investment-grade issuer.

The protection leg. On a credit event, the seller pays $\mathcal{N}(1 - R)$, where R is determined by the auction of Section 25.5.

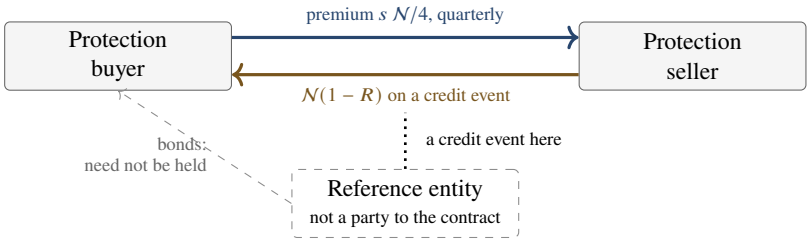


Figure 25.1: The premium and protection legs run between buyer and seller. The dashed link is the one that need not exist.

Figure 25.1 is the contract, and the two dashed elements are all that is peculiar about it. The reference entity, in the dashed box, is named by the contract and is party to nothing. It is not consulted or notified and receives nothing; it may not know that protection on its debt trades at all. A borrower's cost of funds can therefore be quoted in a market it cannot participate in, which is the mechanism behind Section 25.13's question about what a CDS actually insures.

The dashed line, from the reference entity to the buyer, marks a link that need not exist. A protection buyer who holds no bond is not hedging anything, and Section 25.6's index market and Section 25.5's settlement problem both follow from the fact that this link is optional.

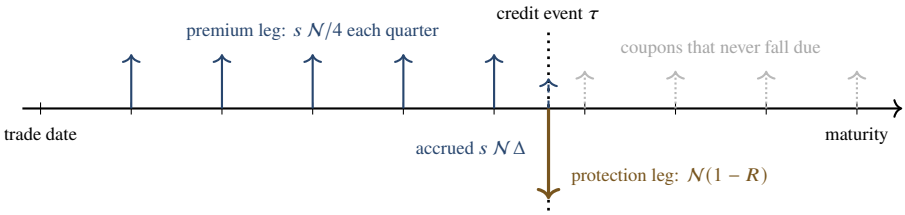


Figure 25.2: Quarterly premium against a single contingent payment. Schematic; the accrual stub is drawn at exaggerated length.

Figure 25.2 puts the same two legs on a calendar. Everything difficult about a CDS is a consequence of the asymmetry in that picture: the premium leg is a stream of small certain payments and the protection leg is one large uncertain one, so the contract's value is dominated by an event whose probability no one observes and whose payout depends on an auction that has not been held.

The buyer of protection is short the credit. They profit if the entity deteriorates, in the same way a put buyer profits if a stock falls, and they need not own the underlying bond—which is the feature that made the instrument useful and the feature that made it controversial.

The CDS completes the market. A bank with a loan it cannot sell can hedge it. An investor who thinks a credit is mispriced can act on that view in either direction. Before CDS, credit was a one-sided market in which the only way to be short was not to own.

The CDS also decouples the size of the exposure from the size of the debt. More protection can be outstanding than there are bonds, which matters at settlement (Section 25.5) and which means the instrument's gross notional does not measure what is paid on a credit event, as Section 1.6 warned about derivatives generally.

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25.2 CREDIT EVENTS, AND WHY THE DEFINITION IS THE CONTRACT

The ISDA definitions specify which events trigger payment. For corporates the standard list is bankruptcy, failure to pay, and, depending on the region, restructuring. For financial reference entities in Europe and Asia-Pacific the 2014 definitions add governmental intervention, a write-down or bail-in of their debt imposed by a resolution authority. For sovereigns the list adds **repudiation or moratorium** (the government disowns its debt or suspends payments on it) and, for emerging-market sovereigns, **obligation acceleration**: a default of some other kind on one of its obligations that lets the holders demand repayment before the debt falls due. Each is defined in detail, and the detail is the contract.

Failure to pay requires a payment actually missed, after any grace period, above a threshold. A payment made late but within the grace period is not a credit event. A payment made by a guarantor is not a credit event.

Bankruptcy is a list of legal states, and different jurisdictions enter them differently. A firm may be economically insolvent for a long time without entering any of them.

Restructuring is the difficult one, and Section 25.10 takes it separately.

Since 2009 whether a credit event has occurred has been decided by a determinations committee of dealers and investors, not by a court or the parties to the trade, and its decisions bind the market. This is efficient, because it prevents thousands of bilateral disputes over the same question. It also concentrates the outcome in a body whose members have positions, a governance problem the market had long noted. ISDA's 2025 reform added an independent governance committee and reasoned decisions, but the members of the determinations committees still hold positions.

The gap between economic distress and a contractual credit event is the chapter's recurring theme. A protection buyer who is right about the credit and wrong about the definition receives nothing.

25.3 STANDARDIZATION: THE BIG BANG

Until 2009 CDS traded at par: the coupon was set to give the contract zero value at inception, so every trade had a different coupon and no two positions could be netted. The 2009 reforms, the "Big Bang," fixed the coupon at a small menu of standard values and moved the difference into an upfront payment:

$$\text{upfront} = \left(s_{\text{par}} - c_{\text{fixed}}\right) \times A \tag{25.1}$$

where A is the **risky annuity**, the present value of one basis point paid until default or maturity.

For North American corporate issuers the menu is 100 basis points for investment grade and 500 for high yield, and those two are by far the most traded; this chapter's examples use them. The menu is wider elsewhere because the upfront has to stay moderate:

- Europe adds 25 for the tightest credits and 300, 750, and 1,000 for wide ones, so that a distressed issuer is not quoted at half its notional upfront;
- sovereign CDS trades mostly on 25 or 100;
- loan CDS uses 100, 250, and 500;
- Japan uses 25 alongside 100 and 500.

A contract's coupon is therefore one of a small number of round numbers chosen to sit near the issuer's spread, and every quoted spread is converted to an upfront against whichever of them the contract carries.

The risky annuity carries the whole valuation, so it is worth building once. A CDS has two legs, and both are expectations over the survival curve Q of Section 24.1. Write $t_1 < \cdots < t_n$

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for the premium dates, Δ_i for the accrual fraction of the period ending at t_i, and $D(t)$ for the risk-free discount factor, the $P(0,t)$ of Section 24.2. The accrual fraction is a day-count fraction, not a quarter of a year: the standard contract accrues Actual/360 between its quarterly payment dates (March 20, June 20, September 20, and December 20, which the CDS market calls IMM dates although they are not the third-Wednesday futures dates of Chapter 6), so Δ_i runs from about 0.2500 to 0.2556 with the calendar and averages about 0.2536, and Example 25.1 measures what setting it to 0.25 costs. The **premium leg** collects the coupon on every date the issuer survives to, plus the part-period accrued if it defaults in between:

$$A = \underbrace{\sum_{i=1}^{n} \Delta_i \, D(t_i) \, Q(t_i)}_{\text{scheduled}} + \underbrace{\frac{1}{2} \sum_{i=1}^{n} \Delta_i \, D(t_i) \big[Q(t_{i-1}) - Q(t_i) \big]}_{\text{accrued on default}} \tag{25.2}$$

per unit of spread. That is the risky annuity. The $1/2$ is the textbook simplification (Hull and White 2000; O'Kane and Turnbull 2003) that default lands, on average, halfway through the period. The market's own model does not: the ISDA CDS Standard Model, through which every quoted spread is converted to an upfront, integrates the accrued premium over each period against the hazard rate. At quarterly frequency the two agree to within a tenth of a percent of the accrual term: given default inside a period, the expected default time falls about $\lambda\Delta/12$ of the period before the midpoint, which shortens the accrual, while paying it at default rather than at the period end discounts it less, slightly more than making up the difference [TN25]. The simplification is kept here for the algebra it makes possible in (25.5). The **protection leg** pays $1 - R$ on default, discounted, like the accrued premium, from the end of the period in which default falls:

$$V = (1 - R) \sum_{i=1}^{n} D(t_i) \big[Q(t_{i-1}) - Q(t_i) \big] \tag{25.3}$$

Setting the two equal defines the par spread:

$$s_{\text{par}} = \frac{V}{A} \tag{25.4}$$

A CDS spread is a ratio of two present values, not a yield, and the denominator is what makes it behave unlike one. Everything else in this chapter is a consequence of that denominator: the upfront of (25.1), the mark-to-market of Section 25.4, the index wedge of Section 25.6.

Why the credit triangle is nearly exact. Under a flat hazard rate, a flat discount curve, and a constant accrual fraction Δ the sums in (25.2) and (25.3) are geometric and cancel term by term, leaving

$$s_{\text{par}} = \frac{2(1 - R)}{\Delta} \tanh\!\left(\frac{\lambda \Delta}{2} \right) \tag{25.5}$$

Two things follow, and both answer questions the book has already raised. The discount rate has vanished entirely, which is why, with a flat hazard rate, a change in the risk-free curve alone does not move a par CDS spread: the survival probabilities and discount factors appear identically in both legs and divide out. With a sloping hazard curve the discount factors survive only as weights, and the spread barely moves. And as $\Delta \to 0$ the expression becomes $\lambda(1 - R)$, so Section 24.2's credit triangle is the continuous-premium limit of this contract rather than an approximation to it. [TN25] does the integrals behind both legs without the half-period rule, derives the closed forms under flat curves, and sets the three payment conventions (at default, half-period, period end) side by side. Example 25.1 takes a single contract from a quoted spread to a cash amount; the table in Example 25.2 is the same calculation repeated across spreads.

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Example 25.1 Valuing a CDS from first principles

Take \$10 million of five-year protection on an issuer quoted at a par spread of 250 basis points, traded against the standard 100 basis point coupon. Recovery is 40%, the risk-free rate is 3% continuously compounded, and premiums are quarterly.

A hazard rate comes first. The credit triangle turns the quote into a default intensity,

$$\lambda = \frac{s}{1-R} = \frac{0.0250}{0.60} = 4.167\%$$

and the survival probability follows as $Q(t) = e^{-\lambda t}$:

Year	1	2	3	5
$Q(t)$	0.9592	0.9200	0.8825	0.8119
Cumulative default	4.08%	8.00%	11.75%	**18.81%**

The premium leg. The buyer pays the coupon quarterly, but only while the issuer survives, and pays the accrued fraction of a period if it defaults part-way through one. Per unit of spread,

$$A = \underbrace{\sum_i \Delta_i D(t_i) Q(t_i)}_{\text{scheduled}} + \underbrace{\sum_i \frac{\Delta_i}{2} D(t_i)\big[Q(t_{i-1}) - Q(t_i)\big]}_{\text{accrued on default}} = 4.1647 + 0.0218 = \mathbf{4.1865}$$

Both sums are multiples of $\sum_{i=1}^{20} D(t_i)Q(t_i) = \sum_{i=1}^{20} e^{-(r+\lambda)t_i} = 16.6588$, since $Q(t_{i-1}) - Q(t_i) = (e^{\lambda\Delta} - 1)\,Q(t_i) = 0.010471\,Q(t_i)$: the scheduled part is 0.25×16.6588, the accrued part $0.125 \times 0.010471 \times 16.6588$. That is the risky annuity, and the 250 bp row of Example 25.2.

The protection leg. The seller pays $1 - R$ on default, discounted from that quarter's end:

$$(1-R)\sum_i D(t_i)\,[Q(t_{i-1}) - Q(t_i)] = 0.60 \times 0.010471 \times 16.6588 = \mathbf{10.4662}\% \text{ of notional}$$

The two legs should reproduce the quoted spread. Dividing the protection leg by the annuity gives the 250 bp back, to within two thousandths of a basis point:

$$s = \frac{0.104662}{4.1865} = 249.998 \text{ bp.}$$

The closed form (25.5) gives the same figure. The near-exactness is a property of the convention rather than a theorem: with the default payment and the accrued coupon both made at the quarter end the discount factors cancel. The accrual term is what makes it work: worth only 0.0218 of the 4.1865, half of one percent of the annuity, but leaving it out puts the par spread at 251.31 bp.

Pay the protection at the default date instead, as the ISDA Standard Model does, and discount it from there. Integrating against the default density (Section 3 of [TN25]) gives $(1-R)\,\lambda\big(1 - e^{-(r+\lambda)T}\big)/(r+\lambda) = 10.5056\%$ instead of 10.4662%; over the same annuity the par spread is 250.94 bp, and the upfront 6.3191% (\$631,908), some \$3,941 more. So the credit triangle is exact for a continuously paid premium and about a basis point light for a quarterly contract.

None of what has been assumed is innocent. Default arrives with constant intensity, so the credit curve is flat. The risk-free rate is flat and independent of the default time, so no wrong-way risk appears. Recovery is known at 40% rather than uncertain and correlated with default. And $\Delta_i = t_i - t_{i-1}$ is taken as exactly 0.25: real contracts accrue Actual/360 between the March, June, September, and December payment dates, making Δ_i nearer 0.2536, which lifts the annuity by 1.44% and, with the quote held at 250 bp, raises the upfront by \$8,154. The conventions are worth more than the third decimal place of the model.

Now the cash changes hands. Back on the quarter-end convention, protection is worth more than the 100 bp coupon stream, and the buyer pays the difference at inception:

$$\text{upfront} = 10.4662\% - 100\,\text{bp} \times 4.1865 = 10.4662\% - 4.1865\% = \mathbf{6.2797}\%$$

or **\$627,966** on \$10 million. Equation (25.1)'s short form, $(250 - 100)\,\text{bp} \times 4.1865$, gives 6.2798%. The short form assumes a par spread of exactly 250 bp; the 0.002 bp by which 249.998 falls short, times the annuity, is the 0.0001% between them: \$9 on \$10 million, \$627,975 against \$627,966.

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Example 25.2 From a spread to an upfront

Five-year contracts on a 100 bp coupon, recovery 40%, rates 3%:

Par spread	Hazard rate	Risky annuity	Upfront
40 bp	0.667%	4.5513	−2.73%
100 bp	1.667%	4.4428	0.00%
150 bp	2.500%	4.3550	+2.18%
250 bp	4.167%	4.1865	+6.28%
500 bp	8.333%	3.8027	+15.21%
800 bp	13.333%	3.4044	+23.83%

The annuity falls from 4.55 to 3.40 as the spread widens, because a contract on a deteriorating credit is expected to terminate early: the risky annuity is the credit market's duration, and default ends the payments rather than only discounting them.

The purpose of the convention is fungibility. With a fixed coupon, standard payment dates, and a standard effective date for the protection, two contracts on the same issuer and maturity differ only in the upfront paid at inception, a cash flow already settled, so they net, they can be cleared, and a position can be closed by trading the offsetting contract. The reform was infrastructure, and it did more for the market's resilience than any pricing insight in this chapter.

25.4 MARKING A POSITION

Under the standard contract a protection position's value is the difference between the current par spread and the fixed coupon, multiplied by the current risky annuity,

$$\text{MTM} = \left(s_t - c\right) A_t \tag{25.6}$$

where c is the standard coupon the contract pays (100 or 500 bp for a North American corporate) and not the par spread at which it was traded. That spread has already been settled: the buyer paid or received the upfront of (25.1) on the trade date, and the cash is gone. What remains is a contract paying c, and (25.6) is what it is worth. The subscripts are easier to miss: s_t and A_t *both* carry t. Example 25.3 shows what getting either wrong costs.

Example 25.3 The annuity moves too

A desk bought five-year protection when the par spread was 120 bp, on the standard 100 bp coupon. The five-year risky annuity at 120 bp was 4.4074, so by (25.1) it paid $(120 - 100)\text{ bp} \times 4.4074 = 0.88\%$ of notional upfront, \$88,148 on \$10 million. A year later the four-year par spread is 300 bp.

What the desk holds is a four-year contract paying 100 bp, and the four-year risky annuity *at the new spread* is 3.4103, so

$$\text{MTM} = (300 - 100)\text{bp} \times 3.4103 = 6.82\% \text{ of notional},$$

or \$682,067, what a dealer would pay to take the contract over. The gain since inception is that less the upfront paid, $6.82\% - 0.88\% = 5.94\%$, or \$593,919, before the year of 100 bp coupons the desk has paid to carry the position.

Two errors are common. Marking against the traded spread, $(300 - 120)\text{ bp} \times 3.4103 = 6.14\%$, mixes a cash flow that has already happened into the value of what remains: the 120 bp was settled in the upfront and does not enter the mark. And using the annuity at the old spread (the four-year annuity at 120 bp is 3.6118) gives $(300 - 100)\text{ bp} \times 3.6118 = 7.22\%$, an overstatement of 5.9%. The annuity must be computed on the current curve, because the premium stream that will actually be paid is the one the market now expects.

The direction of the second error is systematic. As a credit deteriorates the annuity shortens, so protection gains less than a naive duration calculation suggests. The buyer's position is concave in the spread [TN25], and the concavity works against the buyer. A desk sizing a hedge of a fixed

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exposure on duration alone will be under-hedged exactly when the hedge is working.

The standard risk measure is the **CS01**, the change in value for a one-basis-point rise in the spread. It equals the risky annuity times one basis point only when the spread is at the coupon; here the annuity's fall as the spread widens makes the CS01 $3,195 per $10 million, 6% below the $3,410 that the annuity alone suggests, so it must be recomputed at the current spread.

25.5 SETTLEMENT AND THE AUCTION

When a credit event occurs, the contract must be settled, and the mechanism has a history. Physical settlement was the original design: the protection buyer delivers a defaulted bond and receives par. It works only if there are enough bonds, and Section 25.1 noted that protection outstanding can exceed the debt outstanding. When it does, protection buyers bid for deliverable bonds and their price is driven *up*, which cuts the payout on exactly the contracts that should be paying out. This happened, most visibly when Delphi filed in 2005, and it is why the mechanism changed.

The auction replaced physical settlement. Run under separate protocols before 2009 and written into the standard contract by the 2009 reforms, a two-stage auction determines a single recovery rate for the whole market, and all contracts settle in cash against it. The first stage collects dealers' two-way markets and physical settlement requests: the best half of the markets that do not cross average to an **initial market midpoint**, and the requests net to a **net open interest** (the amount of bonds still wanted for purchase or offered for sale once offsetting requests are matched). The second runs a limit-order auction to clear that interest. The final price may move away from the midpoint as far as the open interest pushes it, but no more than a small cap the other way, and the clearing price is the final recovery.

The design solves the squeeze and creates a different problem. A single number now determines the payout on every contract, so many people have a large stake in the auction's outcome and a proportionate incentive to influence it. The mechanism's defenses are that it is open, that participation is broad, and that the two stages make manipulation expensive—not that it is impossible.

The auction recovery is not the eventual recovery. It is the market's price for the defaulted claim a few weeks after the event, which is a forecast made under duress. Realized recoveries after a completed restructuring have differed from auction recoveries substantially in both directions, which matters for Chapter 24's Example 24.3: the R used to convert spreads into hazard rates is a convention, and even the observed R at settlement is a market price rather than a realized outcome.

The Lehman Brothers auction of October 2008 is the standard illustration. It set the recovery at 8.625 cents on the dollar, so sellers of protection owed 91.375% of face, yet the Depository Trust and Clearing Corporation (DTCC) reported about $72 billion of gross notional on Lehman and a net transfer of only about $5.2 billion, against rumors near $400 billion. That is netting, as Section 1.6 said, not a failure of the contracts.

25.6 INDEX CDS

Most CDS risk trades through indices (CDX in North America, iTraxx in Europe), each a standardized portfolio of equally weighted issuers (125 for the investment-grade indices, which the examples assume; 100 for CDX high yield and 75 for iTraxx Crossover), rolling to a new series every six months. The index is built by rule, not by valuation. An administrator selects the constituents at each roll, weights them equally, and fixes the coupon (100 bp for the investment-grade indices, 500 for high yield) for the life of the series. For CDX Investment Grade the constituents are 125 North American issuers rated investment grade and ranked by their CDS trading volume in DTCC data; the high-yield and European indices apply the

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same procedure to a different universe. A new series, which becomes the *on-the-run* series, is launched every March and September; the old one continues to trade as an *off-the-run* series. A constituent that suffers a credit event between rolls is not replaced. Its $1/125$ share is settled through the auction of Section 25.5 (the protection buyer receives $(1 - R)/125$ of the index notional), and it drops out: the contract's notional is scaled by an **index factor**, $124/125 = 0.992$ after one event, the coupon is paid on the reduced notional from then on, and the series is relabeled a new version so that contracts traded before and after the event remain the same instrument. The survivors carry on unchanged to the series' own maturity; the next roll launches a new series and neither alters nor terminates the old one. High-yield indices, with 100 constituents and more frequent events, see the factor move most; investment-grade series often finish their life at a factor of one.

An index CDS is therefore a single contract paying protection on each constituent as it defaults, with the notional reduced accordingly. It is the credit market's most liquid instrument: bid–ask spreads on the index are a fraction of those on constituents, and a desk hedging a portfolio will usually hedge with the index and accept the basis.

Example 25.4 Why the index does not trade at the average spread

Consider a 125-issuer index whose constituents average 80 bp: 115 issuers at 55 bp and 10 distressed issuers at 367.5 bp.

The arithmetic average is 80.0 bp. The spread that makes the index's protection and premium legs balance is **77.3 bp**.

The 2.7 bp difference is arithmetic, not a mispricing. Both legs are additive across constituents. The index's protection leg is the sum of the constituents' protection legs, $\sum_j V_j$, each on its $1/125$ share; and because every constituent pays the same coupon until its own default, the index's premium leg per unit of coupon is the sum of their risky annuities, $\sum_j A_j$. Applying (25.4) to the two sums gives

$$s_{\text{index}} = \frac{\sum_j V_j}{\sum_j A_j} = \frac{\sum_j A_j\, s_j}{\sum_j A_j} \tag{25.7}$$

so the index's fair spread is the annuity-weighted average of its constituents' par spreads, not the arithmetic one. This is what the constituents imply the index should be worth, its *intrinsic* level, rather than how the index is built or quoted: the traded index has its own price, and the standard model converts that quote to an upfront with a single flat hazard curve for the whole index rather than constituent by constituent. A wide-spread issuer has a shorter risky annuity (Example 25.2's effect), so it carries less weight in (25.7) than in a simple mean, and the index trades below it. The wider the dispersion, the larger the gap.

The difference between the level at which the index actually trades and the intrinsic level (25.7) computed from its constituents is the **index skew**. It is a real trading opportunity when it is large enough to cover the cost of trading 125 individual issuers against one index. It widens when the index is used heavily as a macro hedge, because that demand moves the index without moving the constituents. This is Section 19.7's index protection demand appearing in a different market.

Options on a CDS. A **payer option** gives the right to buy protection at a fixed spread s_K on the expiration date T, and a **receiver option** the right to sell it. With the forward risky annuity A as numeraire (the premium leg of (25.2) starting at T, valued today), a payer on a forward spread s_F with volatility σ is worth

$$A\,[\,s_F N(d_1) - s_K N(d_2)\,], \qquad d_{1,2} = \frac{\ln(s_F/s_K) \pm \sigma^2 T/2}{\sigma\sqrt{T}}.$$

This is a payer knocked out by default before T, and one that is not adds the **front-end protection**. With $s_F = 250$ bp on five-year protection starting at $T = 1$, so that $A = e^{-(0.03+0.0417)} \times 4.1865 = 3.897$, $s_K = 300$ bp and an illustrative $\sigma = 60\%$, $d_1 = -0.004$,

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$d_2 = -0.604$ and the payer is worth $3.897 \times (250 \times 0.4985 - 300 \times 0.2730) \approx 166.5$ bp of notional. Macro funds buy payers on the investment-grade index to cap the cost of a widening at the premium; dealers carry the volatility.

25.7 INDEX TRANCHES

Slice the index's loss into layers and sell each separately. A tranche $[a, b]$ absorbs portfolio losses between a and b percent of notional: a is its **attachment point**, the pool loss at which it begins to take losses, b its **detachment point**, the pool loss at which it is exhausted, and the width $b - a$ is its own notional as a fraction of the pool's. The 0–3% **equity** tranche takes the first losses; the 15–30% **senior** tranche suffers only after everything below it is exhausted; the layers in between are the **mezzanine** tranches.

Section 24.13 established the essential point: correlation leaves the portfolio's expected loss unchanged and redistributes it between the layers. The procedure comes before the numbers, because a tranche is valued the same way whatever the pool.

Valuing a tranche. Take a pool of n equally weighted issuers, each with recovery R and the same probability $\pi(t)$ of having defaulted by date t, joined by the one-factor copula of Section 24.13 with correlation ρ. Four steps take a tranche from that description to a par spread, and the model lives only in the second.

Step 1: the tranche's loss as a function of the pool's. Write $K(t)$ for the number of defaults by t and $L(t) = (1 - R)\,K(t)/n$ for the pool's loss fraction. A tranche $[a, b]$ has lost, as a fraction of its own notional,

$$T_{[a,b]}(L) = \frac{\min\{\max(L - a,\, 0),\; b - a\}}{b - a}, \tag{25.8}$$

zero until the pool's loss reaches the attachment point a, then rising one for one until the detachment point b exhausts it. Its remaining notional at t is $1 - T_{[a,b]}(L(t))$.

Step 2: the pool's loss distribution at each date, conditional on the factor. Given the common factor M (the standard normal variable that stands for the state of the economy in Section 24.13's copula), each issuer has defaulted by t independently with probability

$$p_t(M) = N\!\left(\frac{N^{-1}\big(\pi(t)\big) - \sqrt{\rho}\,M}{\sqrt{1-\rho}}\right), \tag{25.9}$$

which is equation (24.12) with the threshold set at date t's default probability, so $K(t) \mid M$ is binomial with parameters n and $p_t(M)$. The tranche's conditional expected loss is therefore a finite sum,

$$\mathbb{E}\big[T_{[a,b]}(L(t)) \,\big|\, M\big] = \sum_{k=0}^{n} \binom{n}{k} p_t(M)^k \big(1 - p_t(M)\big)^{n-k}\, T_{[a,b]}\!\Big(\frac{(1-R)\,k}{n}\Big). \tag{25.10}$$

Step 3: integrate out the factor. The unconditional expected tranche loss at t is

$$E_{[a,b]}(t) = \int_{-\infty}^{\infty} \mathbb{E}\big[T_{[a,b]}(L(t)) \,\big|\, M\big]\, \phi(M)\, dM, \tag{25.11}$$

where ϕ is the standard normal density of the factor M, which is equation (24.13) with the tranche loss as the integrand. In practice the integral is a Gauss–Hermite quadrature (a weighted sum of the integrand at a few dozen fixed values of M) repeated at every payment date.

Step 4: the two legs. On payment dates $t_1 < \cdots < t_m$ with accrual fractions Δ_j and discount factors $D(t_j)$, the protection leg pays the increase in the expected tranche loss over each period, and the premium leg pays the spread on the expected remaining notional, with the half-period

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accrual of (25.2) on the part lost during the period, which is the same as paying it on the expected remaining notional averaged over the period:

$$\begin{aligned} V_{[a,b]} &= \sum_{j=1}^{m} D(t_j)\big[E_{[a,b]}(t_j) - E_{[a,b]}(t_{j-1})\big], \\ A_{[a,b]} &= \sum_{j=1}^{m} \Delta_j\, D(t_j)\Big[1 - E_{[a,b]}(t_j) + \frac{1}{2}\big(E_{[a,b]}(t_j) - E_{[a,b]}(t_{j-1})\big)\Big] \\ &= \sum_{j=1}^{m} \Delta_j\, D(t_j)\Big[1 - \frac{1}{2}\big(E_{[a,b]}(t_{j-1}) + E_{[a,b]}(t_j)\big)\Big], \end{aligned} \tag{25.12}$$

and the par spread is $s_{[a,b]} = V_{[a,b]}/A_{[a,b]}$, exactly as in (25.4) with the tranche's expected loss in the place the default probability occupied for a single issuer. Equity tranches are quoted, like the CDS of distressed issuers, as an upfront against a fixed running coupon of 500 basis points through (25.1); that was the equity convention from the first standardized index tranches, and since the 2009 move to fixed coupons the other tranches have been quoted the same way, as an upfront against 100 or 500 basis points.

Everything in Steps 1, 3, and 4 is bookkeeping that any model must do; Step 2 is where the Gaussian copula enters, and replacing it (with random recovery, a heavier-tailed factor, or a different copula) changes only the conditional probability (25.9). The same is true of the pool's description. Equal weights are exact for the standard indices, but a common $\pi(t)$ is a simplification: a dealer's model gives each constituent its own hazard curve, bootstrapped as in Section 24.3, and the binomial in (25.10) becomes a recursion that adds the constituents one at a time (Andersen, Sidenius, and Basu, 2003; Hull and White, 2004), with nothing else in the four steps moving. Example 25.5 runs the four steps on one pool and tabulates what comes out at the five-year date.

Example 25.5 The same portfolio, five different instruments

Take a 125-issuer index, each issuer with a 7% five-year default probability and 40% recovery. The portfolio's expected loss is $0.07 \times 0.60 = 4.200\%$ of notional at every correlation, because expectation is linear: the pool's loss is the average of 125 indicator variables times $1 - R$, the expectation of each indicator is its own default probability, and no joint probability (hence no ρ) enters the sum.

The tranche numbers come from the copula exactly as Example 24.8 applied it, with a loss given default and a tranche attached. Each default costs $1 - R = 0.6$ of its $1/125$ share, so the pool's loss fraction is $L = 0.6K/125$ for K defaults, and a tranche $[a, b]$ loses $T_{[a,b]}(L)$ of (25.8). This is Step 1. At $\rho = 0.30$ the threshold for a default probability of 7% is $N^{-1}(\pi) = -1.4758$; conditional on the common factor M, defaults are binomial with parameters 125 and $p(M)$, which is Step 2; the conditional pieces (the pool's expected loss and, from (25.10), the expected loss of the equity and senior tranches as fractions of their own notional) are:

M	$p(M)$	$\mathbb{E}[K \mid M]$	$\mathbb{E}[L \mid M]$	$\mathbb{E}[T_{[0,\,0.03]} \mid M]$	$\mathbb{E}[T_{[0.15,\,0.30]} \mid M]$
−3	57.9%	72.4	34.8%	100.0%	99.7%
−2	32.5%	40.6	19.5%	100.0%	30.1%
−1	13.4%	16.7	8.0%	100.0%	0.0%
0	3.9%	4.9	2.3%	71.8%	0.0%
1	0.8%	1.0	0.5%	15.6%	0.0%
2	0.1%	0.1	0.1%	2.1%	0.0%

Integrating each column against $\phi(M)$, which is Step 3 and (25.11), gives the pool 4.200%, the equity tranche 62.59% and the senior tranche 1.88% (the $\rho = 0.30$ row below). The columns show

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why the two tranches move in opposite directions. The equity tranche is wiped out in every state from $M = -1$ down, so it has nothing further to lose in the bad states that correlation makes more likely, while it is spared in the good states that correlation also makes more likely. The senior tranche loses only in states below about $M = -1.6$, which give it a 4.97% chance of any loss; at $\rho = 0$ the factor moves nothing and that chance is essentially zero. Raising ρ moves probability into those states, and that is the whole of the redistribution.

The table gives expected loss as a percentage of each tranche's own notional, with the 30–100% layer that absorbs everything above the senior tranche:

ρ	0–3%	3–7%	7–10%	10–15%	15–30%	30–100%
0.00	95.76%	32.73%	0.59%	0.00%	0.00%	0.00%
0.10	81.81%	33.49%	9.44%	2.16%	0.10%	0.00%
0.20	71.40%	31.51%	13.50%	5.48%	0.78%	0.00%
0.30	62.59%	29.30%	15.21%	7.87%	1.88%	0.03%
0.50	47.36%	24.75%	15.81%	10.40%	4.27%	0.22%
0.80	26.59%	17.11%	13.35%	10.75%	6.96%	1.05%

Weight each column by its layer's width (0.03, 0.04, 0.03, 0.05, 0.15, 0.70) and every row adds to 4.200%: correlation redistributes the loss and cannot create or destroy any of it. The last layer is what makes the accounting close, carrying nothing at low correlation and, at $\rho = 0.80$, about a sixth of the pool's expected loss.

The equity tranche's expected loss falls from 95.8% to 26.6% as correlation rises; the senior tranche's rises from zero to 6.96%. *Selling equity protection is buying correlation; selling senior protection is selling it.*

The senior tranche is where the danger is, and the right panel of Figure 25.3 draws it on its own scale. A senior tranche valued on a correlation of 0.10 (expected loss 0.098%) when the actual correlation is 0.50 (4.272%) has 43 times the expected loss it was priced for. Nothing about the underlying credits has changed: the same 125 issuers, the same 7% default probabilities, the same 4.2% portfolio expected loss. Only a parameter that no market quoted has changed.

Step 4, equation (25.12), turns the expected losses into prices. Take the default probability to build up at a flat hazard rate fitted to the five-year figure, $\lambda = -\ln(0.93)/5 = 1.451\%$, so that $\pi(t) = 1 - e^{-\lambda t}$; quarterly payments; and 3% discounting. At $\rho = 0.30$ the five tranches' par spreads are

	0–3%	3–7%	7–10%	10–15%	15–30%
Five-year expected loss	62.59%	29.30%	15.21%	7.87%	1.88%
Risky annuity $A_{[a,b]}$	2.86	4.02	4.35	4.50	4.60
Par spread (bp)	2,062	671	318	158	37

The spread is not the expected loss divided by five. The equity tranche's annuity is only 2.86 because most of its notional is expected to be gone before maturity, so the running spread is correspondingly high. That is why the market quotes that tranche as an upfront instead: against the 500 bp coupon, $(0.2062 - 0.05) \times 2.86 = 44.7\%$ of notional at inception. Repeating the calculation for the senior tranche at $\rho = 0.10$ and 0.50 gives 1.9 and 85.5 bp against 37 at 0.30: the 43-fold move in expected loss is a move of about 45-fold in the spread, because the annuity barely changes for a tranche that is rarely touched.

The rating on such a tranche is a statement about ρ. Section 24.10 said the letter did not disclose the correlation assumption. The table says what that omission was worth: the difference between an instrument that almost never loses and one that loses 4.3% of its notional in expectation.

And the market's own correlation number is not a correlation. Quoting a tranche in terms of the ρ that reproduces its price (**implied correlation**) produces different values for different tranches on the same portfolio, which would be impossible if ρ were a property of the portfolio. The market moved to **base correlation**: instead of each tranche's own ρ, quote the ρ that prices the **first-loss tranche** $[0, D]$ for each detachment point D (the equity tranche widened to detach

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at D), and price a mezzanine tranche $[a, b]$ as the difference of the two first-loss tranches $[0, b]$ and $[0, a]$, each at its own base correlation. A first-loss tranche's value is monotone in ρ, so its implied correlation is unique and the curve can be interpolated between quoted detachment points; a mezzanine tranche's is not: Example 25.5's 3–7% tranche loses 32.73% at $\rho = 0$, 33.49% at 0.10 and 31.51% at 0.20, so one price can correspond to two correlations or to none. The device removes the inconsistency by construction without removing the fact that generated it. This is Section 19.1's implied volatility smile again: a quoting convention rather than a belief.

Chapter 26 is about what happened when it was treated as a belief.

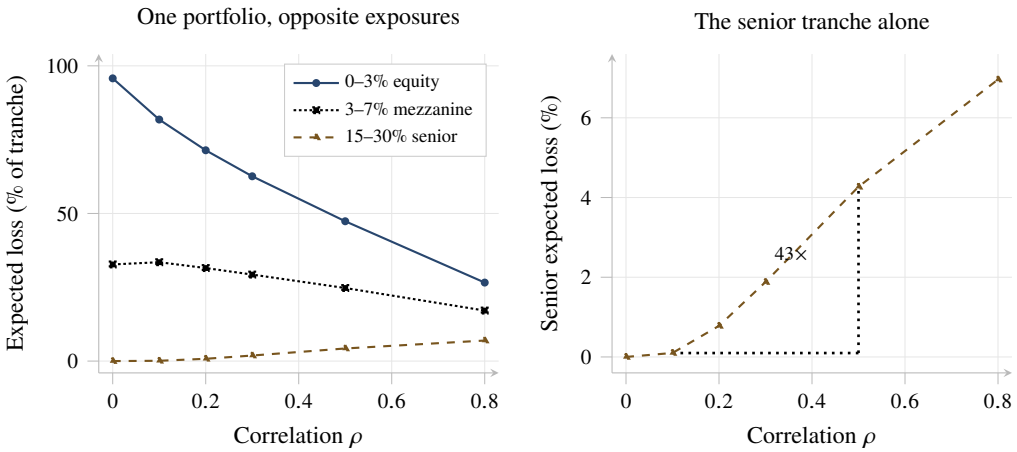


Figure 25.3: Example 25.5's 125-issuer portfolio. The pool's expected loss is 4.200% at every point on both panels.

Figure 25.3 is the picture to carry out of this chapter. The left panel's three curves are one portfolio, and along the horizontal axis only ρ changes: every issuer's default probability and recovery stay fixed, yet the curves go in three different directions; the mezzanine tranche is nearly flat and so says least about ρ. The right panel puts the senior curve on its own scale, where what looked like zero is the forty-three-fold move the example measured.

25.8 BASKET DEFAULT SWAPS

A tranche slices a large portfolio by loss. A **basket default swap** slices a small one (five to ten issuers) by *count*. An nth-to-default basket pays protection on the nth issuer to default and then terminates; the first-to-default basket is the common case, and Section 24.4 has already used it as an instrument whose price, to first order, depends only on the product $\lambda(1 - R)$.

Nothing new is needed to value one. Write $K(t)$ for the number of issuers that have defaulted by t, whose distribution is Section 24.13's equation (24.13). The basket pays exactly when K crosses n, so it is a credit default swap on a synthetic issuer whose survival probability is

$$Q_n(t) = \mathbb{P}\big(K(t) < n\big) \tag{25.13}$$

Substitute Q_n for Q in (25.2) and (25.3) and the par spread is (25.4) unchanged. The basket is a CDS whose reference entity is the nth order statistic of the portfolio's default times, and nothing else needs deriving. Example 25.6 prices all five baskets on one small portfolio.

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Example 25.6 The same five issuers, five different instruments

Take five issuers, each trading at 150 basis points with 40% recovery, discounted at 3%, five years, quarterly premiums. The par spreads, in basis points, are:

ρ	1st	2nd	3rd	4th	5th
0.00	750	133	16	1	0
0.15	669	160	34	5	0
0.30	591	179	56	15	2
0.60	439	196	99	47	17
0.90	269	185	138	102	67

At zero correlation the first-to-default costs the sum of its parts: with independent issuers the first default arrives at the sum of their hazard rates, so the basket is a CDS on one issuer five times as risky, and five issuers at 150 bp give 750 bp. Correlation makes a basket cheaper than the sum because it crowds defaults into the same states: the second and later defaults in a bad state arrive after the contract has terminated.

One entry is worth working in full. Each issuer's hazard rate is $\lambda = 0.0150/0.6 = 2.5\%$ by the credit triangle, so its five-year survival probability is $e^{-5\times0.025} = 0.8825$. At $\rho = 0$ the first-to-default survives only if all five issuers do, $Q_1(5) = 0.8825^5 = 0.5353$. At $\rho = 0.30$, (25.13) is the copula integral $Q_1(5) = \int \left(1 - p_t(M)\right)^5 \phi(M)\, dM = 0.6149$, with p_t from (25.9) at $t = 5$. It is higher because correlation crowds the defaults into the same states and leaves more states with none. Substituting Q_1 for Q at each of the twenty quarterly dates gives a risky annuity of 3.6473 and a protection leg of 0.2157, so $s_1 = 0.2157/3.6473 = 591$ bp, against 3.4665, 0.2600 and 750 bp at $\rho = 0$. The annuity rose and the protection leg fell, and both moved for the same reason.

The opposite movement of the two ends is Section 25.7's, in a different instrument. As correlation rises the first-to-default falls from 750 to 269 bp while the fifth-to-default rises from nothing to 67. A first-to-default is the equity tranche of this portfolio and an nth-to-default with n large is its senior tranche, and the buyer of one and the buyer of the other want opposite things from the same number.

And the five instruments add up. Since $\sum_{n=1}^{5} \mathbf{1}\{K \geq n\} = K$, the five protection legs must sum to the five single-name protection legs at every correlation, and they do, to machine precision. Every default that occurs is paid for exactly once in either portfolio, at the moment it occurs (by the CDS on the issuer that defaulted, or by the basket whose turn it was), so the two sets of cash flows coincide on every path, not merely on average. Correlation shifts protection value among the baskets and cannot create or destroy any of it, which is Section 24.13's statement about tranches, arrived at from the other direction.

25.9 TOTAL RETURN SWAPS

Section 7.12 introduced the total return swap and said Chapter 27 would examine what it made possible in 2021. The instrument belongs here because it is the principal way credit and equity exposure is taken without ownership.

One party pays the total return on a reference asset (coupons, dividends, and price change) and receives a financing rate. The receiver of the total return has the full economics of owning the asset without owning it, and without paying for it: the position is funded by the swap. Three properties follow, and they are the same three that recur whenever an instrument separates economics from ownership.

Leverage is set by the margin, not by the asset. A cash buyer of a bond posts its full price. A total return receiver posts initial margin, which may be a small fraction. The economics are identical and the balance sheet is not.

The exposure is not visible in ownership disclosures. The swap counterparty holds the asset as a hedge and appears as the owner. Positions built this way do not show up where people look for them, which is the quiet accumulation of leverage of Section 7.12, and Chapter 27's

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case study shows what it enabled.

And the risk is counterparty risk presented as market risk. The dealer paying total return is hedged in the market and exposed to the client; if the asset falls, the client owes variation margin, and if the client cannot pay, the dealer holds the asset. The hedge is perfect against the market and worthless against the client, the configuration Chapters 8 and 9 dwelt on.

25.10 RESTRUCTURING

Restructuring is the most contested credit event, and the one where the definition works hardest. A restructuring (a maturity extension, a coupon reduction, a change in seniority, imposed on holders) is economically a loss but not a missed payment. Whether it should count as a credit event has been answered differently in different places and different decades.

The problem it creates is the cheapest-to-deliver problem in its sharpest form. After a bankruptcy, all senior unsecured obligations are worth roughly the same, so which one is delivered hardly matters. After a restructuring, the entity is still alive, and its obligations trade at very different prices depending on maturity: a distressed but living issuer's thirty-year bond is worth far less than its one-year bond. A protection buyer could deliver the cheapest, collecting a payout far exceeding their actual loss.

The market's responses were successive restrictions on what could be delivered (limits on deliverable maturities, then maturity bands matched to the contract's) and, in North America, largely dropping restructuring as a trigger for corporate issuers. Europe kept it, mainly because bank capital rules reward it: a hedge that does not pay on restructuring does not obviously transfer the risk of the loan it hedges. Example 25.7 prices what the clause is worth.

Example 25.7 What the clause is worth

Take an issuer with a 3% annual intensity of bankruptcy or failure to pay, recovering 40%. Suppose including restructuring adds a further 15% to that intensity (0.45 percentage points) but restructuring events recover 70%.

No-restructuring contract	180.0 bp
With restructuring	193.5 bp
Value of the clause	13.5 bp (7% of the spread)

The point is not the number, which depends entirely on the assumed intensities, but that it is small and stable right up until it is neither. When a restructuring is actually negotiated, one contract pays and the other does not, and the gap between them goes from 13 bp to the whole protection payment. A risk system that treats them as substitutes because their historical basis is small has mismeasured a binary exposure as a small one.

25.11 WHEN THE DEFINITION AND THE ECONOMICS COME APART

Three episodes each established something about what a CDS actually insures.

Greece, 2012. The sovereign's debt was restructured through a package that was initially presented as voluntary, and a voluntary exchange is not a credit event. Collective action clauses were then applied retroactively to bind holdouts, and that application—compulsion—triggered the contracts. Protection paid, but only after a period in which it was genuinely unclear whether the largest sovereign restructuring in history would trigger the instrument designed to insure against exactly that.

Manufactured events. In the 2010s several transactions were arranged specifically to trigger or avoid triggering CDS: a distressed borrower persuaded to miss a small payment so that a protection holder would be paid, or refinanced in a way designed to leave protection worthless.

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ISDA amended the definitions in 2019 to address the most obvious form. The episodes are instructive because they show that when a contract references an event rather than a loss, the event can be purchased.

Russia, 2022. Payments on sovereign debt were made in rubles, or blocked by sanctions, or made but unable to reach holders. The question of whether a payment prevented by a third party is a failure to pay had no clean answer, and the determinations process had to resolve a case its drafters had not anticipated. A contract that insures against a borrower's unwillingness or inability is silent about a borrower who is willing, able, and prevented. The committee found a failure to pay in June 2022, on $1.9 million of accrued interest, and sanctions then delayed an auction that set the recovery near 56 cents.

The same gap runs through the corporate market. A liability-management exercise of the kind Section 26.15 describes can cut a lender's recovery while no payment is missed and, in the North American contract, no restructuring counts, so the protection pays nothing, or pays on an obligation whose price has fallen less than the loan's.

25.12 CREDIT SUISSE, MARCH 2023

The most instructive recent case triggered nothing at all, which is the point.

Case Study 25.1 The AT1 write-down and what subordination means

Additional Tier 1 instruments were created after the 2008 crisis to do something specific: absorb losses while a bank is still operating, so that it can be recapitalized without a resolution. They pay a high coupon, and they contain contractual triggers under which they are written down or converted to equity.

In March 2023, as part of the arrangements accompanying Credit Suisse's acquisition, roughly CHF 16 billion of AT1 instruments were written down to zero. Shareholders received approximately CHF 3 billion.

The capital structure was inverted. An instrument senior to equity was extinguished while equity retained value. That is the opposite of the ordering that the word "subordination" is supposed to guarantee.

Three observations follow, and the third is the one for this chapter.

The write-down was contested in law, not only in markets. The Swiss AT1 terms allowed write-down on a viability event determined by the regulator, and holders had been told so. In October 2025, however, the Federal Administrative Court found that no such event had occurred and that the Swiss regulator's order lacked a legal basis; the appeal is pending, and the bonds remain written down. Whichever way it ends, almost no one had priced a write-down ahead of equity, because the instruments had been sold and analyzed as if the ordinary hierarchy applied.

The repricing was immediate and general. AT1 instruments across European banks fell sharply, and several regulators affirmed that in *their* jurisdictions equity would absorb losses first. The market had been pricing a legal ordering that turned out to be jurisdiction-specific.

And no credit event occurred. Credit Suisse did not default; it was acquired. Its senior obligations performed throughout, and when the determinations committee was asked in May 2023 whether the write-down was a governmental intervention, it ruled that it was not: the AT1s ranked below every obligation the contracts reference, subordinated CDS included. So CDS on Credit Suisse did not pay, while holders of an instrument ranking above equity lost everything.

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That last sentence is Section 25.2's thesis in its cleanest form. A CDS insures against a list of events, not against loss. A protection buyer who had correctly identified Credit Suisse as the weakest large European bank, and had bought protection to express it, would have been right about the institution and received nothing—while an AT1 holder with no view at all lost the entire position.

The general lesson extends past this instrument: wherever a payoff is defined by a legal event rather than by an economic outcome, the document is the instrument, and the residual risk is the risk that the document does not say what everyone assumed.

25.13 WHAT A CDS ACTUALLY INSURES

The episodes of Sections 25.11 and 25.12 are usually told as anecdotes. They are more useful as a taxonomy, because a hedger can check each item in advance. A protection buyer who wants to be made whole on a credit loss faces four distinct residual risks, and only the last is normally measured.

Definitional risk. The loss occurs and no listed event does. This is the Credit Suisse case, and it is the largest of the four because it is binary and uncorrelated with anything a risk system tracks. The check is to read the credit event definitions against the ways this borrower could plausibly impose a loss; for a bank, resolution, recapitalization, and default are three routes with three different answers.

Determination risk. An event occurs and the committee rules that it did not, or takes long enough that the position must be marked in the meantime. Greece and Russia are both instances. The check is whether the borrower's plausible distress route is one the committee has ruled on before. Novelty is the risk.

Settlement risk. The event is determined, the contract pays, and the auction recovery differs from the loss suffered on the hedged asset. A hedger holding a long-dated bond who receives an auction recovery struck on the cheapest deliverable has been made whole on a different instrument.

Basis risk. The protection references an entity and the exposure is to a different one: a subsidiary, a guaranteed affiliate, or an unrelated proxy issuer, as in the CVA hedging of Section 9.10. This is the one desks do measure, and for a hedge on the entity itself or a close affiliate it is usually the smallest.

The ordering is the reverse of the attention these risks receive: basis risk is quantified, reported, and hedged, while definitional risk, the one the Credit Suisse case exposed, is not quantified at all.

25.14 CONVERTIBLE BONDS

Section 10.10 deferred convertibles here: they are the other instrument that fuses credit with an option, and the fusion is what makes them interesting. A convertible is a bond that the holder may exchange for a fixed number of shares, the **conversion ratio**. So it is a corporate bond plus a call on the issuer's own equity, and Section 17.4's dilution analysis applies to that call.

Example 25.8 Decomposing a convertible

Face 1000, 2% annual coupon, five years, conversion ratio 20. The stock is at 40, its volatility 35%, rates 4%, and the issuer's credit spread 300 bp.

How the split is made. The floor is the 2% bond discounted at the 4% risk-free rate plus the 300 bp spread, 7% continuously compounded; the option is a European call on 20 shares struck at the conversion price, valued by the Black–Scholes–Merton formula at the 4% risk-free rate with no dividends ($d_1 = 0.362$, $d_2 = -0.421$, 11.86 a share); and the convertible is their sum. The additive

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split is an approximation: it treats conversion as exercisable only at maturity, ignores any issuer call, and lets the option survive default. Example 25.9 measures each omission. Tsiveriotis and Fernandes (1998) is the widely used practitioner model, pricing the debt and equity components on one tree at different discount rates.

Bond floor (straight bond discounted at 7%)	786.14
Conversion value, or *parity* (20×40)	800.00
Conversion price (1000/20)	50.00
Conversion premium over the share price	25.0%
Embedded call on 20 shares struck at 50	237.13
Convertible value	**1023.28**
Equity delta	12.82 shares (64% of the 20)

The instrument's behavior across share prices shows what the issuer sold:

Share price	Convertible	Parity	Delta (shares)
20	827.87	400.00	6.00
30	908.55	600.00	9.95
40	1023.28	800.00	12.82
50	1162.13	1000.00	14.82
70	1484.96	1400.00	17.18

The convertible is bond-like when the stock is low and equity-like when it is high, with the transition governed by the embedded call's delta. That is its selling point to an issuer: a 2% coupon on debt that would otherwise cost 7%, for equity upside above 50, which is Section 17.9's bond-plus-warrant with the components fused rather than separable.

Section 13.7 said why the additive split is not a valuation: conversion and call are node-by-node decisions that interact, so a bond and its embedded options cannot be valued separately and added; and credit belongs on the same tree, since a default ends the conversion right too. Hull's construction puts all three on one tree: each step of length Δt the stock moves up by u, down by $d = 1/u$, or the issuer defaults with probability $1 - e^{-\lambda \Delta t}$, leaving the stock worthless and the bond worth its recovery RF on face value F:

$$u = e^{\sqrt{(\sigma^2 - \lambda)\,\Delta t}}, \qquad p_u = \frac{e^{r\Delta t} - d\,e^{-\lambda \Delta t}}{u - d}, \qquad p_d = \frac{u\,e^{-\lambda \Delta t} - e^{r\Delta t}}{u - d}, \tag{25.14}$$

chosen so that the stock's expected growth, default paths included, is the risk-free rate and its total variance per unit time, jump included, is σ^2. The credit triangle gives $\lambda = s/(1 - R)$. With conversion ratio n and call price K_c, each node's rollback is

$$V = \max\left\{ nS,\ \ \min\left(e^{-r\Delta t}\left[p_u V_u + p_d V_d + (1 - e^{-\lambda \Delta t})\, RF \right],\ K_c \right) \right\} + \text{coupon due}, \tag{25.15}$$

read from the inside out: the bracket is the continuation value; the issuer calls when continuation exceeds K_c; and the holder converts whenever n shares are worth more than what is left. Without a call the min drops out; with conversion only at maturity the outer max acts only there.

Example 25.9 Pricing the convertible on a tree with default

Put Example 25.8's bond on a tree of five annual steps, so that the coupon dates are the nodes. From $s = 300$ bp and $R = 40\%$, $\lambda = 0.03/0.6 = 5\%$; then (25.14) gives $u = 1.3090$, $d = 0.7639$, $p_u = 0.5763$, $p_d = 0.3749$, default probability 0.0488 and discount factor $e^{-0.04} = 0.9608$ per step. In the callable case the issuer may call at 1100 from year 2; Figure 25.4 draws that tree.

One node is worth working in full, through the rollback (25.15). At $t = 4$ with the stock back at 40, the surviving successors are $S = 52.36$, where the bond converts at maturity for 1047.19 plus the final coupon, 1067.19, and $S = 30.56$, where it is redeemed for 1020.00 with the coupon; default

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pays 400. Continuation is $0.9608\times(0.5763\times1067.19+0.3749\times1020.00+0.0488\times400) = 977.09$, against $20\times40 = 800$ for converting now, so the holder holds, and the node is worth 977.09 plus the year-4 coupon, 997.09. The call shows up in two forms. At $t = 2$ with $S = 68.54$ continuation is 1426.81, so the issuer calls, and the holder, offered 1100, converts instead for 1370.77: the call forced conversion three years early and cost the holder 56. At $t = 3$ with $S = 52.36$ continuation is 1148.00, the issuer calls, and this time 20 shares are worth only 1047.19, so the holder takes the 1100 in cash and loses 48 against holding on.

	Five steps	**500 steps**
Bond floor (no conversion right)	793.50	795.10
Conversion at maturity only, or at any time	1046.38	1057.82
Conversion at any time, callable at 1100 from year 2	994.47	988.40
Additive split of Example 25.8, for comparison	1023.28	

Several readings deserve comment. The floor is 795 rather than 786 because recovery is modeled as a payment of 400 at default rather than folded into a 7% discount rate. Conversion at maturity only gives 1058 against the additive 1023: 263 rather than 237 for the option, the rest being the higher floor. The default branch makes the call dearer, not cheaper, because a stock that can drop to zero must grow at $r + \lambda$ conditional on surviving (Merton's 1976 jump-to-ruin result), which outweighs the lower diffusive volatility of 26.9%. The American right is worth nothing here, since without dividends the holder never converts early (Section 11.8). What changes the value is the issuer's call, −52 on five steps and −69 converged, because it ends the bond, by forced conversion or by redemption at 1100, whenever continuing would be worth more than 1100; that is the interaction Section 13.7 said the additive method discards, and it is the largest term.

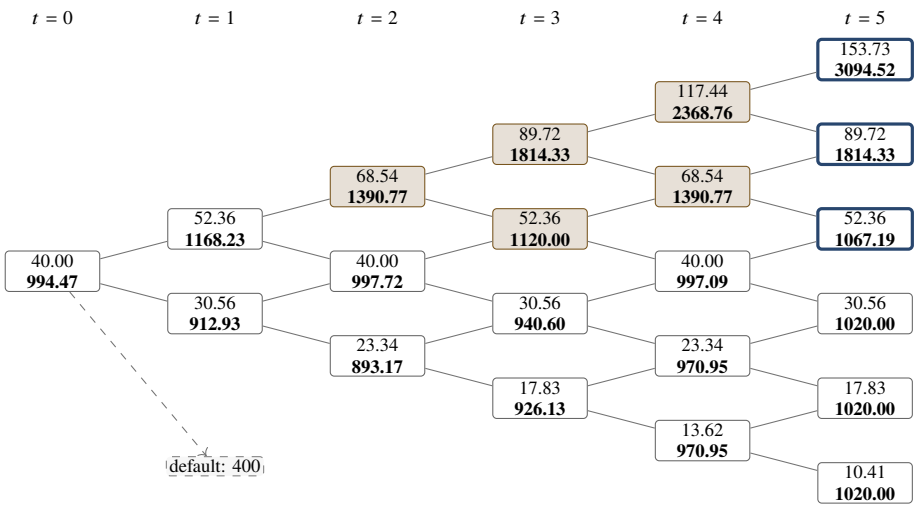


Figure 25.4: Example 25.9's callable convertible on the five-step tree: share price above, bond value (with the coupon paid that date) below. Shaded nodes are called, heavy-bordered nodes converted; the default branch worth 400 exists at every node and is drawn once.

Convertible arbitrage. The classic hedge fund trade buys the convertible and shorts the delta, leaving a position that is:

Long gamma and vega, because the embedded call is now unhedged in second order. This is Chapter 18's dealer position with the sign reversed.

Short credit, because the bond floor is a risky bond.

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And long carry, from the coupon less the cost of the short.

The trade's characteristic failure is that its two exposures go wrong together. A 100 bp widening in the issuer's spread, with the stock unchanged, costs 36.66 on the bond floor (3.58% of the convertible's value). And credit widens precisely when equity volatility spikes, so the position's volatility gain arrives at the same moment as its credit loss.

That would net out if both were realized on the same timetable. They are not. The credit loss is a mark-to-market that arrives immediately; the volatility gain is earned only by gamma-hedging through the move, which requires the fund to still be there. In 2008 many were not, and convertible arbitrage, a strategy that is genuinely hedged in the sense that its Greeks offset, produced one of the worst drawdowns of any hedge fund style. Being hedged in the model and being able to hold the position are different things, which is Section 5.15's limits-of-arbitrage argument in yet another market.

25.15 FUNDED AND UNFUNDED FORMATS

A CDS is **unfunded**: no principal changes hands at inception, and the protection seller's obligation is backed only by their creditworthiness and by collateral under the CSA of Chapter 8. A **credit-linked note** converts the same exposure into a funded one. The investor buys a note, paying cash up front; the issuer invests the proceeds and pays a coupon; if a credit event occurs on the reference entity, principal is reduced accordingly. Economically the investor has sold protection and prepaid the collateral in full. The difference is entirely about counterparty risk, and it runs both ways.

The protection buyer is fully secured. They hold the investor's cash, so there is no exposure to the seller at all. That is why credit-linked notes are the format of choice when the seller is an entity no one wants unsecured exposure to.

The investor acquires exposure to the issuer. Their principal now sits with the note's issuer and is invested in something. A credit-linked note is therefore two credit exposures, not one, and the second is easy to overlook because it is not what the note is named after. Several structures in 2008 delivered losses through the collateral rather than the reference entity.

The same funded-unfunded distinction organizes Chapter 26. A cash securitization is funded: investors buy notes and the proceeds buy the assets. A synthetic one is unfunded: investors sell protection on a reference portfolio that stays where it is. The synthetic version can be created without anyone originating a loan, which is why the outstanding amount of synthetic exposure was not limited by the amount of debt.

25.16 HYBRID CREDIT STRUCTURES

The previous section observed that a credit-linked note is two credit exposures rather than one. The same construction generalizes, and it has become the part of the credit market where the modeling is hardest.

A **hybrid** structure makes a payment contingent on a credit event *and* on the level of something else: an interest rate, an exchange rate, an equity index. A note might pay an equity-linked coupon for as long as a reference issuer survives, redeem at a rate-linked amount unless a portfolio takes losses, or pay in a currency the reference entity does not borrow in. Chapter 35 treats the note wrapper and the distribution question; what belongs here is what the credit leg does to the valuation.

The temptation is to price the two legs separately and multiply, and it fails. Write τ for the default time and X for the other factor. The payoff is $\mathbf{1}\{\tau > T\}\, f(X)$, and its value involves

$$\mathbb{E}[\mathbf{1}\{\tau > T\}\, f(X)] \;\neq\; \mathbb{P}(\tau > T)\, \mathbb{E}[f(X)]$$

unless default and the factor are independent. The two legs are individually observable (survival

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probabilities come from the CDS curve of Section 25.3, the distribution of X from its own option market), and the quantity that joins them is not observable from either. Example 25.10 puts a number on what that costs.

Example 25.10 The correlation a hybrid needs and no market quotes

Take a five-year note paying a single 8% coupon at maturity, and paying it only if the reference issuer has survived *and* an equity index finishes above its initial level. Risk-free rate 4%, dividend yield 2%, index volatility 30%, and a CDS curve implying a five-year survival probability of 0.90.

The index condition is met with probability $N(d_2)$, where

$$d_2 = \frac{(r - q - \sigma^2/2)T}{\sigma\sqrt{T}} = \frac{(0.04 - 0.02 - 0.045) \times 5}{0.30\sqrt{5}} = -0.1863$$

so $N(d_2) = 0.4261$.

Couple the two with a Gaussian copula, Section 24.13's device with two variables rather than a pool: default if the credit factor falls below the 10% point of the standard normal, -1.2816, and the index above strike if its factor exceeds $-d_2 = 0.1863$. The coupon is paid when both conditions hold, which is a bivariate normal probability.

ρ	P(coupon paid)	Expected coupon	Effect on the coupon
−0.4	0.3551	2.8408%	−22.7 bp
0.0	**0.3835**	**3.0678%**	**reference**
0.4	0.4091	3.2727%	+20.5 bp

At $\rho = 0$ the probability is exactly 0.90×0.4261, which is the control on the calculation. At a correlation of 0.4 the expected coupon moves by about **20 basis points**—on a trade whose two legs are each quoted to a fraction of a basis point.

The realistic sign is positive. Equity levels and credit quality move together, so the coupon tends to pay in the states where the issuer also survives, and the note is worth *more* than the independent calculation says. A desk that prices the credit leg off the CDS curve and the equity leg off the volatility surface and multiplies the two is therefore not making a small error in an unknown direction; it is understating the note. Three things follow.

The correlation cannot be implied. Vanilla equity options imply the index's volatility and CDS implies the issuer's hazard rate, and no liquidly traded instrument implies the dependence between them. It is estimated from history, which is Section 26.6's position on default correlation, arrived at in a market that was supposed to have learned that lesson.

The bounds are wide but not useless. Any joint distribution with these marginals pays the coupon with probability between $\max(0.90 + 0.4261 - 1, 0) = 0.3261$ and $\min(0.90, 0.4261) = 0.4261$, so the note's value is confined to a band of $(0.4261 - 0.3261) \times 8\%$, or 80 basis points of coupon, whatever the dependence. That width is $1 - 0.90$ times the coupon and nothing else: for any index probability between 0.10 and 0.90 the two bounds differ by exactly the default probability, so the band is set by the credit leg alone. A reserve sized to the width of that band is defensible without agreeing on a number, which is the same response Section 34.7 offers to model risk in exotics.

And the hedge is cross-gamma. The sensitivity that matters is not delta to the index or spread sensitivity to the issuer but the cross term between them, and there is no instrument whose payoff is that cross term. It is hedged by rebalancing two positions, which means the hedging error is a function of realized rather than implied dependence.

Who trades them, and why. The buyer is an investor who wants a yield the single-factor version cannot produce: attaching a second condition to a coupon raises it, as the range accrual of Section 33.6 does with a rate, because the investor has sold something. The seller is one of the few dealers that can warehouse and partially hedge the cross-risk. Insurers and pension funds buy where the structure lets them reach a yield target within a ratings constraint, the same demand that Chapter 26's senior tranches met.

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What has changed. Section 9.9's wrong-way risk is the same mathematics as this section's correlation: an exposure that grows as the counterparty's credit deteriorates. What has changed is that the difficulty has migrated from the valuation adjustment into the products themselves. A hybrid note needs a model in which credit, rates, and an equity index move together, calibrated to three separately quoted markets, and the number it produces is the price the client pays.

25.17 SOVEREIGN CDS AND THE CURRENCY OF THE PAYOUT

Section 24.12 established that sovereign credit is different in kind. The CDS market adds a feature that has no corporate analog and that connects directly to Chapter 16.

Sovereign CDS is usually denominated in a currency not the sovereign's own. Protection on an emerging-market sovereign trades in dollars; protection on a euro-area sovereign has traded in dollars and euros. The reason is obvious: protection paying in a currency the defaulting sovereign controls is worth less than it appears. The gap is priced.

The two contracts differ by what Chapter 16 would call a quanto effect. A euro-denominated contract on a euro-area sovereign pays euros in a state of the world in which the euro may have depreciated sharply, precisely because that sovereign defaulted. The dollar contract pays the same notional in a currency that has not moved, so it is worth more, and the **quanto spread** between the two is the market's price for devaluation conditional on default.

It is a jump-at-default expectation. If the dollar contract trades meaningfully above the euro one, the market is pricing a substantial currency move conditional on the credit event, not a general currency view, since an unconditional move affects both contracts alike.

During the euro-area crisis it was a market-implied redenomination probability. The quanto spread on peripheral sovereigns widened when the possibility of leaving the currency union was being discussed and narrowed when it receded. A market with no instrument for pricing the breakup of a currency union priced it anyway, in the basis between two CDS contracts.

The pricing requires exactly the machinery of Section 16.7: a correlation between the credit event and the exchange rate, entering as a quanto adjustment. It is this book's plainest case of a credit instrument and a currency instrument that must be priced jointly (the credit-and-currency hybrids of Section 25.16 are the bespoke version).

The seller matters as much as the currency. Section 9.9's textbook case of wrong-way risk is protection on a sovereign bought from a bank in that country: the contract pays in exactly the states in which a bank holding its own sovereign's bonds, and funded in its currency, is least able to pay. In the euro-area crisis of 2010–2012 the case was live rather than hypothetical: Italian and Spanish banks held large amounts of their own governments' bonds, and their CDS spreads rose and fell with their sovereigns', so protection on Italy sold by an Italian bank was weakest exactly when it would have paid. Protection on a sovereign is therefore worth most when it is bought from a seller outside the jurisdiction and fully collateralized, and a desk that buys it from a domestic bank has to price that correlation into its CVA. The main buyers are dealers' CVA desks hedging uncollateralized derivatives exposure to sovereigns, a demand that keeps protection on even the safest governments trading.

25.18 THE MARKET AFTER THE REFORMS

The instrument this chapter has described trades in a market a small fraction of the size of the one that made it famous.

Case Study 25.2 The market that shrank

Gross notional outstanding peaked, on BIS figures, at about $58 trillion at the end of 2007 and fell by roughly an order of magnitude over the following decade; at the end of

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2025 it stood at $11 trillion, 70% of it cleared. Three forces did it, and they are worth separating because only one of them reduced risk.

Compression removed offsetting trades. A dealer with a bought and a sold contract on the same issuer, maturity, and coupon holds two contracts and no risk. The Big Bang's fixed coupons and standard dates made those contracts fungible, and multilateral compression services then tore them up in bulk. This eliminated enormous gross notional and no net exposure at all.

Clearing concentrated and netted the rest. Index CDS moved to central clearing under the mandates of Chapter 2, which nets multilaterally rather than bilaterally and shrinks the reported figure further.

And single-name activity genuinely declined. This is the force that reduced risk, because fewer net positions are now taken, and it reduced the market's usefulness with it. Capital charges on dealer inventory rose, the instrument acquired a reputational cost after the crisis, and several large participants withdrew. Liquid single-name CDS now exists for a few hundred reference entities, which Section 9.10 already flagged as the reason most CVA is hedged by proxy.

The loss is real: a bank hedging a loan to a mid-sized borrower usually cannot buy protection on it, so the risk-transfer function that justified the instrument works only for the largest issuers. The market that remains serves index macro hedging and the CVA desks of Chapter 9 better than it serves the original purpose.

So is the gain: the market that remains is standardized, cleared, compressed, and settled by auction, and none of those was standard in 2007. The instrument that failed in the crisis and the instrument that exists now differ in every respect that mattered then.

The reforms traded breadth for robustness. That is probably the right trade, but it is a trade, not a free improvement.

25.19 WHO TRADES CREDIT DERIVATIVES, AND WHY

Banks buy protection to hedge loan books and to reduce their capital, and this was the instrument's original purpose. Under the substitution rule of Section 27.11 the hedged loan takes the protection seller's risk weight, so the capital it saves can justify a hedge on its own (Example 27.6).

XVA desks buy protection on their counterparties, as Section 9.10 described. This is the largest structural demand for single-name protection, and it is why the market's liquidity is concentrated in the few hundred issuers that are also large derivatives counterparties.

Asset managers use index CDS to adjust credit exposure faster and more cheaply than by trading bonds. Selling index protection is a synthetic long credit position that requires no cash and no bond inventory.

Macro funds buy index protection as a hedge against a general deterioration, and this demand is what moves the index skew of Example 25.4.

Basis traders arbitrage the CDS–bond relationship of Section 24.15, subject to the funding constraint that section identified.

Correlation desks and hedge funds trade the equity and mezzanine tranches against the index or single-name CDS, as in the 2005 trade of Section 26.7; insurers, pension funds, and banks held the senior layers for a yield above their rating, as Chapter 26 records. **Basket default swaps** let an investor earn more than any one issuer's spread for bearing the first default among a handful of issuers; the dealer buying the protection hedges with the constituents and keeps the correlation exposure. **Hedge funds** receive total return to hold financed positions they would otherwise fund and disclose (Section 7.12), and buy most new convertibles, alongside

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long-only convertible funds, for the arbitrage of Section 25.14; issuers are growth companies that would pay far more for straight debt.

The market's structural position is that dealers are net sellers of protection to banks and buyers from asset managers, with the imbalance hedged in the index. A general widening therefore produces a demand for the same hedge from everyone at once. That is Chapter 22's crowding argument transplanted to credit, and Chapter 26 records the last time it operated.

SUMMARY

A CDS pays on a credit event, not on default. When the two differ, the contract's definitions govern, and a protection buyer who is right about the credit and wrong about the definition receives nothing. Since 2009 a determinations committee of market participants, not a court, rules on whether an event occurred.

The 2009 standardization fixed the coupon at a few standard values (100 or 500 basis points for North American corporates) and moved the difference between the coupon and the issuer's spread into an upfront payment. The risky annuity falls as the spread widens, because a deteriorating credit's premium stream is expected to be cut short, so a hedge sized on duration is under-hedged exactly when it is working.

Physical settlement failed because protection outstanding can exceed deliverable debt, and the auction replaced it with a single market-wide recovery rate, a forecast made under duress, not the realized recovery. Slicing an index into tranches redistributes a fixed expected loss: as correlation rises, loss moves out of the equity tranche and into the senior one, so a senior tranche valued at a low correlation when the true one is high carries many times the expected loss it was priced for. Base correlation removes the inconsistency between the tranches' implied correlations without removing its cause. An index trades at the annuity-weighted, not the arithmetic, average of its constituents' spreads, and correlation lowers the price of a first-to-default basket while raising that of the last.

Restructuring is the contested credit event because it leaves the issuer alive with obligations at very different prices. Greece in 2012, the manufactured events of the 2010s, Russia in 2022, and Credit Suisse in March 2023, whose AT1 bonds were written to zero with no credit event, all show that where a payoff is defined by a legal event, the document is the instrument. The market shrank by an order of magnitude after the crisis, trading breadth for robustness.

A credit-linked note is a funded CDS with a second exposure to the note's issuer; a total return swap gives the economics of an asset without ownership; sovereign CDS pays in a currency the sovereign does not control, and the quanto spread is the price of devaluation conditional on default. A convertible is a corporate bond plus a call on the issuer's equity, but default, the conversion right, and the issuer's call interact and must be priced together on one tree. Convertible arbitrage is long gamma and vega, short credit, and long carry, with the credit loss marked immediately and the volatility gain earned only by hedging through the move. A hybrid coupon that needs both survival and an equity level cannot be priced as a product of marginals; the unquoted dependence moves it within a band set by the credit leg alone.

FURTHER READING

The ISDA credit derivatives definitions are the primary source, and Iwashita's OpenGamma note on single-name CDS conventions collects the coupon menus, day counts, and roll dates that the definitions leave to market practice; anyone who intends to rely on a CDS should read the credit event definitions rather than a summary of them. The determinations committee publishes its questions and rulings, and reading a contested one (Greece 2012 or Russia 2022) conveys the character of the problem. On the auction mechanism, the academic work on its design and on whether the auction price is biased relative to eventual recovery is small, and the

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finding of Chernov, Gorbenko, and Makarov (2013) and Du and Zhu (2017) that biases exist and are related to the net open interest is worth knowing before relying on auction recoveries as data. On the Credit Suisse AT1 write-down, the Swiss regulator's own explanation and the subsequent statements from other supervisors are the documents to read, and the Federal Administrative Court's October 2025 judgment, now on appeal to the Federal Supreme Court, is the authoritative legal account so far. For the tranche and basket valuation of Sections 25.7 and 25.8, Hull and White (2004) and Andersen, Sidenius, and Basu (2003) are the sources of the four steps, and O'Kane and Turnbull (2003) is the practitioner treatment of CDS valuation. On convertibles, Tsiveriotis and Fernandes (1998) is the model behind most vendor systems, and the practitioner literature on convertible arbitrage sets out what happened in 2008.

PRACTICE QUESTIONS

25.1 Explain why a protection buyer need not own the reference obligation, giving one benefit and one problem this creates, and explain why gross CDS notional outstanding is not a measure of the risk transferred.

25.2 A five-year CDS has a par spread of 250 bp and trades on a 100 bp coupon with a risky annuity of 4.19. Compute the upfront payment, and explain why the risky annuity falls as the spread widens.

25.3 A desk bought five-year protection on the standard 100 bp coupon when the par spread was 150 bp, paying an upfront of 2.18% of notional. One year later the spread is 400 bp, and the risky annuity on the remaining four years at that level is 3.31. Compute the position's value and its gain since inception as percentages of notional, explain why marking against 150 bp or using the annuity at the original spread misstates them, and state in which direction the annuity error biases a hedge.

25.4 Explain why physical settlement failed, what the auction replaced it with, and why an auction recovery is not the same as the eventual realized recovery.

25.5 Explain why an index CDS trades below the arithmetic average of its constituents' spreads, what the index skew is, and why it widens when the index is used as a macro hedge.

25.6 Explain why restructuring creates a cheapest-to-deliver problem that bankruptcy does not, and why a risk system treating restructuring and no-restructuring contracts as substitutes has mismeasured the exposure.

25.7 State what the Credit Suisse AT1 write-down established about the meaning of subordination, and explain why CDS on Credit Suisse did not pay.

25.8 Explain why an equity tranche's expected loss falls as correlation rises while a senior tranche's rises, given that the portfolio's expected loss is unchanged, and use Example 25.5 to state the factor by which a 15–30% tranche's expected loss grows between $\rho = 0.10$ and $\rho = 0.50$.

25.9 Explain why implied correlations that differ across tranches on the same portfolio cannot all describe that portfolio, and what base correlation does about it.

25.10 A convertible has a conversion ratio of 25 and the stock trades at 30. Compute the conversion value and the conversion price on a face of 1000, and explain why the instrument behaves like a bond at low share prices and like equity at high ones.

25.11 State the exposures of a delta-hedged convertible arbitrage position, and explain why two of them go wrong together and why the timing of their recognition differs.

25.12 Explain why a total return swap position does not appear in ownership disclosures, and

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who bears the risk if the receiver cannot meet a margin call.

25.13 Distinguish the three causes of the CDS market's post-crisis shrinkage, and state which of them reduced risk rather than merely reducing gross notional.

25.14 Explain why a credit-linked note contains two credit exposures rather than one, and explain what the quanto spread between a dollar- and a euro-denominated CDS on a euro-area sovereign measures.

25.15 State the four residual risks a protection buyer faces even when right about the credit, and explain which of them is least measured and why it is the one most likely to produce the largest loss.

25.16 Five issuers each trade at 150 bp. Explain why the first-to-default basket costs the sum of the five spreads at $\rho = 0$ and falls as ρ rises while the fifth-to-default rises, and why the five baskets' protection legs add up to the five single-name legs (Example 25.6).

FURTHER QUESTIONS

25.17 Example 25.2 shows the risky annuity falling from 4.55 to 3.40 as the spread widens. Derive this dependence, show that it makes the upfront a concave function of the spread, and explain what this implies for a desk that hedges a portfolio of wide-spread issuers with a position in a tight-spread one.

25.18 Section 25.5 argues that the auction solved the squeeze and created a different problem. Set out the incentives of a participant with a large protection position, explain what the two-stage design does to constrain them, and propose a modification that would further reduce the incentive to influence the outcome, stating what it would cost.

25.19 Example 25.4 attributes the index-average gap to differences in risky annuity. Derive the exact relation between the index spread and the constituent spreads, show that the gap grows with dispersion, and determine the dispersion at which it becomes large enough to justify trading 125 issuers against the index.

25.20 Example 25.7 finds the restructuring clause worth 7% of the spread. Explain why this number is nearly useless as a risk measure, construct a measure that would capture the exposure correctly, and discuss why a bank hedging a loan book might rationally pay far more than 7% for the clause.

25.21 The Credit Suisse case study argues that where a payoff is defined by a legal event, the document is the instrument. Identify three other instruments in this book whose payoff depends on a legal or administrative determination rather than a market price, and for each state what the residual risk is and who is best placed to bear it.

25.22 Re-price Example 25.9's convertible with a 3% dividend yield on the stock and show that early conversion now has value; then add a holder's put at par at year 3 and state which way each feature moves the value and why.

25.23 Section 25.14 argues that convertible arbitrage was hedged in the model and failed in practice. Distinguish the model risk, the funding risk, and the crowding risk in that failure; explain which of the three the Greeks would have revealed; and relate your answer to Section 5.15 and to Chapter 22's account of 2018.

25.24 Using Example 25.10, show that the 80 bp band on the hybrid coupon does not depend on the index probability for any value between 0.10 and 0.90, and explain what that implies for how a desk should reserve against an unquoted correlation.

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Chapter 26
Securitization, Structured Credit, and the 2007–2009 Crisis

Section 24.13 established that correlation redistributes a portfolio's loss without changing it, and Section 25.7 put numbers on the redistribution: a senior tranche valued at a correlation of 0.10 when the true correlation is 0.50 has forty-three times the expected loss it was priced for. This chapter is about what happened when an industry was built on a number no one could observe.

Every mechanism in this book that produced a loss elsewhere appears here at once: the model risk of Chapter 21, the crowding of Chapter 22, the funding fragility of Chapter 8, the rating-trigger effects of Section 24.10, and the gap between a contract and the risk it was meant to transfer from Section 25.13. The 2007–2009 crisis remains the best available case study because it is the only episode in which all of them fired together.

The chapter runs from the mechanism to the history and back to the present. Sections 26.1 and 26.2 are the structure: what pooling and ranking do, and why a tranche is a call spread on the pool's loss, which makes it convex in every input. Section 26.3 is the counter-example: the agency mortgage market, larger than everything that failed, where the option being sold was known and priced. Sections 26.4 to 26.6 are how the deals were built and modeled: the three sources of credit support and why they shrink when they are needed, re-securitization, and the Gaussian copula and what it was asked to do. Sections 26.7 to 26.12 are the history, ending with the order in which things broke. Sections 26.13 to 26.16 ask what was repaired, which structure survived, and where the same pattern operates now, and Section 26.17 records where the models came from.

26.1 WHAT SECURITIZATION DOES

Pool a set of loans, sell claims on the pool's cash flows, and rank those claims. That is the whole idea, and in its plain form it is useful.

It converts illiquid assets into tradeable ones. A mortgage cannot be sold easily; a security backed by ten thousand of them can. It matches risk to appetite. A pension fund that cannot hold individual mortgages can hold a senior claim on a pool of them. Someone else takes the first-loss piece. And it transfers risk out of banks, which is what the capital frameworks of Chapter 27 are designed to encourage when the transfer is genuine.

None of this is objectionable, and the market for agency mortgage-backed securities (Section 26.3) functioned through the crisis and functions now. What failed was not securitization; it was a specific set of things built on top of it, and keeping the two separate is necessary to draw any useful lesson.

26.2 THE WATERFALL

Cash from the pool is distributed by a **waterfall**: senior claims are paid first, then mezzanine, then equity. Losses arrive in the reverse order, so the equity tranche absorbs the first losses and

the senior tranche suffers only once everything below it is exhausted.

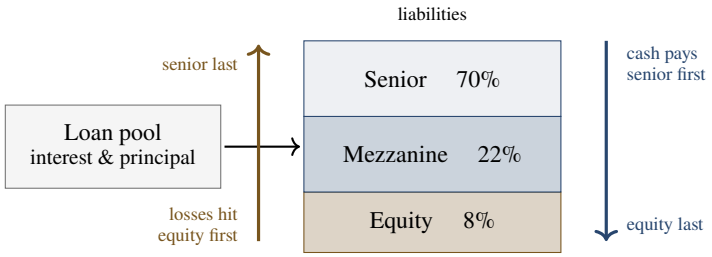


Figure 26.1: Cash flows down by seniority; losses climb from the bottom. Not to scale.

Figure 26.1 is the entire structure, and the two arrows pointing in opposite directions are the reason a single pool can support securities with genuinely different risk. *Nothing has been created*: the pool's total cash and total loss are unchanged, and every currency unit the senior tranche is protected from is one the equity tranche absorbs.

A tranche is therefore an option on the pool's loss. With L the pool's loss as a fraction of its notional, the $[a, b]$ tranche absorbs $T_{[a,b]}(L)$ of equation (25.8) as a fraction of its own notional: a call spread on L struck at a and b, normalized, which is Section 12.2's structure applied to a loss rather than a price. The attachment point a is the strike, the detachment point b the cap, and every property of a tranche in this chapter follows from that.

The family is named after its collateral, and the names are worth fixing because the rest of the chapter uses them. A tranched pool of mortgages is a mortgage-backed security; a tranched pool of bonds or loans is a collateralized debt obligation, the CDO of Section 24.10; when the collateral is leveraged corporate loans it is a **collateralized loan obligation** (CLO), the member of the family that survived, which Section 26.14 treats on its own terms. A CDO whose collateral is itself tranches of other deals is a re-securitization: a CDO of asset-backed securities when the tranches come from mortgage-backed securities, a **CDO-squared** when they come from other CDOs, and Section 26.5 is about what both do. Nothing in the mathematics distinguishes them: $T_{[a,b]}$ is the same function in every case, which is precisely why the differences that matter are in the collateral and in who manages it.

Two immediate consequences follow, and the rest of the chapter rests on them. Adding up the tranches returns the pool: for any points $0 = a_0 < a_1 < \cdots < a_m = 1$, the tranches $[a_{j-1}, a_j]$ satisfy

$$\sum_{j=1}^{m} (a_j - a_{j-1})\, T_{[a_{j-1},a_j]}(L) = L,$$

so securitization redistributes loss and creates none. And a tranche's expected loss is nonlinear in the pool's, because $T_{[a,b]}$ is kinked at both ends, which is where Example 26.1's leverage comes from.

Because a tranche is a call spread, everything Chapter 12 said about spreads applies. A senior tranche is a deeply out-of-the-money option: worth almost nothing in expectation, enormously sensitive to the assumed distribution, and, critically, *convex in the parameters*.

Figure 26.2 shows both halves of the argument: the call-spread payoff on the left, and on the right what that payoff does to an error in the input.

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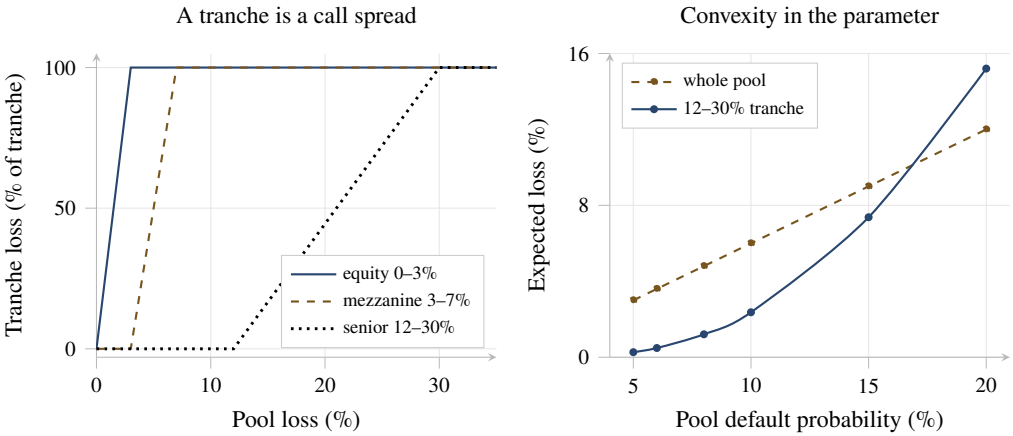


Figure 26.2: Left: tranche loss against pool loss. Right: expected loss against default probability.

Example 26.1 What a tranche does to a parameter error

Take a pool of 100 loans, each with a 5% default probability and 60% loss given default, correlation 0.15. The pool's expected loss is 3.00%, the 12–30% senior tranche's 0.266%.

Now suppose the true default probability was higher than assumed:

True default probability	Pool expected loss	Senior tranche expected loss
5%	3.00%	0.266%
6%	3.60%	0.486%
8%	4.80%	1.208%
10%	6.00%	2.369%
15%	9.00%	7.374%
20%	12.00%	15.205%

The numbers come from the copula exactly as Example 24.8 applied it. The tranche is untouched until 21 of the 100 loans default, so the pool's loss moves with the default probability and the tranche's with the probability of reaching that count.

Tripling the default probability triples the pool's loss and multiplies the senior tranche's by twenty-eight. This is the chapter's central mechanism and it requires no story about fraud or stupidity. A tranche is an option, options are convex, and convexity amplifies parameter error.

The same convexity works in the parameter Section 25.7 identified. Both the default probability and the correlation enter a senior tranche's value with high leverage, and neither was observable. Two unobservable inputs, each amplified by a factor of tens, are no basis for a AAA rating.

26.3 THE SECURITIZATION THAT WORKS

The counter-example comes first, because it is larger than the failures and it did not fail. Agency mortgage-backed securities are issued or guaranteed by Ginnie Mae, Fannie Mae or Freddie Mac, and most are **pass-throughs**: the pool's monthly interest and principal, prepayments included, are passed straight through to investors in proportion to their holdings. They carry no meaningful credit risk: the guarantor absorbs default, and the investor receives principal and interest regardless. What the investor bears instead is prepayment risk, and it is a pure option problem with no correlation parameter in it.

A borrower may repay at any time without penalty. So the investor in a mortgage pool is short a call on the loan, exercised when refinancing is attractive, which is when rates have fallen.

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Example 26.2 Negative convexity, measured

Take a 30-year 5.5% pass-through on a pool of level-payment mortgages (a constant monthly installment over the loan's life), paying monthly, at a 5% yield compounded monthly. The speed is the market's conditional prepayment rate (CPR), the fraction of the outstanding balance repaid early in a year beyond scheduled amortization, applied monthly as $1-(1-\text{CPR})^{1/12}$ of the balance left after scheduled principal; it responds to the refinancing incentive: slow while market rates sit above the coupon, steep once they fall below it, leveling off once those who can refinance have. Prices are per unit of face at $5\% + \Delta y$; "fixed speed" keeps the 12% base rate; the comparison is a seven-year 5.5% semiannual bullet (a bond repaying all its principal at maturity) at a semiannually compounded yield; seven years is about the pool's average life at the base speed (6.6 years), so it isolates the option rather than maturity.

Yield change	CPR	Pass-through price	Price at fixed 12% CPR	7y bullet
−2.0%	35%	1.0534	1.1399	1.1568
−1.0%	22%	1.0496	1.0797	1.0908
0	12%	1.0253	1.0253	1.0292
+1.0%	8%	0.9698	0.9759	0.9718
+2.0%	6%	0.9035	0.9307	0.9181

	Effective duration	Effective convexity
Pass-through, prepayments respond	3.89	**−304**
Same bond, fixed prepayment speed	5.07	+48.5
Seven-year bullet	5.78	+39.9

Effective duration and convexity are read off the unrounded ±1% prices: for the responding pass-through, $(1.0496-0.9698)/(2\times1.0253\times0.01) = 3.89$ and $(1.0496+0.9698-2\times1.0253)/(1.0253\times0.01^2) = -304$.

The convexity is negative, and strongly so. A 200 basis point rally lifts the bullet by 12.4% and the pass-through by 2.7%; a 200 basis point sell-off costs the bullet 10.8% and the pass-through 11.9%. The investor gets a fraction of the upside and all of the downside, because the borrowers refinance when rates fall and stay put when they rise.

The mechanism is exactly Chapter 18's. The investor is short an option, so they are short gamma, and Section 18.12's account of what short gamma does to a hedger applies directly: as rates fall the pool's duration shortens, so a mortgage portfolio hedged to a duration target requires buying duration as rates fall and selling it as they rise. That is mechanical buying into a rally and selling into a sell-off, pushing rates further in the direction they have already moved, by an investor base (banks, money managers, insurers) measured in trillions, and it is one of the largest identifiable feedback flows in any fixed-income market. Two lessons follow for this chapter.

Risk that is genuinely understood can be borne at enormous scale. The agency mortgage-backed securities market is bigger than everything that failed in 2008 put together, its central risk is harder to model than credit default, and it worked. The difference is that the option being sold was known to be an option, priced as one, and hedged as one.

And the failure of Section 26.5 was not about complexity. Prepayment modeling is more technically demanding than a Gaussian copula. What distinguishes the two is not difficulty but whether the model's key input could be checked against anything, and prepayment speeds are observed monthly while default correlation was observed never.

26.4 HOW A DEAL IS ACTUALLY BUILT

A bond's **credit support** is the loss the pool absorbs before the bond takes any, and it comes from three sources. **Subordination** is Section 26.2's tranching: for a senior bond attaching at 20%, the 20% of bonds beneath it. **Overcollateralization** is the excess of assets over liabilities: $980 million of bonds against $1,000 million of loans gives 2% of cushion. **Excess spread** is

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what is left of the pool's coupon after the bonds' coupon and the servicing fees are paid; it is the first thing losses consume, before any tranche is touched, and if it is retained rather than released to the equity it accumulates, so the support grows every year the pool performs.

This is why the structures looked so robust, and the flaw is visible once stated. The support grows *provided the excess spread is actually earned*, and it is earned only while the loans keep paying. A pool that starts defaulting loses its spread at the moment it begins consuming the cushion it already has. This correlation between the buffer and the loss is structurally the wrong-way risk of Section 9.9 and the recovery–default correlation of Section 24.4: a protection that shrinks in the state where it is needed. Example 26.3 runs the two paths side by side.

Example 26.3 Where the protection comes from

A \$1,000 million pool of loans with an 8.5% weighted-average coupon backs \$980 million of bonds costing 5.5% of the pool; servicing and fees take 0.5%, and every percentage below is of the original \$1,000 million. The excess spread is $8.5 - 5.5 - 0.5 =$ **2.5%** a year on the paying balance, \$25 million in a default-free year, and the overcollateralization is 2%.

The promised path. A senior bond attaching at 20% starts with 20% + 2% = 22% of support. With the excess spread retained, each year adds 2.5%: 24.5% at the end of year 1, then 27.0%, 29.5%, 32.0% and 34.5%, the table's "promised" column.

The stressed path. Suppose 8% of the original balance defaults each year at a 40% recovery. The pool then loses $8\% \times 60\% = 4.8\%$ a year, but its performing balance falls by the full 8%: a defaulted loan pays no interest, and what is recovered on it comes back as principal, not as a loan still earning the coupon. Excess spread is earned on the average performing balance of the year, and losses are absorbed by the overcollateralization and the spread earned before they touch subordination:

	Promised	Realized under stress			
Year	Support	Spread earned	Loss	OC left	Support
1	24.5%	2.40%	4.80%	0.00%	19.60%
2	27.0%	2.20%	4.80%	0.00%	17.00%
3	29.5%	2.00%	4.80%	0.00%	14.20%
4	32.0%	1.80%	4.80%	0.00%	11.20%
5	**34.5%**	1.60%	4.80%	0.00%	**8.00%**

Years 1 and 2 are worth working in full. In year 1 the performing balance falls from 100% to 92%, averaging 96%, so the deal earns $2.5\% \times 0.96 = 2.40\%$ of spread against 4.80% of loss. The 2% of overcollateralization and the 2.40% together absorb 4.40%, so the overcollateralization is gone and the remaining 0.40% comes out of subordination: support ends the year at $22.0 + 2.40 - 4.80 = 19.60\%$. In year 2 the balance falls from 92% to 84%, averaging 88%, earning $2.5\% \times 0.88 = 2.20\%$ against another 4.80%; with the overcollateralization gone, the whole 2.60% shortfall comes out of subordination, and support falls to $19.60 - 2.60 = 17.00\%$.

The support does not rise to 34.5%; it falls to 8.00%, a swing of 26.5 points, and through two channels at once: the losses consume it, and the shrinking performing balance earns less of it each year, 1.60% in year five against 2.40% in year one. Cumulative losses of 24.0% have consumed the overcollateralization, all ten points of spread earned, and twelve of the twenty points of subordination; at more than three points a year, a little over two more years exhaust it. The senior bond is on course not to survive.

Figure 26.3 draws this table beside Example 26.2's: on the left the pass-through's price bending the wrong way, on the right the two support paths parting from the first year. Nothing in the example requires a copula, a correlation, or a view about house prices: it requires only that defaults arrive and that the spread be earned on what is left. The models that projected the accumulating column ran only scenarios in which it accumulated. The protection was a

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projection presented in a column that looked like a fact.

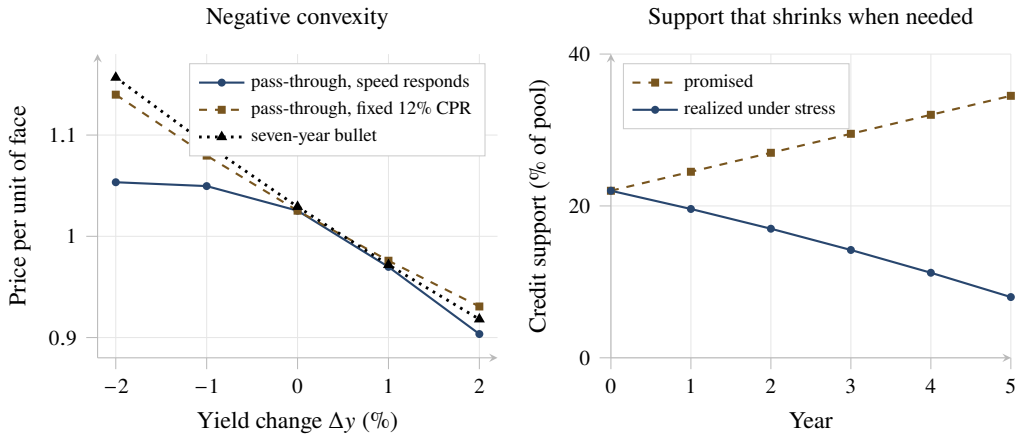


Figure 26.3: Left: the prices of Example 26.2. Right: the promised and stressed support paths of Example 26.3.

26.5 RE-SECURITIZATION

The step that turned a bad decade into a crisis was to securitize twice. Mezzanine tranches are hard to sell: too risky for senior buyers, not levered enough for equity buyers. The solution the market found was to pool them and tranche the pool.

Example 26.4 A CDO of mezzanine tranches

Take the pool of Example 26.1. At a correlation of 0.15 its tranches have expected losses:

Equity 0–3%	64.07%
Mezzanine 3–7%	20.19%
Mezzanine 7–12%	4.45%
Senior 12–30%	0.27%
Super senior 30–100%	0.00%

Now take the 3–7% mezzanine piece from one hundred *different* deals and pool those. Each has an expected loss of 20.19%. Tranche the new pool, and everything depends on the correlation *between* the mezzanine pieces. Model it as the copula applied twice, in Section 24.13's notation. Loan j of deal i has $X_{ij} = \sqrt{0.15}\,M_i + \sqrt{0.85}\,Z_{ij}$, so the loans in a deal share that deal's factor M_i at the correlation 0.15 of Example 26.1; and the hundred deal factors themselves load on one market factor M, $M_i = \sqrt{\rho}\,M + \sqrt{1-\rho}\,Z_i$, where ρ is now the inter-deal correlation. Conditional on M the hundred pieces are independent, each with the loss distribution its own tranche loss function gives it, so the new pool's loss distribution is built exactly as in Section 25.7's Step 2, with a piece's loss distribution in the place a single default occupied. The four layers of the re-securitization then lose:

Inter-deal ρ	"AAA" 30–100%	"AA" 12–30%	3–12%	Equity 0–3%
0.10	1.08%	44.13%	94.36%	99.96%
0.30	3.83%	40.89%	80.19%	97.71%
0.60	7.55%	36.64%	63.34%	86.84%
0.90	11.25%	32.18%	49.18%	69.72%

Weight the columns by width (0.70, 0.18, 0.09, 0.03) and every row adds to 20.19%, the pool's expected loss, as it must. The senior tranche of the re-securitization goes from losing 1.08% of its

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> notional to 11.25% as the inter-deal correlation rises, a factor of ten, driven entirely by a parameter about which no market quoted anything.
>
> And the parameter was not 0.10. The mezzanine tranches of one hundred subprime deals from the same vintage are exposed to one thing: national house prices. They are not a hundred independent bets on a hundred pools; they are one bet, made a hundred times, on whether house prices keep rising.
>
> So the diversification that justified the structure did not exist. Pooling a hundred claims removes idiosyncratic risk only if the risk is idiosyncratic, and the claims being pooled had already had their idiosyncratic risk stripped out by the first securitization, which is precisely what a mezzanine tranche is. *Re-securitizing tranches pools the systematic residual, and the systematic residual does not diversify.*

26.6 THE COPULA AND WHAT IT WAS ASKED TO DO

Section 24.13's one-factor Gaussian copula was the industry standard, and it deserves a more careful verdict than it usually receives.

As a device it is unobjectionable. It is a way of writing down a joint distribution with a single correlation parameter, and it makes the loss distribution computable, which is a legitimate reason to choose a model. Section 25.7 noted that the market's own use of it, quoting tranches by the implied correlation that reproduces their prices, was self-evidently a quoting convention rather than a belief: a single-parameter model producing different parameters at different attachment points is announcing its own misspecification, exactly as the volatility smile does in Chapter 19.

Combining the tranche loss (25.8) with that factor model gives every tranche number in this chapter and the last, and the calculation is Steps 2 and 3 of Section 25.7: conditional on the common factor the defaults are binomial with the probability $p(M)$ of equation (24.12), and one integral over the factor and one binomial sum price the whole capital structure, which is why the model was adopted and could be run on every deal. As the pool grows, its loss fraction becomes $\ell\, p(M)$ (with $\ell = 1 - R$ the loss given default), a function of the factor alone, and the **large-pool** loss distribution is Chapter 24's worst-case default rate (24.15) read in reverse: with each issuer's default probability π and correlation ρ, the pool loses more than $\ell\,\mathrm{WCDR}(q)$ with probability exactly $1 - q$.

That is the formula underneath the Basel capital charge and, read backward, the reason a senior tranche is a statement about ρ. As $\rho \to 0$ the distribution collapses onto $\ell\pi$ and a senior tranche can never be touched; as $\rho \to 1$ it becomes all or nothing: a loss of ℓ with probability π. Rating such a tranche is choosing a point between those two, which is why the letter cannot be read without ρ.

Figure 24.6 drew the matching distribution of defaults for Example 26.1's pool. What is visible there is the middle draining into both tails as ρ rises, the mean pinned at five defaults, a loss of $\ell\pi = 3\%$; what the senior tranche cares about is the part the figure can barely show (the mass beyond the 12% attachment point, 21 defaults, at the right edge of its axis), which the large-pool approximation puts at 1.24% at $\rho = 0.15$, against 1.50% for an exact count over Example 26.1's hundred loans. That mass is the senior tranche. Figure 26.4 draws the probability that the pool's loss exceeds x on the logarithmic scale the tail needs: left of the 12% line the loss is absorbed by the layers beneath; the height of each curve at the line is the probability the senior tranche is touched at all. The curves are nearly straight because this tail is close to exponential in x; what ρ sets is its slope.

The loss distribution is not yet a price. Section 25.7 gave the four steps from one to the other and Example 25.5 ran them on a 125-issuer index portfolio, so the calculation is not repeated; only the principle is restated. The risky annuity is the present value of the notional

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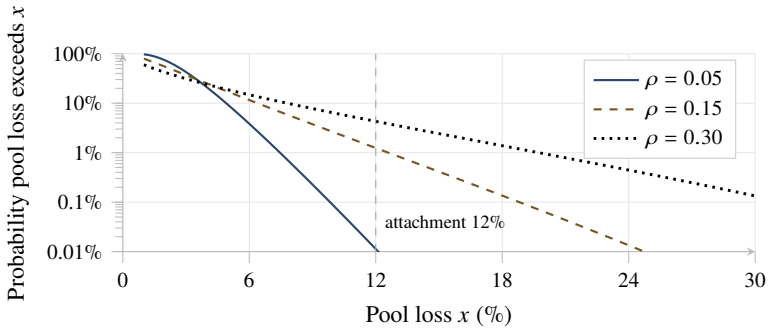


Figure 26.4: Probability that the pool's loss exceeds x, from the large-pool limit of equation (24.14), inverted as in equation (24.15), with $\pi = 5\%$, $\ell = 60\%$, evaluated every 0.25%; logarithmic scale.

that survives and the protection leg of the notional that does not, and a CDS weights by $Q(t_i)$ where a tranche weights by $1 - E_{[a,b]}(t_i)$, the expected fraction still outstanding. Change the product and only the definition of "survives" changes (an issuer's default, a layer's erosion, a pool's amortization for Section 26.4's bonds), which is why the CLO of Section 26.14, whose cash flows are governed by tests rather than by a copula, is priced by projecting the same two quantities through its waterfall, not by a new theory. Three things went wrong with the copula's use, and only one of them is a criticism of the model.

The Gaussian copula has no tail dependence. **Tail dependence** is the tendency of extreme outcomes to arrive together: the probability that one variable is extreme given that another is, as "extreme" is pushed ever further into the tail. Under the Gaussian copula, extreme events are asymptotically independent: conditional on one issuer's latent variable being extreme, the probability that another's is equally extreme falls to zero as the threshold moves out, at any correlation below one. This is empirically false for credit, where defaults cluster in exactly the states the model treats as independent, and it biases senior tranches, the ones that matter only in those states, in the direction of looking safe.

The correlation was calibrated to a period that contained no test. Whatever number was used came from data in which US house prices had not fallen nationally, so the estimate could not contain the event that mattered. This is not a modeling error at all; it is the sampling problem of Section 19.9's pre-1987 discussion, and no choice of copula would have repaired it.

And the output was treated as a measurement. A model that turns an assumption into a rating, run by an institution the issuer pays, produces a number whose precision is inherited from an input no one checked. This is the failure, and it is institutional, not mathematical.

The verdict is that the copula was a bad model used badly, and that the second clause carries most of the weight. A better model calibrated to the same data by the same institutions with the same incentives would have produced approximately the same ratings.

26.7 THE WARNING NO ONE READ: MAY 2005

Two years before the crisis, the correlation market produced a clean, self-contained failure that demonstrated the model's defect precisely. It is worth recounting because everything that mattered in 2007 is visible in it, at a scale small enough that no one was harmed much.

The standard trade was to be long the equity tranche of an index and short the mezzanine, in a ratio that made the package neutral to the index spread. Section 25.7's table explains the position: the equity tranche is long correlation and the mezzanine is much less so, so the package was a way to be long correlation while hedged against the credit market's direction. It carried positively, it had performed for years, and it was widely held.

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In May 2005 two large US automakers were downgraded to junk. The shock was idiosyncratic, and the model had no room for idiosyncratic shocks of that size. In the one-factor copula of Section 24.13, an issuer defaults because the common factor moves or because its own draw is bad, and the second channel is a residual that diversifies away in a 125-issuer index. A downgrade large enough to move two issuers' spreads violently without moving the other 123 is, in the model's terms, an event of negligible probability. The effect went the wrong way on both legs.

The equity tranche fell, because idiosyncratic widening in a few issuers is exactly what an equity tranche is exposed to. *The mezzanine did not widen to compensate*, because the hedge ratio had been computed on a model in which spreads move together: the short mezzanine leg was sized to gain from a widening that never came, and as implied correlation fell the mezzanine spread tightened instead.

The trade lost money on both legs simultaneously, which the model said could not happen. Implied correlation moved in ways no single parameter could reconcile, and the base-correlation apparatus of Section 25.7 was in part a response to the resulting inconsistencies. Three lessons were available in 2005 and were drawn in 2008 instead.

The hedge ratio was model output, not market data. A position hedged by a model is hedged against the model's world, and Section 21.11's measurement of model risk is the same observation with a barrier option in place of a tranche.

A parameter that must be inferred from the prices it explains is not a parameter. The market was calibrating ρ to tranche prices and then using ρ to hedge tranches, which is circular in a way that produces sensible numbers right up until the relationship it assumes fails.

And crowding turned a model error into a market event. The trade was widely held in the same direction, so the unwind moved the prices being unwound—the mechanism of Chapters 22 and 23 in a market where no one had yet thought to look for it.

The episode was small, contained, and correctly diagnosed at the time. That diagnosis was never carried over to the much larger business of using the same model, on much less observable portfolios, to rate instruments AAA.

26.8 RATINGS: THE SAME LETTER, DIFFERENT RISK

Section 24.10 said the rating on a structured tranche depended on a correlation assumption the letter did not disclose. Here is what that meant in practice. A AAA corporate bond and a AAA senior tranche were notionally calibrated to the same idealized default probability; their loss distributions in bad states were entirely different.

A corporate bond's rating moves a notch at a time. If the issuer deteriorates it becomes AA, then A, and a holder can sell into a market that reprices gradually.

A tranche's rating is closer to binary. Example 26.1 took a senior tranche from 0.27% to 15.2% expected loss on a default-probability move that raised the pool's own expected loss only fourfold, from 3% to 12%. There is no gradual path; the tranche is fine until the attachment point is threatened and then it is not.

Two structural features made this worse.

Rating triggers were embedded in contracts. Downgrades forced sales by mandate-constrained holders and triggered collateral calls, which is Section 24.10's mechanism operating on an instrument whose downgrades, when they came, came in blocks of many notches at once.

And the ratings were used as a substitute for analysis. An investor holding a AAA tranche was, in practice, holding an agency's correlation assumption. The rating was not a summary of an analysis the investor could have done themselves; it was the entire analysis, which is a different thing and was not disclosed as such.

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26.9 THE INCENTIVE CHAIN

The technical story above is sufficient to produce large losses. It is not sufficient to produce a crisis, and the additional ingredient was that no one in the chain retained the risk they created.

A mortgage was originated by a broker paid a fee per loan, sold to a bank that pooled it, tranched by an arranger paid a fee per deal, rated by an agency the arranger paid, and sold to an investor who trusted the rating. At every step the pay was for volume and the risk moved on.

Originate-to-distribute is not inherently wrong. It is the same separation of origination from risk-bearing that makes any capital market work, and Section 24.16's discussion of banks distributing credit describes the same thing functioning correctly.

What made it fail was that the distribution was not real. Banks retained **super-senior** tranches (the top layer of the capital structure, ranking above the notes sold as AAA) because they were unsellable and appeared riskless, kept them on their own books at thin capital charges, and placed other tranches in off-balance-sheet vehicles funded with short-term commercial paper, to which they provided liquidity backstops. The risk came full circle, and the capital framework of the time did not capture it because the exposure was contingent and the vehicle was not consolidated. That is the mechanism Chapter 27 exists to address, and Section 26.13 records the response.

26.10 THE EUROPEAN ALTERNATIVE

Europe funded mortgages differently, and the contrast isolates the feature that did the damage. A **covered bond** is a bank's bond secured on a pool of mortgages that *stays on its balance sheet*. The holder has **dual recourse**: a claim on the bank and, if it fails, a preferential claim on the ring-fenced pool. The pool is dynamic: the issuer must replace loans that deteriorate. Banks, insurers and central banks hold covered bonds for their safety and the favorable regulatory treatment that safety earns.

No risk is transferred at all, which is exactly the point. Covered bonds have a very long history and a default record close to nil, through the crisis included. The reasons map one-to-one onto the failures of this chapter.

The originator keeps the credit risk, so Section 26.9's chain has no one to pass it to. An issuer that writes bad loans must replace them out of its own assets.

There is no tranching, so Example 26.1's convexity cannot arise.

And there is no re-securitization, because there are no tranches to re-pool.

The trade-off is equally clear. Covered bonds do not free up bank capital, because the risk never leaves. They are a funding instrument, not a risk-transfer instrument, and a banking system financed entirely this way would lend less per unit of capital.

So the comparison is not that Europe was wiser. It is that two systems made different choices about whether risk transfer was the objective, and the system that did not attempt it did not suffer the failures that attempting it produced. Securitization's benefits and its failure modes come from the same feature, and a reader deciding what to think about the instrument should hold both halves of that sentence at once.

26.11 SYNTHETIC CDOS AND THE SUPPLY PROBLEM

Section 25.15 established the distinction. A cash CDO buys the assets; a synthetic one sells protection on a reference portfolio via the CDS of Chapter 25.

The synthetic format removed the constraint that there had to be a loan. Demand for AAA-rated yield exceeded the supply of mortgages, and a synthetic structure could reference the same pool as many times as there were counterparties willing to take the other side. This has two consequences, and they should be separated because they are often conflated.

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It multiplied the exposure without multiplying the risk of default. A hundred synthetic deals referencing the same mortgages do not make them more likely to default; they create a hundred times the notional of contracts whose value depends on whether they do.

And it therefore multiplied the interconnection. The loss on the mortgages was bounded; the transfer of that loss through the system was not. A bounded shock propagated through an unbounded web of contracts is a different problem from a large shock, and it is why the crisis's transmission was through counterparty exposure rather than through mortgage losses alone.

A fair counter-argument runs as follows. Synthetic structures also let the risk be held by whoever wanted it, including investors who correctly thought subprime overvalued and used the format to be short. The instrument was symmetric; the understanding was not.

26.12 WHAT ACTUALLY FAILED

Case Study 26.1 The order in which things broke

The sequence matters, because the popular account compresses it and loses the mechanism.

2006–2007: the pool assumptions were wrong. Subprime delinquencies rose above anything in the calibration sample, for reasons (loosening underwriting, adjustable-rate resets, house prices flattening) visible in the data before they reached the ratings.

Mid-2007: tranches repriced nonlinearly. Example 26.1's convexity worked: bad pool losses became catastrophic tranche losses, mezzanine and re-securitized tranches first. In stylized round numbers, not the record of any one pool: if falling house prices double cumulative defaults from 15% to 30% of a pool's balance, and a foreclosed loan loses half its balance, the pool's loss doubles from 7.5% to 15%. A mezzanine tranche attaching at 8% and detaching at 20% loses nothing at 7.5% and $(15-8)/12 = 58\%$ of its principal at 15%.

Late 2007: the funding withdrew. The off-balance-sheet vehicles of Section 26.9 funded long-dated tranches with short-dated paper. When the paper could not be rolled, the sponsoring banks took the assets back onto their balance sheets: Chapter 8's funding-fragility problem, arriving by a route the capital framework had not anticipated.

2008: the counterparty channel opened. Institutions that had sold protection on senior tranches faced collateral calls as the marks fell. The largest case, AIG (Section 40.9), was an insurer that had written protection without capital against the mark-to-market exposure, reasoning that the tranches would never default; they did not need to, because the collateral calls on the marks and on its downgrade sufficed.

And then valuation itself failed. With no trades, the marks came from models, and Chapter 21's model-risk warning arrived at the worst moment: institutions marking the same instrument differently, disputing collateral calls, unable to say who was solvent. Chapter 9's XVA machinery and Chapter 8's collateral standards exist because of it.

Three observations follow for a reader who was not there.

The mortgage losses alone were survivable. Realized US subprime losses were large but below the banking system's capital. The crisis was in the transmission, not the shock.

The instruments performed as written. Senior tranches of the mortgage securitizations largely did pay, eventually; those of the re-securitizations mostly did not. Holders who could fund them to maturity mostly recovered. The losses were concentrated in those who had to sell, Section 5.15's limits of arbitrage at systemic scale.

And the risk had not been transferred. That is the finding that should be carried forward from this chapter, because it is the one that generalizes: a structure that appears to move risk out of a system, while the system retains the funding, the liquidity backstop,

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and the counterparty exposure, has not moved the risk.

26.13 WHAT WAS FIXED

More was fixed than is usually acknowledged, and a fair account matters for Section 26.16.

Risk retention. Sponsors must retain an economic interest (typically 5%) in what they securitize, though US open-market CLO managers have been exempt since a 2018 court ruling. This addresses Section 26.9 directly, and its weakness is that the expected loss on a 5% slice can be smaller than the fee income from arranging the deal.

Consolidation and capital. Off-balance-sheet vehicles are consolidated; liquidity backstops carry capital; re-securitizations carry punitive risk weights that have made CDO-squared structures uneconomic rather than merely unfashionable.

Rating agency reform. Methodologies must be published, ratings on structured products are labeled distinctly, and many regulatory rules no longer rely on ratings. That is the most important change, since the mechanical use of ratings is what made downgrades destructive.

And the infrastructure of Chapters 2 and 8. Central clearing, initial margin on uncleared trades, and standardized collateral terms address the counterparty channel of the crisis.

The structure that failed cannot easily recur. A CDO of subprime mezzanine tranches, rated by an issuer-paid agency, held in an unconsolidated vehicle funded with commercial paper, is blocked at each link: the re-securitization by risk weights, the vehicle by consolidation, the paper funding by the capital its backstop carries, and the rating by rules that no longer use it. What has not been fixed is the general pattern, the subject of Section 26.16.

26.14 THE STRUCTURE THAT SURVIVED

A CLO is worth setting out on its own terms rather than only by contrast, because it is the member of the family a reader is most likely to meet. The loans come first. A manager accumulates them in a **warehouse**, a credit line from the arranging bank with the future equity investors bearing the price risk; at closing the deal issues perhaps $400–500 million of notes from AAA down through BB with an unrated equity piece beneath, repays the warehouse, and completes the portfolio in a short ramp-up: typically 150–300 leveraged corporate borrowers, floating rate, senior secured. During a **reinvestment period** of about five years, principal received is reinvested rather than returned, subject to eligibility and diversity constraints written into the indenture. The manager trades within those constraints and usually holds part of the equity.

The feature that matters most is the one the 2007 structures lacked: a set of **coverage tests** that redirect cash automatically as the portfolio deteriorates. The senior **overcollateralization test** compares collateral par with the notes it supports,

$$\text{OC} = \frac{\text{collateral par}}{\text{par of this tranche and everything senior to it}} \tag{26.1}$$

and if it breaches its trigger, interest due to the equity amortizes the senior notes until it passes.

Example 26.5 The test that deleverages the deal automatically

Take a $400 million CLO with $256 million of AAA notes and a senior overcollateralization trigger of 120%. The opening ratio (26.1) is $400/256 = 156.25\%$. The test fails once collateral par falls below $1.20 \times 256 = 307.2$, after $92.8 million of erosion, or 23.2% of the pool.

Suppose it does. At $300 million of par the ratio is 117.2%, a breach of 2.8 points of overcollateralization. Restoring it requires paying down the AAA by X where $300/(256-X) = 1.20$,

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so

$$X = 256 - \frac{300}{1.20} = 6.0$$

Six million dollars, one and a half percent of the deal, cures a breach that looked like 2.8 points. The diversion works on the *denominator*: paying down senior notes shrinks the ratio's base.

At $280 million of par the ratio is 109.4% and the required pay-down is $22.7 million, so the cure gets rapidly more expensive as the collateral erodes.

Two consequences follow. The deal deleverages itself with no one deciding to, and Chapter 8's funding spirals show that deleveraging which needs a decision under stress does not happen. And *the equity holder pays for it* (the diverted cash is the equity's distribution), so the party with the most information and the most incentive to gamble for resurrection is the one whose cash is taken first. Compare Example 26.1, where nothing stood between a parameter error and the senior tranche.

A CLO is structurally the same object as the instruments in Section 26.5 (a pool, a waterfall, a AAA senior piece), and CLOs are now the largest structured credit market. They performed well through 2008 and through 2020, and the reasons are specific enough to be diagnostic.

The collateral is not exposed to one factor. Loans to a hundred companies in many industries carry real idiosyncratic risk; the subprime mortgages of one vintage did not: Example 26.4's point. The diversification a CLO claims is real.

There is no re-securitization. CLOs hold loans, not other deals' tranches; Section 26.5's compounding cannot start.

The liabilities are term-funded and match the assets. A CLO issues notes with maturities matched to the loan portfolio and cannot be forced to sell when spreads widen. Section 26.12's funding withdrawal has no analog—and this, rather than the credit quality, is probably the decisive difference. The instruments that failed in 2008 were largely fine on a hold-to-maturity basis; what killed their holders was having to sell.

And a manager has capital at risk. A CLO is managed within constraints, and the manager holds equity: Section 26.9's alignment problem addressed by construction, not regulation.

What is worth watching. Loan documentation has weakened over the cycle: **covenant-lite** loans, which carry no maintenance covenant (a financial ratio the borrower must keep meeting while the loan is outstanding), are now standard, so the early warnings a covenant breach used to provide arrive later. Recoveries on leveraged loans have been trending below historical averages (Moody's 2023), which Section 24.4's identification problem makes easy to underweight. And the investor base for the AAA tranches is concentrated.

26.15 THE LEVERAGED LOAN MARKET

The previous section described the structure and named its collateral in passing: leveraged corporate borrowers, floating rate, senior secured. That collateral is a market in its own right, and the two can no longer be understood separately: CLOs are the largest single buyer of leveraged loans, and leveraged loans are very nearly the whole asset side of CLOs.

The scale is the first thing to state. The US institutional leveraged loan market stood at about **$1.55 trillion** at the end of 2025 on the Morningstar LSTA index, more than three times the $465 billion it measured at the end of 2010, with Europe at more than €300 billion; CLOs own about **two-thirds** of the US total. The market has therefore grown past the US high-yield bond market, which stood at about $1.48 trillion of par, and it did so while the credit derivatives market of Chapter 25 shrank to about a third of its 2010 size. Any statement about where corporate credit risk is traded has to account for both movements.

Why it is a credit market and not a loan book. A leveraged loan is originated by a bank and then distributed. The arranging bank underwrites the whole facility, syndicates it to institutional

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investors, and retains little or none of it, so the instrument is priced by a market rather than held to maturity by its originator. It trades in a secondary market on standardized documentation, it is quoted daily, it settles through an assignment process, and index providers publish total-return indices on it. The economics are those of a floating-rate senior secured bond with a call at par, and the borrower is typically a company owned by a private equity sponsor, which is what supplies the leverage the name refers to.

Who is on each side, and why. The borrower is a sponsor-owned company financing an acquisition or a recapitalization, and it wants a floating-rate instrument it can prepay without penalty, because the sponsor intends to exit. The lender is a CLO buying assets for a portfolio it has already sold liabilities against, a loan mutual fund or exchange-traded fund offering retail investors a floating-rate credit exposure, or a separately managed account. The arranging bank wants the fee and not the asset. Each of those preferences is visible in the instrument: the prepayment option at par exists because the borrower wanted it, the floating rate exists because the largest buyer funds itself floating, and the covenant package is light because the arranger must clear the deal in a competitive market.

That last point is where the interesting arithmetic is. The previous section observed that recoveries on leveraged loans have been running below their historical averages, and the natural reading, that the borrowers have become weaker, is not the main explanation. Example 26.6 separates what changed from what did not.

Example 26.6 Why a senior claim recovers less without the borrower getting worse

Take three capital structures on the same company, with enterprise value at origination of $1,200 million and total debt of $1,000 million in each; only the split between first-lien debt (the senior secured loan, with the first claim on the collateral) and junior debt differs.

Structure	First lien	Junior	Decline that first impairs the first lien
Traditional	400	600	66.7%
Intermediate	700	300	41.7%
Loan-only	**1,000**	**0**	**16.7%**

A first lien is impaired when enterprise value falls below its own par, so what protects it is everything beneath it, junior debt and equity alike, measured against its own size, and historically the senior claim was a small slice of the capital structure. With total debt fixed, shrinking the junior debt to zero makes the first lien the whole of the debt and moves the enterprise-value decline that first touches the first lien from **66.7% to 16.7%**.

Now apply one distress outcome to all three. Enterprise value falls to $700 million, a decline of 41.7%:

Structure	First-lien recovery	Junior recovery
Traditional	100.0%	50.0%
Intermediate	100.0%	0.0%
Loan-only	**70.0%**	—

The same enterprise outcome pays the first lien 100% or 70% according to how much debt sits beneath it, and nothing about the borrower differs between the three structures. An average first-lien recovery computed across a market that has shifted from the first row to the third will fall even if every borrower is identical and every enterprise outcome is identical. Two further mechanisms push the loan-only figure down from there, and they compound rather than substitute.

Covenant-lite defers the event. A maintenance covenant forces a restructuring while enterprise value is still comparatively high; without one, the borrower continues until it cannot pay, and the recovery is measured on a smaller enterprise. A further 10% of erosion before the event takes the 70.0% to **63.0%**.

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A liability-management exercise moves the collateral. Loan documents that permit transfers to unrestricted subsidiaries (affiliates outside the lenders' collateral and covenants), or that allow new money to prime the existing lenders (rank ahead of them), let a distressed borrower and some of its lenders act against the rest. If 20% of enterprise value leaves the collateral package, the recovery falls to **50.4%**.

A claim can therefore be senior, secured, and first in the waterfall, and still recover half, without any of the three words having become false. Section 24.4's point about recovery rates was that they are used with a confidence the data do not support; this is the mechanism behind that unreliability, and it is a documentation variable rather than a credit variable.

What this does to the structure above it. A CLO built on loan-only borrowers is holding collateral whose recovery distribution is worse and wider than the historical series its ratings were calibrated on, which is Section 26.8's problem arriving by a new route: the letter is the same and the loss behind it is not. The coverage tests of Example 26.5 still work (they are measured on par rather than on recovery, and they still divert cash automatically), but they trigger later relative to the eventual loss than they would have on a traditional capital structure. Nothing here undoes the four structural defenses the previous section set out; it sharpens one of the weaknesses that section named, the falling recoveries.

26.16 WHERE THE SAME STRUCTURE IS NOW

Two current markets share enough of the pattern to be worth examining, not to predict a crisis but to ask which of the specific failures each shares.

Significant risk transfer. Banks buy protection on the junior tranches of their own loans (the first-loss and mezzanine pieces), retaining the senior exposure, to reduce regulatory capital; the market has grown fast.

What is better. The bank keeps the loans and the senior tranche, so Section 26.9's incentive problem inverts: in 2007 the originator kept none of what it made, while here it keeps every loss beyond the junior tranches it sold, and so most of the exposure to its own underwriting. Protection is usually funded, so Section 25.15's counterparty problem is largely absent, and the loans are the bank's own, which it underwrote and can monitor.

What is the same. The capital relief depends on a supervisory judgment that the transfer is genuine, and the history of this chapter is a history of transfers that appeared genuine and were not. And the buyers are concentrated: a few specialist funds, often themselves levered.

Private credit. Section 24.16 raised it and this is the natural place to finish the argument.

What is better, and it is substantial. The institutional holders are closed-end funds with locked-up capital that cannot be run on, so there is no maturity transformation of the kind that broke the vehicles in Section 26.12. The retail "semi-liquid" vehicles are the exception: they promise periodic redemptions, usually capped at 5% of net asset value a quarter, and several had to ration them in 2026. The loans are held whole, not tranched, so Example 26.1's convexity is absent.

What is worth watching. Valuations are infrequent and produced by the holder, so the marks that would signal deterioration may not exist. That is not the same as the risk being absent, and Section 26.12 records that the last time marks became unreliable was the moment the crisis became unmanageable. Leverage has entered at the fund level. And the banks that no longer make these loans often finance the funds that do, which rebuilds the exposure at one remove.

Neither is a repeat. Both differ from the 2007 structure in ways that address its failures; both make the risk's location harder to observe. The chapter's lesson is not that any structure is dangerous but that a system loses the ability to see a risk before it loses the ability to bear it.

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26.17 WHERE THIS CAME FROM

Portfolio credit went wrong step by step, each step defensible. Vasicek's one-factor loan-portfolio model (published 2002, circulated far earlier) underlies Basel's capital formula and is sound within its assumptions. Li (2000) then applied a Gaussian copula to default times, and the tranche market was built on it. Andersen and Sidenius (2004) had shown how to relax the assumptions that mattered (random recovery and random factor loadings), and Hull and White (2004) had given a valuation without simulation. Coval, Jurek and Stafford (2009) stated this chapter's convexity argument formally, once the losses had arrived. Gorton and Metrick (2012) supplied the funding-run account Section 26.12 rests on.

SUMMARY

Securitization pools loans and ranks claims on them, making illiquid assets tradeable and matching risk to appetite. The agency market worked through the crisis and works now; what failed was what was built on top.

A tranche is a call spread on the pool's loss, so it is an option, and options are convex in their parameters: raising the pool's default probability raises the pool's expected loss in proportion and multiplies the senior tranche's many times over. Two unobservable inputs, each amplified that strongly, are not a basis for a AAA rating. Agency mortgage-backed securities carry pure prepayment risk, harder to model than a Gaussian copula, and they worked. The difference is not difficulty but whether the key input can be checked: prepayment speeds are observed monthly, default correlation never.

Re-securitization compounded the amplification. Re-tranching mezzanine tranches produced a "AAA" piece whose expected loss depended almost entirely on the correlation between deals, which no one could measure—and the subprime mezzanines of one vintage are one bet on national house prices.

The Gaussian copula has no tail dependence, but the failure was institutional: a model calibrated to a sample with no national house-price decline, its output treated as a measurement by issuer-paid institutions. May 2005 had staged the rehearsal. The risk that was distributed came back: banks kept super-senior tranches on their books, and other tranches in unconsolidated vehicles funded with paper they backstopped. Mortgage losses were survivable; the crisis was in the transmission.

Collateralized loan obligations are the same object and performed through 2008 and 2020: diversified collateral, whole loans, term funding and a manager holding equity remove most of the crisis's mechanisms, and coverage tests deleverage the deal at the equity's expense. The loans beneath a CLO are a market of their own, and a senior secured claim there can recover far less than its history suggests without any of its three adjectives having become false, because recovery depends on how much debt sits beneath it, a documentation variable. Significant risk transfer and private credit share parts of the pattern, and what they share is a risk whose location is harder to observe. A system loses the ability to see a risk before it loses the ability to bear it.

FURTHER READING

The Financial Crisis Inquiry Commission's report is the public account of record and its dissents are worth reading alongside it, since the disagreements are about causation, not facts. Gorton's work on the repo run treats Section 26.12's funding mechanism and corrects the impression that the crisis was mainly about mortgage losses. On the copula, Li's original paper is short and makes none of the claims its critics attribute to it. The May 2005 episode of Section 26.7 is set out as it happened in Belsham, Vause and Wells's Bank of England *Financial Stability Review* article on credit correlation, a diagnosis two years early. Coval, Jurek and Stafford's analysis of

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structured finance is the formal statement of Example 26.1's convexity argument. On what was fixed, the Basel documents on securitization capital are specific, and the right source for judging whether a current structure shares the defects. On private credit and significant risk transfer, the supervisory literature is where the structures and their risks are documented, and both markets are moving.

PRACTICE QUESTIONS

26.1 State three legitimate purposes of securitization, and explain why the failure of 2007–2009 does not condemn all of them.

26.2 Explain why a tranche is a call spread on the pool's loss, identify the strikes, and use Example 26.1 to state the factor by which the pool's and the senior tranche's expected losses each rise when the default probability triples.

26.3 Explain why an agency mortgage pass-through has negative convexity, identify the option and who holds it, and use Example 26.2 to give the effective convexity of the pass-through and of the same bond at a fixed prepayment speed.

26.4 Explain why a mortgage investor hedging to a duration target buys duration as rates fall, and relate this to Section 18.12.

26.5 Name the three sources of credit support in a securitization, explain which is consumed first, and explain why the accumulating-support calculation of Example 26.3 overstates protection in exactly the scenario where protection is needed.

26.6 Explain why mezzanine tranches were pooled and re-tranched rather than sold directly, why doing so did not diversify, and use Example 26.4 to state the range of the "AAA" tranche's expected loss across inter-deal correlations.

26.7 Explain what tail dependence means, why the Gaussian copula's lack of it matters specifically for senior tranches, and why calibrating the correlation to pre-2006 data could not have captured the event that mattered.

26.8 Explain why a AAA tranche's rating is closer to binary than a AAA corporate bond's, and why rating triggers made downgrades destructive rather than merely informative.

26.9 Explain why the May 2005 episode caused losses on both legs of a trade that was supposed to be hedged, and why inferring a correlation from tranche prices and then using it to hedge tranches is circular.

26.10 Describe the originate-to-distribute chain, identify at which point the risk was supposed to stop, and explain how the risk that banks appeared to have distributed returned to their balance sheets.

26.11 Explain why a synthetic CDO multiplies notional without multiplying the underlying default risk, and what this does to interconnection.

26.12 State the sequence in which the crisis unfolded, identify the step at which mortgage losses stopped being the binding problem, and explain why senior tranches largely paid in the end and yet produced enormous losses.

26.13 State four respects in which a CLO differs from a 2007 subprime CDO, say which of them is probably decisive, and use Example 26.6 to explain what is nonetheless worth watching in its collateral.

26.14 A \$500 million CLO has \$320 million of AAA notes and a senior overcollateralization trigger of 125%. Use Example 26.5's method to find the collateral par at which the test fails, and the pay-down of the AAA notes that restores it when par has fallen to \$390

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million and to $360 million. Explain why the cure is smaller than the breach appears, and who pays for it.

26.15 Explain what dual recourse means in a covered bond, why it removes the incentive problem, why covered bonds free up no bank capital, and what that implies about lending capacity.

26.16 Explain why the incentive problem in significant risk transfer is inverted relative to 2007, and state one respect in which private credit is safer than the pre-crisis structure and one that warrants monitoring.

FURTHER QUESTIONS

26.17 Example 26.1 shows a senior tranche's expected loss rising twenty-eight-fold when the pool's rises threefold. Derive the elasticity of a tranche's expected loss with respect to the pool's default probability, show how it depends on the attachment point, and determine where in the capital structure the elasticity is largest.

26.18 Example 26.4 argues that re-securitization pools the systematic residual. Formalize this: decompose a mezzanine tranche's loss into systematic and idiosyncratic components, show what pooling does to each, and derive the condition under which re-securitization adds diversification rather than concentration.

26.19 Section 26.6 argues that the copula was a bad model used badly, with the second clause carrying most of the weight. Construct the counter-argument that a model with tail dependence would have produced materially different ratings; identify what data would settle the question; and state what your answer implies about the value of model improvement relative to institutional reform.

26.20 Section 26.13 says risk retention addresses the incentive problem but that the expected loss on a 5% slice can be small relative to arranging fees. Model the arranger's decision, derive the retention level at which incentives are aligned, and discuss why regulators chose a number below it.

26.21 Section 26.11 states a fair counter-argument that synthetic structures let those who understood the risk be short. Evaluate whether the existence of informed short sellers made the market more or less efficient in this case, and reconcile your answer with the fact that prices did not adjust until the losses arrived.

26.22 Section 26.16 claims that a system loses the ability to see a risk before it loses the ability to bear it. Apply this to private credit: specify what would have to be observed to detect deterioration early, explain why infrequent holder-produced valuations obstruct it, and propose a disclosure regime that would restore observability without forcing mark-to-market on holders who need not sell.

26.23 Compare the 2007–2009 episode with the three episodes of Chapters 18, 22 and 23 along the dimensions of crowding, mechanical hedging, leverage, funding fragility, and observability. Identify which features are common to all four, which are unique to the crisis, and use the comparison to state what a supervisor should monitor that is not specific to any instrument.

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Chapter 27
Counterparty Credit Risk and Capital

Chapter 9 priced counterparty risk. This chapter capitalizes it, and the two are not the same activity.

CVA is a price: what the risk is worth, computed with risk-neutral probabilities and hedgeable in principle. Regulatory capital is a buffer: what must be held against the possibility that the price was wrong. *Capital in this chapter always means regulatory capital*: equity and other loss-absorbing instruments a bank is required to hold, set by the Basel framework of Section 9.7 and written into national rules by each jurisdiction's regulator, which is why the same standard binds differently in Europe, the United States, and China. Section 9.7 flagged that a firm can reduce one without reducing the other, and the reason is that they are computed by different rules for different purposes: one by the firm's own model against market prices, the other by a standardized formula against a supervisory scenario.

The chapter runs from the exposure to the framework that measures it, and then to what the framework misses. Sections 27.1 and 27.2 are the object itself and the legal clause that reduces it most. Sections 27.3 to 27.5 are the standardized approach and the capital it feeds, and Sections 27.6 to 27.9 the charges around it, including wrong-way risk. Sections 27.10 to 27.12 are the leverage backstop, a bank's whole requirement, and what both cost a clearing intermediary. Sections 27.13 to 27.17 are the limit: what a bilateral framework cannot see, and who bears it.

Section 7.10 said residual counterparty risk is capitalized under this framework and a 2021 episode showed how quickly it returns; Section 7.12 said total return swaps are the instrument through which leverage accumulates quietly and pointed here. Section 8.7 deferred wrong-way risk in general. Section 27.14 is where all three arrive together.

27.1 WHY IT NEEDS ITS OWN FRAMEWORK

A loan's exposure is known: it is the amount lent. A derivative's is not.

The exposure is an option on the market. A swap that is worth nothing today may be worth $50 million in two years, and the counterparty's failure matters only in the states where it is worth something to you. Section 9.1 made this point and Figure 9.1 drew the profile; the consequence for capital is that the amount at risk must be modeled rather than looked up.

Three features follow, and each requires machinery a loan book does not need.

It is bilateral. Either party may end up owing the other, so exposure can change sign, and both parties hold credit risk on the same contract at different times.

It is portfolio-dependent. The exposure to a counterparty is a property of the whole netting set, not of any trade in it. A new trade can reduce total exposure, which no loan ever does.

And it is mitigated by mechanisms that can themselves fail. Netting depends on enforceability, collateral on operational performance and on the value of what is posted, which Chapters 1 and 8 addressed and which the capital framework largely assumes will work.

Figure 27.1 draws that option over the life of a swap, as the expected exposure and the 95th-percentile PFE that Section 9.1 defined. Section 9.1 explained the hump in words; in symbols, per unit of notional the PFE is about $(T - t) \times 1.645\,\sigma\sqrt{t}$ at a normal rate volatility σ, and the expected exposure is the same product with 0.40 in place of 1.645, so diffusion and the shrinking number of payments balance at a peak about a third of the way through. The lowest curve is the same swap under daily variation margin with a ten-day margin period of risk. Margin settles the mark-to-market each day, so the only move that can accumulate is the one over the ten days it takes to close out a defaulted counterparty: the curve replaces $\sqrt{t}$ by $\sqrt{10/250}$, which is why it has no hump and simply declines with the payments left. It is also positive even at inception, when the swap is worth nothing and no margin is posted: a counterparty that fails on the first day still leaves ten days of moves uncovered. [TN27] derives all three curves. The gap between the margined curve and the others is what the maturity factor of Section 27.3 is about.

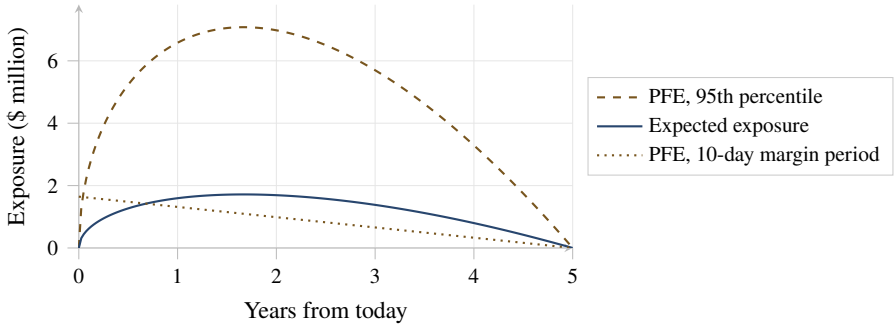


Figure 27.1: Exposure on a five-year par swap, $100 million notional, at a normal rate volatility of 100 basis points a year. Illustrative, not market data.

27.2 WHAT NETTING IS WORTH

Before any formula, Example 27.1 measures the single largest exposure reducer in the framework.

Example 27.1 The netting benefit

Take the book of Example 1.1: five trades with one counterparty, current mark-to-market values in millions:

$$+40, \quad -35, \quad +20, \quad -15, \quad -8$$

Without an enforceable netting agreement, the exposure is the sum of the positive values: the counterparty's administrator would enforce the trades you owe on and pay only a recovery fraction on the trades owed to you. That is the cherry-picking whose cost Example 1.1 computed, and here it gives

$$40 + 20 = \$60 \text{ million}$$

With netting, the exposure is the net value:

$$\max(40 - 35 + 20 - 15 - 8,\ 0) = \$2 \text{ million}$$

That is a 97% reduction, from a legal clause rather than a hedge, and it is only today's exposure. The formula of Section 27.3 adds a second component for how much worse it could become before a default is closed out, and netting reduces that too, because offsetting trades partly cancel before it is computed.

This is why Section 1.5 called the enforceability of close-out netting a precondition for a derivatives market existing at all. The whole framework assumes netting is enforceable, and where it is not (no close-out netting statute, or a counterparty type carved out), exposure and capital

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revert to the gross figure. A jurisdiction's **netting opinion** (the legal opinion, commissioned by the industry for each jurisdiction, that close-out netting will be upheld in a counterparty's insolvency there) is therefore worth more to a dealer than any risk model.

27.3 SA-CCR, AND WHAT MARGINING BUYS

The **standardized approach for counterparty credit risk** (SA-CCR) computes the exposure at default (EAD, Section 24.14), the amount the capital rules treat as lent to the counterparty, for each netting set (Chapter 8), the trades with one counterparty that a single enforceable netting agreement covers:

$$\text{EAD} = \alpha \times (\text{RC} + \text{PFE}), \qquad \alpha = 1.4 \tag{27.1}$$

The two components are computed as follows; the scalar α is Section 27.5's subject.

Replacement cost. RC is what the counterparty's default would cost today. Write V for the current market value of the netting set to the bank, the sum of its trades' values, and C for the net collateral the bank holds against it after haircuts. Without a margin agreement,

$$\text{RC} = \max(V - C,\ 0) \tag{27.2}$$

With one, RC is the largest exposure that would *not* trigger a margin call:

$$\text{RC} = \max(V - C,\ \text{TH} + \text{MTA} - \text{NICA},\ 0) \tag{27.3}$$

where TH is the threshold below which no variation margin is called, MTA the minimum transfer amount, and NICA the net independent collateral amount (initial margin and other independent collateral received, less any posted that is not held in a segregated, bankruptcy-remote account). The middle term is there because a margin agreement does not hold the exposure at zero: no call is made below the threshold, and then only above the minimum transfer, so the exposure can sit at TH + MTA with the counterparty in full compliance, and only initial margin in hand offsets that (Figure 27.2, left). With zero threshold and minimum transfer amount and no initial margin, the term vanishes and RC is the uncollateralized mark-to-market as before.

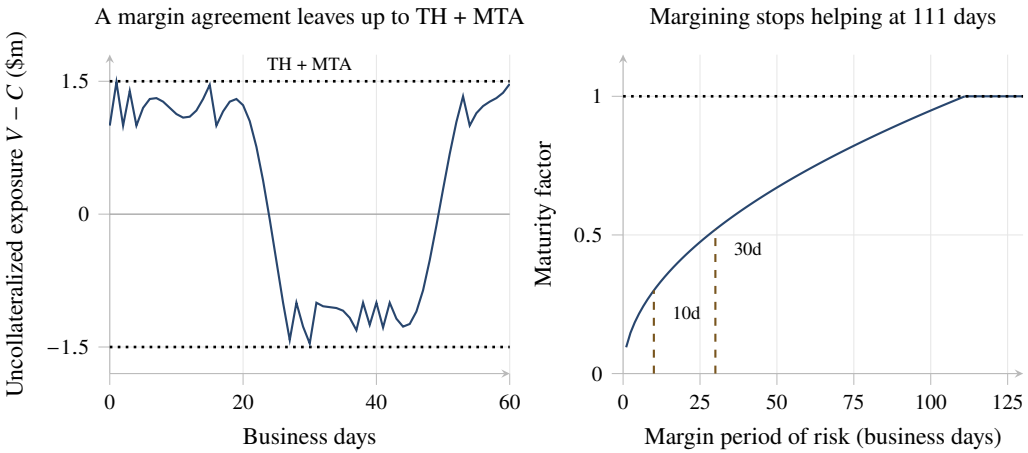


Figure 27.2: Left: uncollateralized exposure under a margin agreement (TH = \$1m, MTA = \$0.5m), illustrative path; each jump back toward zero is a margin call. Right: equation (27.7), capped at one.

Potential future exposure. PFE is a supervisory estimate of how much worse the exposure could become before a default is closed out. Despite the shared name, it is not the 95th

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percentile of Figure 27.1 but an average, as the end of this section shows:

$$\text{PFE} = \text{multiplier} \times \sum_{\text{asset classes}} \text{AddOn} \tag{27.4}$$

The **add-on** for an asset class stands in for how far that class's positions can move over the horizon that matters. Within interest rates, the class Example 27.2 uses, trades are grouped into **hedging sets**, one per currency, and within a hedging set into three maturity buckets: under one year, one to five years, over five years. Each trade i contributes an **effective notional**

$$D_i = \delta_i \times d_i \times \text{MF}_i \times N_i \tag{27.5}$$

where N_i is the notional; δ_i is the **supervisory delta**, $+1$ for a position that gains when rates rise (paying fixed), -1 for one that loses, or a Black–Scholes delta for an option; d_i is the **supervisory duration**, the present value at a fixed 5% of one unit a year from the trade's start date t_i to its maturity T_i,

$$d_i = \frac{e^{-0.05t_i} - e^{-0.05T_i}}{0.05} \tag{27.6}$$

and MF_i is the **maturity factor**, which scales the position to the horizon over which it is exposed:

$$\text{MF} = \sqrt{\min(M, 1)} \ \text{ unmargined}, \qquad \text{MF} = 1.5\sqrt{\text{MPOR}/250} \ \text{ margined} \tag{27.7}$$

with M the remaining maturity in years, floored at ten business days, and MPOR the margin period of risk in business days, the time assumed to pass between the last margin exchange and close-out: ten days for a daily bilateral agreement, five for a clearing member's exposure to its clients, twenty above 5,000 trades or with illiquid collateral or a derivative that cannot easily be replaced in a stressed market, and double after repeated margin disputes. The 250 is the framework's business days in a year, so MPOR/250 is that window in years.

The product is not a notional in any economic sense. The first three factors are the trade's sensitivity to the rate ($\delta_i d_i N_i$ is its signed annuity, so Example 27.2's $D = 442.40$ million says the swap gains or loses \$44,240 per basis point), and MF_i carries the square root of the horizon over which that sensitivity is exposed. The volatility that turns the two into money sits in the supervisory factor below. So D_i is the amount of a reference trade with unit duration and a full one-year horizon that carries the same exposure, which is what *effective* means in the name.

Effective notionals for the three maturity buckets $k = 1, 2, 3$ are summed within each, $D_k = \sum_{i \in k} D_i$, so offsetting trades cancel, and the buckets then combine with partial offset into the hedging set's effective notional EN:

$$\text{EN} = \sqrt{D_1^2 + D_2^2 + D_3^2 + 2 \times 0.7\, D_1 D_2 + 2 \times 0.7\, D_2 D_3 + 2 \times 0.3\, D_1 D_3} \tag{27.8}$$

Equation (27.8) is the standard deviation of a sum of correlated normal variables, with correlations of 0.7 between adjacent buckets and 0.3 between the outer two, and those correlations are what *partial* means. Take a payer of $D_2 = +100$ in the one-to-five-year bucket against a receiver of $D_3 = -100$ in the over-five-year bucket. Then (27.8) returns $\sqrt{100^2 + 100^2 - 2 \times 0.7 \times 100 \times 100} = 77$, not the 0 of a perfect hedge or the 141 of none: the position counts as mostly, not wholly, offset. Within a bucket the offset is complete, because there the D_i are added before anything is squared.

The hedging set's add-on is then

$$\text{AddOn}_{\text{hedging set}} = \text{SF} \times \text{EN} \tag{27.9}$$

with the **supervisory factor** SF = 0.50% for interest rates, and the asset class's add-on is the sum over its hedging sets. The other classes follow the same pattern with their own factors and

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hedging sets (4% per currency pair for foreign exchange, 32% for a single stock with partial offset across issuers for equity), and the Basel standard has the details. The approach covers derivatives of every kind, cleared as well as bilateral (which is what lets Example 27.3 put three regimes in one table, since clearing changes the risk weight and not the measure), but not securities financing.

The **multiplier** in (27.4) recognizes collateral in excess of the current exposure, or a netting set that is out of the money, both of which cushion future moves:

$$\text{multiplier} = \min\left\{1,\ 0.05 + 0.95\,\exp\left(\frac{V - C}{2 \times 0.95 \times \sum \text{AddOn}}\right)\right\} \tag{27.10}$$

It equals one whenever $V \geq C$, so for a netting set at or in the money the PFE is simply the sum of the add-ons, and it falls toward a floor of 5% as $C - V$ grows relative to the add-on.

None of this is arbitrary. The formula is a closed-form approximation of the internal-model quantity of Section 27.5: underneath it the netting set's value is normal with no drift, as in the model behind Figure 27.1, and the target is the effective expected positive exposure over one year (an average of the expected exposure over that year, defined precisely in Section 27.5), which α scales as it scales a bank's own model. Each piece corresponds:

- the supervisory factor is the expected exposure, averaged over one year, of a unit at-the-money position at a stressed *normal* volatility, a move in rate rather than a percentage return (0.50% for rates corresponds to about 190 basis points a year);
- the maturity factor is the average of $\sqrt{t}$ over the horizon, discussed below;
- the bucket correlations behind (27.8) are a supervisory judgment that maturities move together but not identically;
- the multiplier is a floored exponential fitted to the exact normal expectation for a position that starts out of the money.

[TN27] works each correspondence through.

Example 27.2 A single swap, and what margin does to it

Take a five-year interest rate swap, \$100 million notional, in which the bank pays fixed. It is currently at zero mark-to-market, with no collateral and no margin agreement.

Replacement cost. $V = 0$ and $C = 0$, so by (27.2) RC = 0.

Effective notional. Paying fixed gains when rates rise, so $\delta = +1$. The trade starts now and runs five years, so $t = 0$ and $T = 5$, and by (27.6)

$$d = \frac{1 - e^{-0.25}}{0.05} = 4.4240$$

Unmargined with five years remaining, $\text{MF} = \sqrt{\min(5, 1)} = 1$, so $D = 1 \times 4.4240 \times 1 \times 100 =$ \$442.40 million. It is the only trade in its hedging set and sits in the one-to-five-year bucket, so EN = 442.40.

Add-on, PFE, and EAD. AddOn = $0.50\% \times 442.40 =$ \$2.212 million. With $V - C = 0$ the multiplier is one, so PFE = 2.212 and

$$\text{EAD} = 1.4 \times (0 + 2.212) = \$3.10 \text{ million}$$

In the money. If the swap were instead worth \$5 million to the bank, RC = 5, the multiplier stays at one, and EAD = $1.4 \times (5 + 2.212) =$ \$10.10 million.

Out of the money. If it were worth −\$5 million, RC = 0 and the multiplier does its work:

$$\text{multiplier} = 0.05 + 0.95\,e^{-5/(2 \times 0.95 \times 2.212)} = 0.339$$

so PFE = $0.339 \times 2.212 =$ \$0.75 million and EAD = $1.4 \times 0.75 =$ \$1.05 million. A swap the bank owes money on still carries exposure, because rates can move far enough to swing its value more than \$5 million in the bank's favor, but only about a third of the at-the-money amount.

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Now add a margin agreement with daily exchange, zero threshold and minimum transfer amount, and a ten-day margin period of risk. At zero mark-to-market (27.3) still gives RC = 0, and the maturity factor becomes MF = $1.5\sqrt{10/250} = 0.30$, so $D = 0.30 \times 442.40 = 132.72$, the add-on falls to $0.50\% \times 132.72 = \$0.664$ million, and

$$\text{EAD} = 1.4 \times 0.664 = \$0.93 \text{ million}$$

That is a 70% reduction in the add-on, purely from margining. This is the framework's central incentive and it is deliberate: the margin period of risk is the only window in which an unhedged move can hurt, so shortening it shortens the exposure.

Example 27.2's 70% reduction came from the maturity factor alone, and (27.7) contains a limit that is not obvious. The margined factor is capped at the unmargined value of one, and setting $1.5\sqrt{\text{MPOR}/250} = 1$ gives $\text{MPOR} = 250/1.5^2 = 111$ business days: beyond about five months of close-out risk, margining provides no add-on relief at all. The formula's logic is that exposure grows with the square root of time, so a shorter close-out window means a smaller possible move. But at 111 days the margined exposure, put on the unmargined scale by the 1.5 explained next, is as large as the unmargined one, and there is nothing left to relieve.

The 1.5 is a ratio of two averages. The supervisory factors are calibrated to an unmargined exposure averaged over one year, and an exposure growing as $\sqrt{t}$ averages to two-thirds of its one-year value, since $\int_0^1 \sqrt{t}\,dt = 2/3$; a margined exposure is the same ten-day move at every date, so it averages to the full $\sqrt{\text{MPOR}/250}$. Dividing by $2/3$ puts the two on one scale, and $1/(2/3) = 1.5$.

The practical content is in the middle of the curve. A position that genuinely takes thirty days to unwind (a concentrated equity book, an illiquid credit portfolio) keeps 52% of the unmargined add-on even under a daily margin agreement. The right panel of Figure 27.2 shows where the relief comes from and how quickly it runs out.

Section 27.14 breaks the ten-day assumption. The regulatory ten days is a claim about liquidity, not the calendar; a dealer that applies it to a position worth many days of trading volume has used an input it did not verify (the twenty-day floor for hard-to-replace derivatives leaves that call to the bank), which is Chapter 21's lesson, that a model's assumptions decide what it may be used for, arriving in a capital formula rather than a pricing model.

27.4 CAPITAL UNDER THREE REGIMES

EAD is not capital. Multiplied by the counterparty's risk weight RW it gives **risk-weighted assets** (RWA), and those times the required capital ratio CR give the capital the trade consumes:

$$\text{RWA} = \text{EAD} \times \text{RW}, \qquad \text{capital} = \text{RWA} \times \text{CR} = \text{EAD} \times \text{RW} \times \text{CR} \tag{27.11}$$

Neither factor is the bank's to choose. The supervisory risk weights are prescribed by counterparty type:

- 0% for a high-grade sovereign;
- 2% for a trade against a **qualifying CCP** (one licensed and supervised under rules consistent with the international standards for clearing houses);
- 20% for a highly rated bank;
- 100% for a typical unrated corporate;
- more for a speculative-grade one.

CR is the minimum ratio Section 1.12 described, 8% plus the buffers a jurisdiction imposes. A bank with model approval estimates some risk weights itself, within supervisory floors, which is the internal-ratings route of Section 39.10. Put the steps together and the framework's effect on market structure becomes a single table.

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Example 27.3 Why everything moved to clearing

Take Example 27.2's swap at zero mark-to-market, against a corporate at a 100% risk weight, with a 10.5% capital requirement (the 8% minimum plus the 2.5% capital conservation buffer, a layer a bank may draw down under stress only at the price of limits on its dividends). The EAD is the \$3.10 million computed there, carried to a third decimal so that the capital column can be checked.

	EAD	Risk wt.	RWA	Capital
Uncleared, unmargined	\$3.097m	100%	\$3.097m	\$0.3252m
Uncleared, margined (10d)	\$0.929m	100%	\$0.929m	\$0.0975m
Cleared at a qualifying CCP (10d)	\$0.929m	**2%**	\$0.019m	\$0.00195m

The ratio is 167 : 50 : 1: clearing costs 0.6% of the capital of leaving the trade uncleared and unmargined.

Two changes drive it, and they compound. Margining replaces the unmargined maturity factor of one by $1.5\sqrt{10/250} = 0.30$, so EAD falls 70% to \$0.929m, and clearing keeps that ten-day floor (the five days of (27.7) are for a clearing member's exposure to its own clients, Section 27.12); the 2% weight for trade exposures against a qualifying CCP is then a further factor of fifty.

Turning capital into a running charge at a 10% hurdle rate (Section 9.7's KVA calculation) gives the number a client actually sees:

	bp per year of notional
Uncleared, unmargined	3.25
Uncleared, margined	0.98
Cleared	0.02

Three basis points a year sounds small, and on a swap that trades at a fraction of a basis point of bid–ask it is decisive. A dealer quoting the same swap to a cleared and an uncleared client is quoting from different cost bases, and the gap is larger than the spread.

This is the answer to a question Chapter 2 raised and did not settle. The clearing mandate covered standardized products; it did not reach the rest of the market, and much of that cleared anyway. The table is why. Capital rules did more to move the market than the mandate did.

The table omits one qualification. A clearing member also holds capital against its default fund contribution, which the 2% line above excludes. The charge is set for the member, not the trade: it is the member's share, in proportion to its contribution, of the clearing house's own hypothetical capital requirement, computed as though the clearing house were a bank, and it can be no lower than that of a 2% risk weight on the contribution. For an illustrative \$100 million contribution the floor alone is $2\% \times 100 \times 10.5\% = \$0.21$ million; spread over 10,000 cleared swaps like this one it adds \$21 to each, against the \$96,000 that clearing saves on each margined uncleared swap. The risk-sensitive charge is larger than its floor, but it would have to be thousands of times larger to change the ordering. It is the price of the mutualization of Chapter 2, a charge against a loss caused by another member's default.

27.5 THE MEANING OF $\alpha = 1.4$

The scalar α in (27.1) is the most misread number in the framework. It looks like a measurement—exposures are 40% larger than they appear—and it is not one.

Under the internal model method, a bank computes an exposure profile by simulation, reduces it to **effective expected positive exposure** (the running maximum of expected exposure, averaged over the first year) and treats the result as a loan-equivalent amount. But a loan equivalent computed this way understates the capital a full portfolio credit model would require, because it ignores the correlation between exposures and defaults across counterparties and the granularity of the portfolio: a book concentrated in a few large counterparties is less diversified than the loan formula assumes.

α is the supervisory bridge over that gap. The industry studies behind it, led by ISDA in

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2003, found about 1.1 for diversified dealer books; supervisors set 1.4 as a conservative default (Basel Committee 2005), with a floor of 1.2 for banks permitted to estimate their own. So the 40% uplift is a buffer for model error, not a property of any trade; reading it as a real 40% higher exposure leads a desk to try to hedge something that does not exist.

The two routes meet on a plain swap: at the supervisory factor's 188 basis points of normal volatility, Example 27.2's swap has an effective expected positive exposure of $2.20 million ([TN27]), which α takes to $3.08 million against SA-CCR's $3.10 million. A bank with supervisory approval keeps the internal model method because it simulates the actual netting set, margin terms and collateral, where SA-CCR applies prescribed factors. The benefit has a limit. Under the output floor, a bank on the internal model method computes counterparty exposure under SA-CCR as well, for the standardized base of the floor, so its total risk-weighted assets cannot fall below 72.5% of what that base gives.

27.6 THE CAPITAL CHARGE FOR CVA

A separate charge covers the risk that CVA itself moves: that a counterparty's credit spread widens, marking down the value of the derivative book, without anyone defaulting. The Basel Committee's 2011 announcement of the charge put roughly two-thirds of the crisis's counterparty credit losses in these mark-to-market movements, which the earlier framework had not capitalized. Section 9.7 explained why the charge exists and why an action that lowers the CVA price need not lower the charge.

The framework offers two approaches, and a simpler option for small banks. The **basic approach**, BA-CVA, is the default: a formula on exposures and on supervisory risk weights set by the counterparty's sector and rating. A bank whose non-centrally-cleared derivatives fall below a notional of €100 billion may instead set the charge equal to its counterparty credit capital. The standardized approach for CVA, SA-CVA, is a sensitivity-based calculation, delta and vega against credit spreads and the market factors driving the exposure, and is closer to how a desk actually hedges; it requires supervisory approval. It is a different standardized approach from Section 27.3's, which measures the exposure this charge sits on.

27.7 HEDGING CVA, AND THE EXEMPTIONS

The standardized CVA charge is sensitivity-based, which means eligible hedges reduce it. Which hedges are eligible, and by how much, turns out to determine what the desk of Section 9.10 actually trades.

A single-name CDS on the counterparty is recognized almost in full under the standardized approach, because the charge is computed on that counterparty's spread and the CDS is written on that same counterparty: buy the protection and the charge falls by what the protection removes. The basic approach caps the relief: however complete the hedge, it removes at most three-quarters of the charge.

An index CDS is recognized in part. The framework imposes a correlation between the index and the issuer, so some of the hedge's benefit is disallowed however well it works. A proxy (a CDS on a different issuer in the same sector and region) is recognized less still. This matters because for most counterparties there is no single-name CDS at all. Chapter 25's case study of the market that shrank established that liquid single-name CDS now exists for a few hundred issuers, while a dealer has thousands of counterparties.

So the instruments a desk can actually buy are the ones the framework recognizes least. The consequence is a genuine dilemma rather than a rule to memorize: an index hedge removes most of the economic risk, earns partial capital relief, and introduces a basis (the index-to-single-name gap of Section 25.6) that the CVA charge still capitalizes, because the index is recognized only in part, and that shows up in earnings every quarter. Hedging the risk

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and hedging the capital charge are different trades, and the desk that does both holds a position that neither aim, pursued alone, would have chosen.

Then there are the exemptions, and they are large. The EU framework exempts transactions with non-financial counterparties, sovereigns, and pension funds from the CVA capital charge, on the argument that these clients cannot post collateral and would be priced out of hedging, and its 2024 revision of the rules kept the exemptions. The United Kingdom's Basel 3.1 rules remove them, and the United States and Canada did not adopt them. There the relief comes only from the framework's own carve-out, under which a trade cleared through a qualifying CCP is outside the CVA charge altogether, so what matters is how a trade is cleared rather than who is on the other side of it.

Whatever one thinks of the policy, the mechanical consequence should be stated. *The exposure is unchanged; only the capital changes.* An EU dealer's CVA on its corporate book is a real, volatile, unhedged number producing earnings swings and no capital requirement—and a dealer that *does* hedge it takes on market risk that is capitalized, while the CVA capital charge it is hedging against was already zero and cannot fall further.

The rule therefore discourages hedging the very exposures it chose not to charge for. That is an unusually clean example of the problem this chapter keeps returning to: a capital framework does not merely measure behavior, it selects it, and leniency in one place reappears as distortion in another.

27.8 DVA, AND WHY CAPITAL REFUSES IT

Section 9.3 established that a firm's own default risk produces a debit valuation adjustment (a gain when the firm's own credit deteriorates), that accounting standards require it to be recognized, and why regulators refuse it: a gain that can be monetized only by defaulting is not a buffer against default. The rule is that DVA gains and losses are filtered out of **common equity tier 1**, the highest-quality layer of capital (ordinary shares and retained earnings), so a bank whose spread widens may book a large accounting profit of which none counts as capital. Fair value and capital are answering different questions about the same liability, and the same reasoning is why Section 27.6's capital charge is computed on CVA gross of DVA while the accounting number is net.

27.9 WRONG-WAY RISK

Section 9.9 defined wrong-way risk (exposure that grows precisely where the counterparty is least able to pay) and showed that it makes the amount CVA must insure the exposure conditional on default, not the average. Example 27.4 puts a size on that substitution for CVA.

Example 27.4 What the correlation costs

A counterparty with a 3% annual default probability, 60% loss given default, and an average exposure of \$10 million has, under independence, a one-year CVA $= 0.60 \times 0.03 \times 10 =$ \$0.18 million. Putting $E[\text{exposure} \mid \text{default}]$ in place of the \$10 million multiplies that by the ratio of the two, so a correlation strong enough to double the conditional exposure doubles the charge, to \$0.36 million. The effect is linear in the conditional exposure ratio, and it is not small.

The calculation contains no correlation parameter. The conditional exposure ratio is the only thing that matters, and it is one number rather than a joint distribution, which is convenient, and also the reason the quantity is so hard to estimate. It is a conditional expectation on an event that has never occurred for this counterparty.

Wrong-way risk divides in two, and the distinction determines what can be done about it. *Specific wrong-way risk* is structural, like Section 9.9's sovereign case: taking a counterparty's

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own shares as collateral, or entering a commodity swap with a producer whose solvency depends on that commodity's price. These are identifiable in advance from the trade's economics. Where a derivative's underlying or reference entity is legally connected to the counterparty, as with an equity swap or a CDS on the counterparty itself, the regulatory response is to require the exposure to be computed as if that entity had already defaulted (own-share collateral falls under eligibility rules instead): a jump to default, with no correlation estimated. It is a rule rather than a model, because the model would need a correlation no one can estimate.

General wrong-way risk is milder: the counterparty's credit is correlated with the market factors driving the exposure. It is handled through the α of Section 27.5, whose calibration allows for that correlation, and by calibrating the exposure model to a period of stress.

The recurring lesson from three chapters generalizes. Section 24.4's recovery-default correlation, Section 26.4's excess spread that vanishes with losses, and this section's exposure-default correlation are the same failure: a protection whose value is negatively correlated with the need for it. Once named, it is checkable: for any mitigant, ask what it is worth in the state that makes it necessary.

27.10 THE LEVERAGE RATIO AS A BACKSTOP

Everything so far is risk-based: exposure is scaled by a risk weight, so a safer counterparty costs less capital. Alongside it sits a deliberately risk-*insensitive* constraint. The **leverage ratio** is tier 1 capital (common equity tier 1 plus the perpetual instruments that rank just behind it) divided by total exposure (assets, derivative exposure, securities financing and off-balance-sheet items, none risk-weighted), and the ratio, written LR, must clear a minimum, so every trade costs capital whatever its quality.

With two constraints on the same trade, one of them binds, and which one is arithmetic. Both sides are capital held against the trade. The leverage rule asks for $\text{EAD} \times \text{LR}$, since the trade adds its EAD to total exposure whatever its quality; the risk-based rule asks for the $\text{EAD} \times \text{RW} \times \text{CR}$ of (27.11). A bank must satisfy both ratios at once, and the same capital counts toward each, so a trade costs the larger of the two and never their sum. The first is larger when

$$\text{EAD} \times \text{LR} \;>\; \text{EAD} \times \text{RW} \times \text{CR} \qquad \Longleftrightarrow \qquad \text{RW} \;<\; \text{LR}/\text{CR}$$

so with a 10.5% capital ratio:

Leverage requirement	Leverage binds below a risk weight of
3% (baseline)	29%
5% (US G-SIB holding companies, before April 2026)	48%
6% (their insured banks, before April 2026)	57%

Sovereigns at 0%, qualifying CCPs at 2%, highly rated banks at 20%: all of the highest-quality exposures fall below the crossover. For exactly those businesses, capital is set by a measure that does not care that the risk is low.

The crossover is only a first approximation, and both refinements move it the leverage ratio's way. The leverage exposure measure uses its own version of EAD, recognizing collateral more grudgingly than (27.10) does, so a well-collateralized book looks larger to it than to the risk-based rule. And the two ratios are measured against different capital: tier 1 for leverage, total capital for the 10.5%, and since tier 1 is the narrower and costlier layer, a point of leverage requirement weighs more than a point of the risk-based one.

This is intended. A backstop that varied with risk weights would inherit whatever error is in them, and Chapter 26's pre-crisis experience is a story about risk weights being wrong. The leverage ratio bounds what the risk-based framework can be talked into.

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But the cost is specific and has been observed. The businesses it makes expensive (Treasury market-making, repo intermediation, client clearing) are low-margin, high-volume, and low-risk. A constraint that is insensitive to risk pushes a dealer out of its safest activities first, because those are where the constraint bites hardest relative to the revenue.

In March 2020 this became visible. Treasury market functioning deteriorated under a surge in volume, dealer balance sheets were constrained, and the supervisory response included temporarily excluding Treasuries and central bank reserves from the leverage exposure measure. The exemption expired in March 2021 and was not renewed; the recalibration of the US enhanced leverage ratio, finalized in November 2025 and effective April 1, 2026, lowered the requirement on all exposures instead, replacing the largest banks' fixed 2% buffer with half of each bank's Basel-method global systemically important bank surcharge. The underlying question, whether a backstop should be relaxed at exactly the moment the activity it constrains becomes systemically important, has no comfortable answer. Chapters 6 and 8 both rest on dealers intermediating at low margin, so a rule that taxes intermediation by volume rather than risk determines market liquidity and should be read as doing so.

27.11 HOW A BANK'S CAPITAL REQUIREMENT IS BUILT

Everything so far has priced one trade's capital through (27.11). A bank's requirement is assembled from many such pieces, and seeing the whole explains both what a derivative's capital competes with and which lever a desk can actually pull. The assembly has a numerator, the capital the bank holds, and a denominator, the risk-weighted assets it holds it against.

The numerator. Capital comes in layers, ranked by how readily they absorb loss, and each layer must be at least a set share of RWA. These are the **Pillar 1** minimums:

- *Common equity tier 1* (CET1), the ordinary shares and retained earnings of Section 27.8: at least 4.5%.
- *Tier 1*, which adds perpetual instruments that can be written down or converted: at least 6%.
- *Total capital*, which adds subordinated debt: at least 8%.

On top sit buffers, all of them met with CET1:

- the **capital conservation buffer**, 2.5%;
- the **countercyclical buffer**, 0 to 2.5%, which each jurisdiction sets according to its credit cycle;
- for a **global systemically important bank** (G-SIB), a surcharge of 1% to 3.5%, set by the bank's size, interconnection and complexity.

A bank that dips into the buffers is not in breach, but its dividends and bonuses are restricted until it rebuilds them. In the United States the conservation buffer of the largest banks is replaced by the stress capital buffer of Section 39.12, which is at least as large. Two further layers complete the framework:

- **Pillar 2**, the supervisor's bank-specific add-on for risks the formulas miss, such as concentration and interest rate risk in the banking book;
- **Pillar 3**, the public disclosure that lets the market check the arithmetic.

The denominator. Total RWA is the sum of one term per risk type, each computed by its own rule:

$$\mathrm{RWA} = \mathrm{RWA}_{\mathrm{credit}} + \mathrm{RWA}_{\mathrm{CCR}} + \mathrm{RWA}_{\mathrm{CVA}} + \mathrm{RWA}_{\mathrm{market}} + \mathrm{RWA}_{\mathrm{op}} + \mathrm{RWA}_{\mathrm{sec}} \tag{27.12}$$

- *Credit*: loans and bonds in the banking book, weighted by Table 27.1 below or by the internal-ratings formula of Section 39.10. It is usually by far the largest term.

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- *Counterparty credit*: this chapter's SA-CCR exposure times the counterparty's risk weight.
- *CVA*: the charge of Section 27.6, times 12.5.
- *Market*: the trading-book charge of Chapter 39, times 12.5.
- *Operational*: losses from failed processes, people and systems, fraud and litigation among them. The charge is a standardized formula on a **business indicator** built from the bank's income, multiplied by coefficients of 12% to 18% that rise with size and, where the supervisor allows, scaled by the bank's own loss history; times 12.5.
- *Securitization*: a framework of its own, sketched below.

The factor 12.5 is the reciprocal of 8%: it turns a capital charge into the RWA that would require it. Under the output floor of Section 39.10, total RWA from internal models may not fall below 72.5% of what the standardized rules give for every term together.

Example 27.5 From risk-weighted assets to a capital requirement

A G-SIB in the 1.5% surcharge bucket, in a jurisdiction with a 0.5% countercyclical buffer, reports RWA in billions of dollars:

Risk type	RWA ($bn)
Credit (banking book)	600
Counterparty credit (SA-CCR)	40
CVA	30
Market (Chapter 39)	60
Operational	100
Securitization	20
Total	**850**

The buffers add 2.5% + 1.5% + 0.5% = 4.5% of CET1 to every layer, so the requirements are

	Ratio	Capital ($bn)
CET1	4.5% + 4.5% = 9.0%	76.50
Tier 1	6.0% + 4.5% = 10.5%	89.25
Total capital	8.0% + 4.5% = 12.5%	106.25

before any Pillar 2 add-on. The bank's leverage exposure is $2,500 billion, and a G-SIB's leverage requirement is the 3% minimum plus half its surcharge, 3% + 0.75% = 3.75%, or $93.75 billion of tier 1. That exceeds the risk-based tier 1 figure of $89.25 billion, so the leverage ratio binds. The reason is Section 27.10's crossover: the bank's average risk weight is 850/2,500 = 34%, below the 3.75%/10.5% = 35.7% at which the two rules cost the same.

Read the table for what it says about derivatives. Counterparty and CVA risk together are 70 of 850, about 8%, and the market term another 7%: the capital a derivatives business consumes is a minority of the bank's, which is why its marginal cost depends on what the rest of the balance sheet is doing. Here, with leverage binding, a desk that lowered its risk weights would save nothing, and one that shrank its exposure would.

Standardized credit risk weights. A bank without model approval, and every bank for the output floor, weights its banking-book exposures by Table 27.1. The weight depends on the type of borrower and, in the Basel text, on its external rating.

An off-balance-sheet item enters through a **credit conversion factor**, the share of it treated as drawn:

- 100% for a guarantee that substitutes for a loan;
- 40% for an undrawn commitment;
- 10% for a commitment the bank can cancel without notice;

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Table 27.1: Standardized risk weights for credit exposures in the Basel framework (%), by external rating. Jurisdictions that do not use ratings, the United States among them, substitute their own schedules.

	AAA to AA	A	BBB	BB	B	Below B	Unrated
Sovereign	0	20	50	100	100	150	100
Bank	20	30	50	100	100	150	30–150
Corporate	20	50	75	100	150	150	100
Regulatory retail 75; residential mortgage 20 to 70 by loan-to-value; equity 250; subordinated debt 150							

- 20% for a short-term trade letter of credit.

A \$200 million undrawn revolving facility to a BBB corporate therefore carries $200 \times 0.40 \times 0.75 = \$60$ million of RWA before a cent is lent.

Credit risk mitigation. The weights apply to the exposure the bank actually bears, and three things reduce it:

- *Collateral* reduces the exposure itself. Under the comprehensive approach the exposure after mitigation is

$$E^{*} = \max\{0,\ E(1 + H_e) - C(1 - H_c - H_{fx})\},$$

  where the supervisory haircuts H grow with the collateral's volatility (zero for cash, a few percent for a short government bond, 20% or more for equities), with a further 8% if the collateral is in another currency.
- *Guarantees and credit derivatives* work by substitution: the protected part of the exposure takes the risk weight of the protection seller instead of the borrower's.
- *Netting*, the subject of Section 27.2, reduces exposure for derivatives and securities financing.

Substitution is how a bank's purchase of CDS protection (one of the uses Chapter 25 lists) turns into capital relief (Example 27.6), and it comes with conditions:

- the protection must be direct, irrevocable and unconditional;
- if the CDS does not count restructuring as a credit event, only 60% of it is recognized, because the loss a restructuring imposes falls on the lender unprotected;
- if the protection runs out before the loan, it is recognized only in proportion, as $P_a = P\,(t - 0.25)/(T - 0.25)$ for protection P with t years left against a loan with T (capped at five), and not at all if it has three months or less to run.

Example 27.6 What a CDS hedge does to a loan's capital

A bank has a \$100 million five-year loan to an unrated corporate, weighted 100%, at the 10.5% requirement of Example 27.3. It buys five-year protection from a bank rated AA, weighted 20%.

	RWA (\$m)	Capital (\$m)
Unhedged	100.00	10.50
Hedged, CDS covers restructuring	20.00	2.10
Hedged, CDS without restructuring	52.00	5.46
Hedged with a three-year CDS	53.68	5.64

The full hedge moves the whole loan to the protection seller's 20%. Without restructuring, 60% of it is recognized: $60 \times 0.20 + 40 \times 1.00 = 52$. With a three-year contract against a five-year loan, the recognized amount is $100 \times (3 - 0.25)/(5 - 0.25) = 57.895$, and $57.895 \times 0.20 + 42.105 \times 1.00 =$

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53.68.

The full hedge saves $8.40 million of capital. At the 10% hurdle rate of Example 27.3 that is worth $0.84 million a year, 84 bp of the loan. So for a borrower whose CDS trades below 84 bp the capital saved covers the hedge's cost on its own, and for one above it the hedge is bought for the risk it removes, with the capital relief covering only part of its cost. The comparison also leaves out two costs the hedge adds back: the CDS is itself a derivative, with a counterparty exposure under SA-CCR and a CVA charge, both small for a single bank-sold contract. A $100 million loan secured by $60 million of cash, by contrast, needs no counterparty at all: $E^* = 100 - 60 = 40$, and RWA fall to $40 million.

Securitization. A securitization position has its own risk-weight function, because a tranche's risk depends on where it sits in the waterfall of Section 26.2 as much as on the pool. The framework computes the capital the pool would need if held whole, K, and charges each tranche according to how much of that it absorbs:

- a tranche below K is weighted 1,250%, the reciprocal of 8%, so it costs capital dollar for dollar;
- a tranche above it is charged by a formula that falls with the attachment point, down to a floor of 15% for a senior tranche (10% for one meeting the simple, transparent and comparable criteria).

Those two numbers explain the significant risk transfer trades of Chapter 26. A bank that sells the first-loss and mezzanine tranches of a loan book, and keeps the senior one at or near the floor, replaces the loans' risk weights with a fraction of them. The regulator's test for whether the transfer is real is exactly how much risk has left the bank.

The pieces fit one way. A desk's capital is a share of a requirement whose binding constraint (the risk-based ratios, the leverage ratio, the output floor, or a stress test) is set by the whole bank. Which one binds decides which lever moves capital: risk weights matter only while a risk-weighted rule binds, exposure only while the leverage ratio does, and only the standardized weights while the floor does.

27.12 CLIENT CLEARING AND THE COST OF BEING AN INTERMEDIARY

Example 27.3 showed the capital benefit of clearing to the party whose trade is cleared. Somebody has to provide the access, and that role's economics are worse than the table shows.

A client does not face the clearing house directly. A clearing member stands between them, facing the CCP on one side and the client on the other, and the two legs do not cancel: the member owes the CCP whatever the client fails to pay, and holds the client's margin against it. Three consequences follow, and together they explain a surprising market structure.

The member's exposure to the client is a derivative exposure and capitalized as one, even though the member has no market risk on it: each leg's market moves offset the other's, but the two face different counterparties and so cannot be netted.

Client margin is segregated, so the member cannot use it to fund. Chapter 8's segregation requirement, which exists so that the client's collateral survives the member's failure, means the member funds its own contribution to the CCP without the benefit of what it holds. That is Section 9.5's MVA arriving as an operating cost rather than as a charge on a trade.

And the leverage ratio counts the exposure. Section 27.10's crossover applies with force here: the risk weight against a qualifying CCP is 2%, so the risk-based charge is negligible and the leverage requirement binds. Whether segregated client initial margin may be recognized as reducing the leverage exposure measure has been changed more than once and remains among the most consequential technical questions in the framework, because it determines whether

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client clearing is a viable business.

The observable result is concentration. Client clearing in the major currencies is provided by a small number of banks, and several large institutions have exited. A framework designed to reduce concentration of counterparty risk has concentrated access to the mechanism that reduces it—a reminder that capital rules allocate activity as well as loss-absorbency.

Concentration also interacts with Chapter 2's central concern. Porting a defaulted member's client positions requires another member able to take them, so the fewer members there are, the less credible porting becomes—and porting is what the whole client-protection story rests on.

27.13 CONCENTRATION AND THE LIMITS OF BILATERAL MEASUREMENT

Everything above measures a dealer's exposure to a counterparty. Nothing above measures a counterparty's exposure to the market, and that gap is the framework's most important weakness.

The framework is bilateral by construction: each firm computes its own exposure to each of its counterparties, using data it has. A client's positions at other dealers are not data it has.

So a counterparty that is prudently margined at every dealer individually can be catastrophically levered in aggregate, and no participant in the system is in a position to observe it. The only parties who could are the client, who has no obligation to disclose, and the regulators, whose reporting captures cleared and reported trades but not, in useful time, a consolidated view of one client's leverage across brokers. In March 2021 that gap cost the banks more than $10 billion.

27.14 ARCHEGOS

Case Study 27.1 What total return swaps made possible

In March 2021 a family office defaulted on positions built through total return swaps at several prime brokers. Losses to the banks totaled over $10 billion, with a single institution losing more than $5 billion—one of the largest single-counterparty losses in the derivatives market's history, and it involved no exotic instrument and no pricing error. A stylized version shows the arithmetic; the actual exposure, by the SEC's account, peaked near $160 billion.

Take a fund with $10 billion of capital taking five times leverage through total return swaps, split across six prime brokers:

Total gross exposure	$50bn
What each broker sees	$8.3bn
Initial margin held by each, at 15%	$1.25bn

The 15% is a prime brokerage number for a diversified equity book. Now the underlying positions, concentrated in a small number of issuers, fall:

Move	Loss on gross	
−10%	$5.0bn	survivable
−20%	$10.0bn	client's capital exhausted
−30%	$15.0bn	client insolvent; loss beyond the margin falls on the brokers

The client's capital is exhausted at a 20% move, which a portfolio concentrated in a few stocks can make in one bad week. At 30%, each broker holds $1.25 billion of margin against a $2.5 billion loss: a shortfall of $1.25 billion each. This assumes a broker can look only to the margin it holds, since the client's other $2.5 billion of capital is not

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posted. Were all of it collected, the total shortfall would be $5 billion rather than $7.5 billion. Four failures stand out, and they are worth separating because only one is about the instrument.

Invisibility. Section 7.12 said total return swaps are how leverage accumulates quietly, and this is what that meant. Each broker's exposure was individually unremarkable. The swap format also kept the positions out of ownership disclosures, so the concentration was not visible to the market either.

Margin that was not risk-based. Initial margin was set at levels reflecting competition for the client's business rather than the concentration and liquidity of the underlying. The margin period of risk assumption of Example 27.2 (that a portfolio can be closed out in ten days) was false for a position worth many days of trading volume in each issuer.

Concentration measured against the wrong thing. An $8.3 billion exposure is modest against a large bank's balance sheet and enormous against the free float of a mid-cap stock. The relevant denominator is liquidity, not capital, which is the same lesson Section 23.12's single-stock snapshot drew from the other direction.

And a liquidation that moved the price. When the brokers sold, they sold the same stocks into the same market, which is Chapter 22's crowding and Chapter 23's feedback in the least sophisticated form: six firms with the same collateral trying to sell it in the same week. The brokers that sold first lost least, which converted a credit problem into a race.

What the framework did and did not catch. SA-CCR would have registered each broker's exposure and required capital against it; no bilateral framework registers that five other institutions hold the same exposure to the same client. Even a correct measurement of each slice would have left the aggregate invisible, which is the chapter's conclusion.

Supervisory responses have focused on margin practices for concentrated positions and prime brokerage risk governance, gathered in the Basel Committee's 2024 guidelines; a US proposal for disclosing large security-based swap positions was withdrawn in 2025. None solves the aggregation problem, which needs either a consolidated view no institution has or a disclosure obligation on the client.

27.15 INITIAL MARGIN AS THE REAL ANSWER

Chapter 8 described the uncleared margin rules as documentation and operations. What they buy is capital, and the two margin types buy different parts of it.

Variation margin removes current exposure: it is settled daily and brings the mark-to-market back to zero, which is the RC of (27.2). Initial margin removes the gap risk: the move between the last margin call and the close-out, which is precisely the margin period of risk that sets the maturity factor in Example 27.2. The two differ in kind. Variation margin is a transfer of value that has already happened; initial margin is a buffer against value that has not, which is why it must be segregated and cannot be rehypothecated.

Its cost is Section 9.5's MVA, and its benefit is the EAD it removes: initial margin held enters (27.3) as NICA and, once it exceeds the current exposure, lowers the multiplier of (27.10). A dealer's decision about how much initial margin to demand is therefore a trade between a funding cost it pays and a capital cost it avoids, and Archegos is what happens when competition pushes that trade toward the client.

The structural point is that initial margin is the only mitigant that works without information. Netting requires legal certainty; wrong-way risk requires identifying the correlation; concentration limits require knowing the aggregate. Initial margin protects against a move of a given size whatever its cause, which is what makes it right when what you cannot see is what hurts you.

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27.16 WHAT THE FRAMEWORK DOES NOT CAPTURE

The framework leaves four things unmeasured, and each has cost someone money.

Aggregate leverage. It is Section 27.13's gap, which nothing bilateral sees.

Liquidity of the underlying. SA-CCR's supervisory factors are by asset class, not by the liquidity of the specific position. A position worth thirty days of an issuer's trading volume and one worth an hour's carry the same add-on per dollar of notional.

Correlated liquidation. The framework asks what a firm loses if one counterparty fails. It does not ask what it loses if that failure forces five other firms to sell the same assets.

And the model risk in the exposure itself. The exposure profile is a simulation output, and Chapter 21 established that model choice moves such outputs materially; SA-CCR's standardization is partly a response, at the cost of being wrong, conservative for some portfolios and lenient for others, wherever a portfolio differs from those it was calibrated to.

None of this is an argument against the framework. The framework substantially improved on what preceded it, and the CVA charge in particular addressed a loss that had gone entirely uncapitalized. The argument is that a capital framework measures what a firm can see, and the losses that matter most have repeatedly come from what the whole system could not.

Capital is a buffer, not a control. It fixes what must be held once a position exists; it does not decide whether to take the position. The chapter's central episode turned on judgments made inside firms (how much initial margin to demand of a client, which limit to override, how fast a concentrated holding could be sold), and no calibration reaches those. Regulation can make imprudence expensive, which is most of what the post-crisis framework achieved; declining to be imprudent is left to the institution, and Section 27.14 is what it costs when a firm does not.

27.17 WHO BEARS WHAT

Dealers bear counterparty risk, capitalize it under this chapter, price it under Chapter 9, report it in their accounts, and hedge it in the credit market of Chapter 25; Section 9.11 warned that the first three numbers would not reconcile, and the hedge matches none of them exactly.

Clearing houses bear it mutualized, under the default waterfall of Chapter 2. Moving a trade to a CCP converts a bilateral exposure into a contribution to a shared pool, which is a real reduction in system-wide exposure and a real concentration of it in one place.

Clients bear the cost, through the charges Chapter 9 enumerated, seen or not.

And the system bears the residual, which is the part this chapter says cannot be measured bilaterally. Whoever bears that is decided after the fact, and in 2008 and 2021 it favored whoever sold first.

SUMMARY

CVA is a price and regulatory capital is a buffer, computed by different rules for different purposes. A derivative's exposure is an option on the market, bilateral, belonging to the whole netting set, and dependent on mitigants that can fail. Netting is the largest reducer of it: a portfolio whose trades largely offset can carry a net exposure that is a small fraction of its gross, a reduction obtained from a legal clause, so a jurisdiction's netting opinion is worth more to a dealer than any risk model.

SA-CCR sets EAD = $1.4 \times$ (RC + PFE), with RC the collateralized mark-to-market (or the exposure a margin agreement's threshold leaves uncalled) and PFE built from add-ons by asset class. Daily margin cuts a swap's exposure sharply through the maturity factor alone, which scales the add-on to the margin period of risk rather than to the trade's maturity, yet a position that takes weeks to unwind keeps much of its add-on. The capital a swap consumes falls by a large multiple from uncleared to margined and again from margined to cleared, which is why

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capital rules moved more of the market to clearing than the mandate did. The α of 1.4 is a buffer for model error, not a property of any trade.

A separate CVA charge covers spreads widening without default, two-thirds of the crisis's counterparty losses and uncapitalized before it; hedge recognition is uneven, and the EU's exemptions discourage hedging exactly the exposures the charge was built for. Capital filters DVA out of common equity tier 1 because a gain realizable only in default is not a buffer. Wrong-way risk, recovery–default correlation, and vanishing excess spread are one failure mode: protection whose value falls as the need for it rises.

The leverage ratio binds wherever the risk weight is below LR/CR, so it makes the safest, thinnest businesses expensive first: Treasury market-making, repo, and client clearing. A bank's whole requirement stacks minimums of 4.5%, 6% and 8% of risk-weighted assets for common equity, tier 1 and total capital, then conservation, countercyclical and systemic buffers. Which rule binds for the whole bank (risk-based, leverage, floor or stress test) decides whether a desk saves capital by cutting its own risk weights, the standardized ones, or its exposure. And the framework is bilateral: in 2021 a family office that looked prudently margined at each of its prime brokers, and was heavily levered across them through total return swaps, cost them over $10 billion with no exotic instrument and no pricing error. Even a correct measurement of each slice leaves the aggregate invisible. Initial margin is the one mitigant that works without information, and a capital framework measures only what a firm can see.

FURTHER READING

The consolidated Basel Framework, published online by the Bank for International Settlements, is the primary source for Section 27.11: its credit risk chapters hold the standardized weights, the conversion factors and the mitigation rules, and the Basel Committee's *Basel III: Finalising Post-Crisis Reforms* (2017) is the reform that set them. The Basel standards on SA-CCR and on CVA risk are the primary sources and are specific enough to compute from; anyone implementing either should read them rather than a summary, since the hedging-set definitions carry most of the practical content. Gregory's book on counterparty credit risk and XVA is the standard practitioner treatment and covers the pricing of Chapter 9 and the capital of this chapter in one place. On Archegos, the independent report commissioned by the largest loss-making bank sets out how risk limits were set and overridden, and the Basel Committee's *Guidelines for Counterparty Credit Risk Management* (2024) are the supervisory response; the supervisory reviews that followed say more about what changed than about what happened. On wrong-way risk, the literature is small, and the regulatory rule (compute as if default had occurred) remains more widely used than most of the models. On the aggregation gap of Section 27.13, the work on position transparency and on consolidated leverage reporting is active and unresolved.

PRACTICE QUESTIONS

27.1 Explain the difference between CVA as a price and CVA capital as a buffer, and why reducing one need not reduce the other.

27.2 Four trades with a counterparty mark at +30, −12, +8, −20. Compute the exposure with and without netting, and explain why a jurisdiction's netting opinion is described as worth more than any risk model.

27.3 Compute the SA-CCR exposure at default for a portfolio with a replacement cost of $8 million and a PFE add-on of $3 million. Then compute the maturity factor for a ten-day margin period of risk, state what it does to the add-on, and state what assumption the margin period embodies.

27.4 Compute the margin period of risk at which the maturity factor reaches one, and state

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what this means for an illiquid portfolio.

27.5 A bank pays fixed on a three-year $100 million swap, unmargined, with no collateral. Compute its supervisory duration, effective notional and add-on. It then adds a seven-year receive-fixed swap of $47.2 million, chosen so that the two have equal and opposite dollar duration. Compute the hedging set's effective notional and add-on, and explain why a position with no net rate sensitivity still carries most of the add-on, and what would change if the second trade had a four-year maturity instead.

27.6 Using Example 27.3, state the ratio of capital across the three regimes and the two changes that produce it, then convert a capital requirement of $0.20 million on $100 million of notional into a running charge in basis points at a 12% hurdle rate.

27.7 Explain why the clearing mandate is described as less important than the capital rules in moving the market to clearing.

27.8 Explain what $\alpha = 1.4$ is calibrated to bridge, and why it is not a statement about any individual trade.

27.9 Explain what the CVA capital charge covers that the default charge does not, why this loss went uncapitalized before the charge existed, and why the hedges a desk can actually buy are the ones the framework recognizes least.

27.10 Explain why regulatory capital filters DVA out of common equity tier 1 while accounting requires it to be recognized.

27.11 A counterparty has a 4% default probability, 60% loss given default, and an average exposure of $20 million. Compute CVA under independence, and again if the exposure conditional on default is twice the average. Distinguish specific from general wrong-way risk and state how each is treated.

27.12 Give three examples from this book of a protection whose value is negatively correlated with the need for it.

27.13 Derive the risk weight below which the leverage ratio binds at a 10.5% capital requirement and a 5% leverage requirement, and explain why a risk-insensitive constraint makes a dealer's safest businesses expensive first.

27.14 Explain why a clearing member's capital requirement is not eliminated by the fact that its client-facing and CCP-facing legs offset, and how the resulting concentration weakens the porting mechanism of Chapter 2.

27.15 In the Archegos arithmetic, state the move at which the client's capital is exhausted and the shortfall per broker at a 30% move, taking margin as all a broker can collect, and explain why the relevant denominator for concentration is liquidity rather than the dealer's capital.

27.16 Explain why initial margin is described as the only mitigant that works without information, and state four things the capital framework does not capture.

27.17 In Example 27.5, the countercyclical buffer rises from 0.5% to 2.0%. Compute the new CET1, tier 1 and total capital requirements, and state whether the leverage ratio still binds and why the buffer change does not affect the leverage requirement.

27.18 A bank has a $50 million loan with four years to run to a BBB-rated corporate and hedges it with a two-year CDS, covering restructuring, bought from a bank rated A. Using Table 27.1, compute the loan's RWA before and after the hedge, and explain why protection shorter than the loan is recognized only in part.

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27.19 Explain why a securitization tranche below the pool's own capital requirement is weighted 1,250%, and why the floor on the senior tranche's weight is what makes a significant risk transfer trade worth doing.

FURTHER QUESTIONS

27.20 Section 27.5 shows SA-CCR matching α times the internal-model effective expected positive exposure on Example 27.2's swap. Name a portfolio for which SA-CCR would be conservative and one for which it would be lenient, and explain why.

27.21 Section 27.9 argues that recovery-default correlation, disappearing excess spread, and exposure-default correlation are one failure mode. Formalize the common structure, propose a single diagnostic that would detect all three, and apply it to two mitigants used elsewhere in this book.

27.22 Section 27.13 identifies the aggregation gap. Design a reporting regime that would have revealed the Archegos position: specify what would be reported, by whom, to whom, and at what frequency; then state the costs (competitive, operational, and in terms of what the disclosure itself might cause).

27.23 The Archegos snapshot notes that brokers who sold first lost least. Model this as a game between the brokers, identify the equilibrium, and explain why a coordinated liquidation would have produced a better aggregate outcome and why it was not achievable. Relate your answer to Chapter 2's default waterfall, which is designed to solve exactly this problem for cleared trades.

27.24 Section 27.15 argues initial margin works without information. Construct the counter-argument that margin levels themselves require information about volatility, liquidity, and concentration, and reconcile the two positions by distinguishing what must be known to set the level from what must be known for the protection to function.

27.25 Section 27.16 says a capital framework measures what a firm can see. Compare this with Chapter 26's conclusion that a system loses the ability to see a risk before it loses the ability to bear it, and with Chapter 23's observation that 0DTE hedging demand is computable in advance. Use the three to argue either that observability is the binding constraint on financial stability or that it is not.

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